

Local Kummer Realizations and Verdier Duality

Continuous cohomology, nearby cycles, and finite-root traces

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Abstract

For a fixed semistable node, with optional smooth directions and chosen relative Kummer data, we compare ordinary root, nearby-cycle and tube cohomology on a selected underlying-perfect integral adic coefficient source. The common value is continuous inertia cohomology, and the analytic realization is fully faithful on this source. For supported cohomology we use fixed finite-root vector-bundle carriers of the geometric dimension. Their geometric trace system is identified with the dual of continuous cochains, with the dimension shift and Tate twist, and hence with nearby costalks and analytic compact support through the specified Verdier pairings. The supported comparison is constructed through this common dual model; it does not require a morphism from a carrier to the tube or define an exceptional direct image on the bare infinite-root fibre. For the fixed data, the comparisons are natural in coefficients and preserve reduction and module actions. For a node without extra smooth directions, the rational unipotent subsource is equivalent to perfect modules over the continuous cochain algebra.

1 Introduction

The vanishing tube of a semistable node $xy = \pi$ is an open annulus. Its relative Kummer inertia is $\Gamma = \mathbf{Z}_\ell(1)$, and continuous cochains describe the ordinary cohomology of its Kummer coefficients. We relate this description to the relative root fibre and nearby cycles, then identify the corresponding supported objects.

The main result includes smooth disc directions. Write $R_r = \mathbf{Z}/\ell^r$. A coefficient is a derived-cartesian system $V_\bullet$ of continuous Γ -complexes, each perfect over R_r , with a common finite Tor interval; its precise source and realization are given in Definition 2.1.

Theorem 1.1 (Local star comparison and finite-carrier trace). *Let R be a complete strictly henselian discrete valuation ring with algebraically closed residue field k , uniformizer π , and $\ell \neq \text{char } k$. Fix $C = \widehat{\text{Frac } R}$. Put*

$$X = \text{Spec } R[x, y, z_1, \dots, z_e]/(xy - \pi), \quad Y = V(x, y, z_1, \dots, z_e), \quad N = 1 + e.$$

Use the vertical log structure, compatible base roots, and the relative Kummer tower of x ; these fix the splitting $P^{\text{gp}} = \mathbf{Z}\rho \oplus \mathbf{Z}e_x$. Fix the compactification $\bar{X} : X_0Y_0 = \pi X_1Y_1$ in $(\mathbf{P}_R^1)^2 \times (\mathbf{P}_R^1)^e$ and the carriers $g_n : Z_n = \mathbf{A}_{BK_n}^N \rightarrow \text{Spec } k$, where $K_n = \mu_{\ell^n}$ and $\Gamma = \mathbf{Z}_\ell(1)$. Write $p_n : Z_n \rightarrow BK_n$ for the bundle projection and $q_{m,n} : Z_m \rightarrow Z_n$ for the quotient-induced map with the same carrier.

Let $V_\bullet \in \mathcal{S}_{\text{ad}}(\Gamma)$, with notation as in Definition 2.1. The coefficient source is the associated-sheaf construction on these Kummer torsors, with the induced tame transport on its actual log realization and no additional wild base-inertia representation. Its finite-precision extensions and coefficient duals are log-lisse and underlying perfect near Y . The

compactification boundary misses Y and

$$T =]Y[_C = \{ |\pi| < |x| < 1 \} \times (\mathbb{D}_C^\circ)^e$$

is separated, smooth and boundaryless of dimension N . These properties follow from this explicit model and construction. Write $\mathcal{E}_r$ for the generic realization, $j_\eta : X_{\bar{\eta}} \hookrightarrow \bar{X}_{\bar{\eta}}$, $i : Y \hookrightarrow \bar{X}_s$, and $D_\ell = R\mathrm{Hom}_{\mathbf{Z}_\ell}(-, \mathbf{Z}_\ell)$.

- (i) The realization $\mathcal{R}_{\mathrm{ad}} : \mathcal{S}_{\mathrm{ad}}(\Gamma) \rightarrow \mathcal{P}_{\mathrm{ad}}(T)$ is fully faithful. Its ordinary images satisfy

$$Rg_{*,\mathrm{ad}}^{\mathrm{act}} V_\bullet \simeq N_{*,\mathrm{ad}}(V_\bullet) \simeq R\Gamma_{\mathrm{ad}}(T, \mathcal{R}_{\mathrm{ad}} V_\bullet) \simeq K_\Gamma(V_\bullet),$$

where $N_{*,r} = i^* R\Psi_{\bar{X}}^t Rj_{\eta,*} \mathcal{E}_r$.

- (ii) At each precision choose a finite presentation $W_{n_0(r),r}$ of V_r as in Definition 6.3, and inflate it for $n \geq n_0(r)$, setting $E_{n,r} = p_n^* W_{n,r}$. The fixed-precision functor of Lemma 6.4 is computed by

$$H_{!,r}(V_r) = R \varprojlim_{n \geq n_0(r), \mathrm{Tr}^!} Rg_{n,!} E_{n,r}, \quad G_{!,\mathrm{ad}}^{\mathrm{tr}}(V_\bullet) = R \varprojlim_r H_{!,r}(V_r).$$

The root transitions are geometric counits. The coefficient limit uses that lemma's natural maps b_r , followed by the given source reductions. Lemma 6.5 proves their precise perfect-dual reduction square and cartesianness. This construction is presentation-independent and satisfies

$$G_{!,\mathrm{ad}}^{\mathrm{tr}}(V_\bullet) \simeq N_{\mathrm{ad}}^c(V_\bullet) \simeq R\Gamma_{c,\mathrm{ad}}(T, \mathcal{R}_{\mathrm{ad}} V_\bullet) \simeq D_\ell K_\Gamma(V_\bullet^\vee)(-N)[-2N],$$

where $N_r^c = i^! R\Psi_{\bar{X}}^t Rj_{\eta,!} \mathcal{E}_r$. This supported functor is defined by the fixed finite-carrier trace system.

All limits are derived, and the integral outputs recover their cartesian finite-precision systems. The comparisons preserve the star-algebra module actions and the specified coefficient reductions. The trace row is an equivalence in $D(\mathbf{Z}_\ell)$ and its corresponding star-algebra module category. The actual logarithmic star comparisons retain their induced tame transport; additional supplied finite semilinear actions are treated in Proposition 5.6. For input Tor-amplitude $[a, b]$, their perfect output amplitudes are $[a, b+1]$ and $[a+2N-1, b+2N]$. For the unit, $A = K_\Gamma(\mathbf{Z}_\ell)$ has trace partner $D_\ell A(-N)[-2N]$ with its evaluation module structure. Canonicity is relative to the stated model, neutralization, compactification, carrier and enhanced coefficient categories.

The supported comparison is the central construction. Finite root stacks have unbounded group cohomology, whereas continuous inertia has a finite Koszul model. We first form the geometric trace system using the full finite group bars. Pairing with the ordinary root system identifies its limit with the continuous dual. The same pairing identifies the coefficient-reduction arrows b_r and the module actions. Verdier duality then relates this trace limit to nearby costalks and analytic compact support.

Geometric and algebraic inputs. Logarithmic local acyclicity and the log/ordinary nearby comparison come from Nakayama and the exposition of Illusie [12, 13]. Berkovich's formal comparison and analytic duality identify the tube realizations [8, 11]. We use the natural maps in these constructions, so evaluation and trace provide the supported comparison together with its module structure.

Infinite-root geometry and its Kummer interpretation are developed in [14, 15, 16]. Talpo–Vistoli’s topos equivalence [14, Theorem 6.22, arXiv v2 numbering] and the profinite-shape comparisons of [23, §§5–6] supply the surrounding framework. Here the relative construction fixes a base-root neutralization and the ℓ -power tower; finite-presentation descent tracks its structural coefficient functor over the base.

For finite G , Laszlo–Olsson identify compact support on BG with the dual of the full group-cochain complex [17, Example 4.9.2]. This finite duality and the formal passage from a dual colimit to a limit are inputs to our trace construction. The work here is the geometric assembly: identifying its counits, its original coefficient arrows and its comparison with the two supported realizations. The continuous calculation uses the pseudocompact framework of Brumer [22]. Esnault–Kerz [24, Appendix A.2] identify rank-one continuous actions with Iwasawa coefficients and relate derived invariants and coinvariants by the Iwasawa twist and shift. Their construction uses pro-etale constructible coefficients and an artinian coefficient localization in that subsection; here we work with finite-precision and cartesian adic models on the specified underlying-perfect source. We verify the specified coefficient maps and reductions in this model, then assemble the fixed-carrier counits with the actual supported geometric realizations.

Scope and organization. All geometric comparisons use the fixed model, base roots, compactification and carriers in the theorem. Its supported row is a point-valued module comparison. The unipotent corollary supplies the coefficient interface for semistable refinement; cross-stratum links and Gysin maps require their geometric constructions. Model independence and convolution are further questions.

Sections 2–4 construct the coefficients and ordinary realizations. Section 5 proves the supported geometric comparison, and Section 6 constructs the finite-root trace system. The ordinary root and geometric star comparisons supply the mapping comparison in Section 4; the geometric supported comparison and finite-root trace meet in Section 7, which completes the main theorem and gives examples. The appendices collect adic recovery, descent over a noetherian base, the relative root site, the continuous dualizing module, and the general geometric comparison tools. Their constructible coefficient and star results use the full selected source; general-base coefficient duality uses the lisse-perfect domain specified there.

2 The node and its Kummer coefficients

Fix the data of Theorem 1.1. The chart is $\mathbf{N} \rightarrow P = \mathbf{N}^2$, $1 \mapsto \rho = (1, 1)$. The coordinate x identifies $P^{\text{gp}}/\mathbf{Z}\rho$ with $\mathbf{Z}$. Fixing compatible roots of π leaves the relative groups $K_n = \mu_{\ell^n}$ and $\Gamma = \varprojlim_n K_n$. The cover $a^{\ell^n} = x$ on the generic fibre is a K_n -torsor; its associated-sheaf functor constructs the coefficients used below.

A point of the generic fibre specializes to Y precisely when $|x|, |y|, |z_i| < 1$. Thus

$$T = \{ |\pi| < |x| < 1 \} \times (\mathbb{D}^\circ)^e.$$

We compactify by $X_0 Y_0 = \pi X_1 Y_1$ in $(\mathbf{P}_R^1)^2 \times (\mathbf{P}_R^1)^e$; the boundary misses Y . The finite root carriers are $\mathbf{A}_{BK_n}^N$, with $N = 1 + e$. Their vector-bundle trace fixes the factor $(-N)[-2N]$.

Throughout, $R_r = \mathbf{Z}/\ell^r$, n is root level (root order ℓ^n), and r is coefficient precision. General-base coefficient conventions appear in Definition A.1. We take the continuous root (co)limit at fixed r , recover the integral coefficient system, and finally invert ℓ if desired. Write $V^\vee = R\text{Hom}_{\mathbf{Z}_\ell}(V, \mathbf{Z}_\ell)$ with contragredient action. This coefficient dual differs from the geometric source dual $V^\vee(N)[2N]$.

Definition 2.1 (The source and analytic target of adic realization). Let $\mathcal{A}_{\Gamma,r}$ be discrete continuous Γ -modules with coefficients R_r , and set

$$\mathcal{S}_r = \{V \in D(\mathcal{A}_{\Gamma,r}) : V \text{ is perfect over } R_r\}, \quad \mathcal{P}_r(T) = \text{Perf}_{\text{lisse}}(T_{\text{ét}}, R_r).$$

The latter is the full subcategory of module-sheaf complexes locally represented by bounded finite free complexes, with finite locally constant cohomology. Both are full subcategories of the stable localizations at quasi-isomorphisms. Scalar reduction is the *derived* functor $-\otimes_{R_{r+1}}^L R_r$ in the ambient categories. Write $\mathcal{S}_{\text{ad}}(\Gamma)$ and $\mathcal{P}_{\text{ad}}(T)$ for the full subcategories of their ambient category limits consisting of cartesian systems in the displayed classes with one common finite Tor interval. This includes *all* such systems. Proposition B.14 proves their classical Iwasawa-admissibility at a point. No common finite root level or trivializing cover across all r is required. The point realization of the underlying system of $V_{\bullet}$ is $V = R\varprojlim_r V_r$; its perfectness is a conclusion of Lemma A.9.

Tensor, the coefficient dual, and the internal coefficient Hom are formed levelwise: $V_r^{\vee} = R\text{Hom}_{R_r}(V_r, R_r)$ with contragredient action, and $\underline{\text{Hom}}(V_r, W_r) = V_r^{\vee} \otimes_{R_r}^L W_r$. These are distinct from the global equivariant mapping complex. The same conventions apply on T . Completed tensor means the resulting cartesian system; no ordinary integral sheaf is substituted. Set

$$K_{\Gamma}(V_{\bullet}) = R\varprojlim_r C_{\text{cts}}^*(\Gamma, V_r).$$

We also write this as $R\text{Hom}_{\mathbf{Z}_{\ell}[[\Gamma]]}^{\text{cts}}(\mathbf{Z}_{\ell}, V_{\bullet})$, computed by the finite augmentation Koszul complex in the adic module model of Theorem B.13. Proposition B.14 constructs classical pseudocompact models for the entire point source, and Proposition B.15 identifies the resulting calculation with classical continuous derived Hom. The definition of the source and its mapping spectra is the cartesian-system one.

Point outputs and the three action data. The point-valued adic target is $\varprojlim_r D(R_r)$; its uniformly perfect cartesian systems recover in $D(\mathbf{Z}_{\ell})$. The star algebra is $A_{\bullet} = (K_{\Gamma}(R_r))_r$, and the supported row is compared in its module category, hence after recovery in $\text{Mod}_{K_{\Gamma}(\mathbf{Z}_{\ell})}(D(\mathbf{Z}_{\ell}))$. The relative Γ -action belongs to the coefficient source. External tame transport belongs to the actual geometric log source and is retained by its specialization and star comparisons. An additional finite semilinear descent diagram is separate data, handled by Proposition 5.6. The finite-carrier trace row is stated after forgetting external base transport; a full infinite tame-equivariant trace row is not asserted by its point-valued proof. Forgetting that action takes no invariants under base tame inertia.

3 Root coefficients and continuous cohomology

On the chosen logarithmic fibre, finite root reductions are BK_n . Finite nilpotent thickening does not change their étale coefficient topoi. The following descent results identify the actual root structural functor, including its maps, with continuous invariants.

Proposition 3.1 (Finite derived action descent over a point). *Fix r . Let $\mathcal{D}_{n,r}^{b,\text{fin}}$ be the full subcategory of $D(R_r[K_n]\text{-Mod})$ with bounded finite underlying cohomology, and let $\mathcal{D}_{\Gamma,r}^{b,\text{fin}}$ be the analogous subcategory of the derived category of discrete continuous Γ -modules. Inflation induces an equivalence of stable categories*

$$\text{colim}_n \mathcal{D}_{n,r}^{b,\text{fin}} \simeq \mathcal{D}_{\Gamma,r}^{b,\text{fin}}.$$

It restricts to the full subcategories whose underlying R_r -complex is perfect. A finite simplicial diagram, including its specified higher simplices, descends to a common level. This does not assert descent of an entire infinite diagram at one level.

Proof. First consider finite modules M, N with action through K_{n_0} . Resolve M by a bounded-above complex P of finitely generated free $R_r[K_{n_0}]$ -modules. On forgetting the action this is a resolution by finite free R_r -modules. The equivariant bar construction applied to this resolution computes derived equivariant Hom: its double complex has terms

$$C^a(K_n, \operatorname{Hom}_{R_r}(P^{-b}, N)), \quad a, b \geq 0.$$

Use continuous cochains for Γ . This computation follows by taking the bar resolution relative to the forgetful functor to R_r -modules; the free underlying resolution makes each vertical Hom derived-correct. Each diagonal $a + b = t$ is finite. Every continuous map from the compact space Γ^a to the finite coefficient module factors through some K_n^a : its finitely many clopen fibres have a common open normal subgroup on which they are invariant. Consequently the filtered colimit of the finite double complexes is the continuous double complex, with all differentials, and exactness of that colimit gives

$$\operatorname{colim}_{n \geq n_0} R\operatorname{Hom}_{R_r[K_n]}(M, N) \simeq R\operatorname{Hom}_{\Gamma, \operatorname{disc}}(M, N).$$

There is no finite-group truncation in this argument.

Finite Postnikov filtrations in both variables extend the identity to bounded objects with finite cohomology. In particular it identifies mapping spectra in all degrees. Every finite continuous module has action through a finite quotient. Induct on the number of nonzero cohomology degrees of an object on the continuous side. Present its lower truncation and its highest cohomology module at finite levels; the mapping identity presents the Postnikov attaching morphism at a common larger level. Its fibre presents the object. This proves essential surjectivity without asserting that the cohomology modules of a perfect complex are perfect. Inflation leaves the underlying R_r -complex unchanged, so the equivalence restricts to the underlying-perfect class. For the last assertion use the filtered-quasicategory model and finite-diagram argument in Lemma B.3, applied to the category equivalence just proved. It descends the finitely specified simplices themselves, not only the actions on cohomology. $\square$

Corollary 3.2 (The actual root structural functor over a geometric point). *Over an algebraically closed point, the actual relative root site has the generating site of finite continuous Γ -sets. Its sheaf topos is the topos of discrete continuous Γ -sets, structural pullback is trivial action, and actual derived star on the selected coefficients is $C_{\operatorname{cts}}^*(\Gamma, -)$.*

Proof. For the node, at order $a = \ell^n$ the rigidified atlas is explicitly $\operatorname{Spec} k[u, v]/(u^a, v^a, uv)$ with μ_a -weights $(1, -1)$. Its monomial reduction is the point, and refinements send u_n, v_n to the corresponding powers of u_m, v_m . Lemma C.4 verifies the nerve and test-torsor properties used in the limit argument (and gives the formula for the other split log-point charts). Theorem C.9 descends etale fp objects, maps and coverings, allowing fpqc-stack sources. At a finite level, nil reduction gives BK_n . An etale fp algebraic space over an algebraically closed field is a finite set of points. Descent along the torsor atlas equips it with a K_n -action; covering families are jointly surjective maps. The filtered site is therefore finite continuous Γ -sets. The generating-site argument, with the smoothification just specified, identifies its sheaves with smooth sets. Structural pullback is trivial action on finite base sets, which generate the base topos, hence on all base sheaves. The resulting exact equivalence of module-sheaf abelian categories identifies injectives and the right adjoint of structural pullback. Deriving in their complex localizations gives continuous invariants with its actual unit. Point finite-action descent and the finite augmentation resolution prove selected boundedness and reduction. This point argument uses neither general-base constructible descent nor Δ -rectification. $\square$

Put $\Lambda = \mathbf{Z}_\ell[[\Gamma]]$. A generator γ gives $\Lambda = \mathbf{Z}_\ell[[\gamma - 1]]$ and

$$K_\Gamma(V) = R\mathrm{Hom}_\Lambda^{\mathrm{cts}}(\mathbf{Z}_\ell, V) \simeq [V \xrightarrow{\gamma-1} V].$$

The two displayed copies have relative degrees 0, 1. In rank d the corresponding complex is the Koszul complex of $\gamma_1 - 1, \dots, \gamma_d - 1$. Naturality is expressed by derived continuous cohomology; the coordinates only supply a finite model.

For the following calculation let M be a finite free lattice, $\Gamma_M = M \otimes \mathbf{Z}_\ell(1)$, and $\Lambda_M^{\mathrm{Iw}} = \mathbf{Z}_\ell[[\Gamma_M]]$. The node uses $M = \mathbf{Z}$. Write $\mathbf{K}_M^{\mathrm{Iw}}$ for the augmentation Koszul resolution in (16), and $\mathbf{B}_M^{\mathrm{cts}}$ for the completed normalized bar resolution of the same trivial module. Admissibility is the point coefficient condition of Definition A.1; Proposition B.14 verifies it for our entire point source.

Lemma 3.3 (Continuous bar–Iwasawa comparison). *There is an augmentation-preserving equivalence*

$$\mathbf{K}_M^{\mathrm{Iw}} \xrightarrow{\sim} \mathbf{B}_M^{\mathrm{cts}}$$

in the derived category of complete Λ_M^{Iw} -modules. For an Iwasawa-admissible coefficient V_{cont} , both sides compute continuous group cohomology:

$$R\Gamma_{\mathrm{cts}}(\Gamma_M, V_{\mathrm{cont}}) \simeq R\mathrm{Hom}_{\Lambda_M^{\mathrm{Iw}}}^{\mathrm{cts}}(\mathbf{K}_M^{\mathrm{Iw}}, V_{\mathrm{cont}}). \quad (1)$$

The comparison is natural in continuous lattice maps and finite groups of lattice automorphisms. The cup product is the canonical multiplication on the derived endomorphism object of the augmentation module; it does not depend on a strict choice of comparison map.

Proof. After the chosen basis, $\Lambda_M^{\mathrm{Iw}} = \mathbf{Z}_\ell[[T_1, \dots, T_d]]$ and the augmentation module is the quotient by the regular sequence $(T_1, \dots, T_d)$. Hence (16) is a finite free resolution. The completed bar complex is a topologically free resolution of the same augmentation module. The comparison theorem for projective resolutions gives the displayed equivalence, uniquely up to contractible homotopy in the enhanced derived category. Applying continuous internal Hom gives (1). Naturality and equivariance follow by using the basis-free derived object $R\mathrm{Hom}_{\Lambda_M^{\mathrm{Iw}}}^{\mathrm{cts}}(\mathbf{Z}_\ell, -)$; the chosen Koszul complex is only a finite model for it. $\square$

4 Analytic realization and nearby cycles

The Kummer torsors of the node define both the generic coefficient and its logarithmic extension. We first compute its nearby and tube cohomology, then prove full faithfulness by comparing internal Hom. Write $R\Theta$ for the specialization functor of Definition E.1. The general log-smooth comparison used in this calculation is Theorem E.6 in Appendix E.

Corollary 4.1 (Kummer coefficients on a semistable node). *Let the residue field be algebraically closed, put*

$$X = \mathrm{Spec} R[x, y, z_1, \dots, z_e]/(xy - \pi), \quad Y = V(x, y, z_1, \dots, z_e),$$

and give X the log structure of its special fibre. Take the full generic fibre near Y and no extra descent group. For every selected bounded derived continuous $\Gamma = \mathbf{Z}_\ell(1)$ coefficient, there is a compatible actual Kummer realization on the geometric tube

$$]Y[_{\bar{\eta}} = \{ |\pi| < |x| < 1 \} \times (\mathbb{D}^\circ)^e.$$

For this source the actual root star, actual analytic star, and tame nearby star are identified by Theorem E.6, including algebra/module structures and selected adic recovery. The analytic coefficient realization is fully faithful on the selected underlying-perfect derived category.

Proof. The chart $\mathbf{N} \rightarrow \mathbf{N}^2$, $1 \mapsto (1, 1)$, is vertical, log smooth and saturated, with primitive base element and relative rank one. Its boundary-free neighbourhood satisfies Theorem E.6. At precision r , Proposition 3.1 presents the coefficient and its maps over some $K_n = \mu_{\ell^n}$. The fs Kummer cover

$$\mathrm{Spec} R[a, y, z]/(a^{\ell^n} y - \pi) \longrightarrow X$$

is a K_n -torsor: in its fs self-fibre product the ratio of the two roots is an ℓ^n th root of unity, giving the deck copies. Associated sheaves are exact and refinement agrees with inflation. They therefore construct the log-lisse extension on the entire selected derived source.

This covering scheme is fs log smooth but need not be saturated over R ; the local-acyclicity input only needs fs log smoothness. After adjoining compatible π_n , a saturated model of its selected geometric component is $ab = \pi_n$, with $x = a^{\ell^n}$, $y = b^{\ell^n}$ and deck action $(a, b) \mapsto (\zeta a, \zeta^{-1} b)$. Refinement raises both coordinates to ℓ^{m-n} . This gives the stated annulus and discs.

Apply Theorem E.6 to the constructed coefficient. For perfect E, F their realized internal Hom is the associated sheaf of $R\mathrm{Hom}_{R_r}(E, F)$ with conjugation action. Its star comparison is the realization-induced mapping map, as verified in Lemma 4.3; its source is $C_{\mathrm{cts}}^*(\Gamma, R\mathrm{Hom}_{R_r}(E, F))$. This proves full faithfulness. Natural reduction and the uniform Koszul bounds give the cartesian adic statements. $\square$

Theorem 4.2 (Realization of local Kummer coefficients). *Let $Y = \mathrm{Spec} k$ be a geometric face of a vertical fs saturated log-smooth algebraization, with the primitive split chart $\mathbf{N}\rho \rightarrow P$ of Definition E.2, and suppose the generic boundary misses Y . Fix the chart's relative base-root neutralization, an integral splitting $P^{\mathrm{gp}} = \mathbf{Z}\rho \oplus Q$, and no additional descent group. Canonicity below is relative to these data. Every selected bounded derived continuous coefficient for $\Gamma = \mathrm{Hom}(P^{\mathrm{gp}}/\mathbf{Z}\rho, \mathbf{Z}_{\ell}(1))$ has an actual lisse Kummer generic realization. On this source,*

$$Rg_*^{\mathrm{act}} E_r \simeq C_{\mathrm{cts}}^*(\Gamma, E_r) \simeq i_Y^* R\Psi_X^t \mathcal{E}_r \simeq R\Theta(\mathcal{E}_r^{\mathrm{an}}|_{|Y|}).$$

The same holds for the selected cartesian adic systems, with uniform bounded recovery, and after rationalization. The comparisons preserve the unit algebra and coefficient modules and all coefficient maps. The analytic realization is fully faithful on this selected source. The coefficient extension is constructed in the proof.

Proof. Here is the source construction in arbitrary rank. Primitivity allows an integral splitting $P^{\mathrm{gp}} = \mathbf{Z}\rho \oplus Q$; fix one compatible with the chosen neutralization. The absolute chart root cover adjoining P/ℓ^n is a torsor in the Kummer etale topology with group

$$A_n = \mathrm{Hom}(P^{\mathrm{gp}}, \mu_{\ell^n}) = K_n \times \mu_{\ell^n}.$$

The second factor is the base-root direction. Extend a K_n -module along the projection $A_n \rightarrow K_n$ and form its associated sheaf on X_{ket} . Over the chosen geometric base-root lift the relative action is the original K_n -action. This is an actual finite Kummer torsor construction, even though its underlying scheme map is not etale on the special fibre. Its generic restriction is ordinary etale because the log structure is vertical.

Associated sheaves are exact in the module variable. Compatible chart root covers make this construction commute with inflation. Pointwise finite derived-action descent therefore extends it to all selected bounded derived continuous coefficients, with their morphisms and finite diagrams. The result is an actual log-lisse extension of the kind required by Theorem E.6. Apply that theorem, Proposition C.23 and Corollary C.21 to obtain the displayed comparisons. The analytic term means $R\nu_* \mu^*$, as specified in Definition E.1.

For full faithfulness apply the same comparison to the realized internal Hom of two underlying-perfect coefficients, as in Corollary 4.1. Its source is the continuous equivariant

mapping complex; the comparison is the map induced by the same associated-sheaf realization; the classifying-map and unit verification is spelled out in Lemma 4.3. Proposition 4.4 gives the adic functor and full faithfulness; uniform Koszul bounds give its point-valued cohomology recovery. The construction is relative to the fixed neutralization; descent between arbitrary choices or different characteristic strata is not asserted here. $\square$

Lemma 4.3 (The local mapping comparison is induced by realization). *Fix the geometric-face chart, relative splitting and base-root lift of Theorem 4.2, and write $\mathcal{R}_r$ for its actual associated-sheaf realization on $T =]Y[_{\bar{\eta}}$. For underlying-perfect continuous coefficients V, W , the canonical map induced by that functor is an equivalence*

$$R\mathrm{Hom}_{\Gamma, \mathrm{disc}}(V, W) \longrightarrow R\mathrm{Hom}_{T_{\acute{e}t}}(\mathcal{R}_r V, \mathcal{R}_r W).$$

This identifies the realization-induced map of mapping complexes. Its finite-precision reduction squares induce the adic assertion of Proposition 4.4.

Proof. Use the derived model of Lemma E.17. First make the analytic site comparison explicit. Over a geometric face, Theorem E.6 takes values in $R\Gamma(T_{q\acute{e}t}, \mu^* F)$. For the bounded constructible R_r -torsion complexes used here, start with their ordinary etale cohomology sheaves F . The canonical map

$$R\Gamma(T_{\acute{e}t}, F) \longrightarrow R\Gamma(T_{q\acute{e}t}, \mu^* F)$$

is an equivalence by [9, Theorem 3.3(i)–(ii)], with $U = X = T$. Finite truncation triangles extend this to our bounded complexes. This is the natural map of the tensor inverse image μ^* , so it respects evaluation and tensor pairings. More explicitly, μ^* is exact and strong symmetric monoidal; the displayed global-sections map is its adjunction-unit map. The ordinary/quasi-etale site square for a finite etale Kummer torsor commutes on inverse images. Naturality of the unit in that square gives compatibility with torsor pullback. Thus the canonical comparison map preserves these structures. It justifies using ordinary analytic etale global sections in this proof; it neither supplies $\mu^!$ nor realizes an arbitrary quasi-etale sheaf on the ordinary etale site.

At a finite level, the associated-sheaf functor for the actual Kummer torsor is an exact tensor inverse image. It preserves derived tensor products: equivariant free resolutions are underlying-flat, and their associated sheaves are locally flat. Its compatible inflation maps therefore give a tensor realization after finite derived-action descent. In particular, for underlying-perfect V, W , it identifies

$$\mathcal{R}_r(R\mathrm{Hom}_{R_r}(V, W)) \simeq R\mathcal{H}om_T(\mathcal{R}_r V, \mathcal{R}_r W),$$

with the conjugation action on the coefficient on the left. Here perfectness supplies the evaluation identity $V^\vee \otimes W$; the tensor realization preserves this evaluation.

Let $U = R\mathrm{Hom}_{R_r}(V, W)$ and present it at a finite level K_n . Its actual torsor classifying map supplies

$$\kappa_n : C^*(K_n, U_n) \longrightarrow R\Gamma(T, \mathcal{R}_r U),$$

by the pullback/direct-image adjunction, as in Proposition E.10. These maps commute with inflation. In the relative logarithmic fibre the same classifying map is induced by $\Gamma \rightarrow K_n$: the absolute chart torsor has first been restricted along the fixed geometric base-root lift. Consequently its map there is the canonical inflation $C^*(K_n, U_n) \rightarrow C_{\mathrm{cts}}^*(\Gamma, U)$.

Naturality of the log-specialization unit and of completion for this actual torsor classifying map gives the commuting triangle

$$\begin{array}{ccc} C^*(K_n, U_n) & \longrightarrow & C_{\mathrm{cts}}^*(\Gamma, U) \\ & \searrow \kappa_n & \downarrow \beta\gamma \\ & & R\Gamma(T, \mathcal{R}_r U). \end{array}$$

To see the map-level assertion, construct both diagonals by inserting the adjunction unit for the finite torsor and then the specialization unit. Transitivity of these units gives the two composites, while the completion map is their mate. Thus the triangle compares the same classifying morphism on the two actual sites. It does not follow from merely having the same deck group, and does not assert that κ_n is an equivalence at finite n .

The middle vertex and the homotopy in that triangle can be retained explicitly. Put $L(U) = R\Gamma(Y, i_Y^* R\Psi_X Rj_* \mathcal{R}_{\eta, r} U)$. For each finite presentation the diagram is

$$\begin{array}{ccccc} C^*(K_n, U_n) & \xrightarrow{\text{inf}} & C_{\text{cts}}^*(\Gamma, U) & \xrightarrow{\gamma} & L(U) \\ \downarrow \kappa_n & & \downarrow \beta\gamma & & \downarrow \beta \\ R\Gamma(T, \mathcal{R}_r U) & = & R\Gamma(T, \mathcal{R}_r U) & = & R\Gamma(T, \mathcal{R}_r U). \end{array}$$

Construct its left rectangle on the augmented Čech nerve of the actual torsor. In degree q use the $(q+1)$ -fold fibre product of that torsor, its log restriction, and its formal completion. All face and degeneracy maps are the pullback projections and diagonals of this one torsor. The specialization unit is natural for these maps; the completion mate is obtained from the same inverse-image square in each degree. The unit identities therefore give a map of augmented cosimplicial diagrams. Totalization supplies the displayed homotopy, with compatibility under refinements from the morphisms of these nerves. The coherent composition cells are those of these actual functors and their adjunctions. Applying this construction to evaluation and tensor products proves compatibility with composition of internal Homs. This describes the homotopy used above and does not identify any finite-level bar with the continuous output.

Take the filtered colimit of the source of this triangle at fixed r . Point finite-action descent identifies it with continuous cochains; hence its diagonal is precisely the realization-induced map. Theorem E.6 proves that $\beta\gamma$ is invertible. Applying the internal-Hom identity above now proves the displayed mapping equivalence, compatibly with evaluation, identities and composition. The finite group bars have remained full throughout.

For adic systems, use the coefficient-compatible tensor realizations and their actual cartesian outputs. Mapping spectra in the limit category of such systems are the inverse limits of the finite precision mapping spectra, so the just proved equivalences give adic full faithfulness. This is a statement about adic systems, not an identification with an ordinary inverse-limit sheaf on the small étale site. No common finite action level for all r is used. $\square$

Proposition 4.4 (Adic realization and its mapping spectra). *The finite Kummer associated-sheaf functors induce an exact tensor functor $\mathcal{R}_{\text{ad}} : \mathcal{S}_{\text{ad}}(\Gamma) \rightarrow \mathcal{P}_{\text{ad}}(T)$. It preserves coefficient duals and is fully faithful. For two systems its mapping map is*

$$\text{Map}_{\mathcal{S}_{\text{ad}}}(V_{\bullet}, W_{\bullet}) = \varprojlim_r \text{Map}_{\mathcal{S}_r}(V_r, W_r) \xrightarrow{\sim} \varprojlim_r \text{Map}_{\mathcal{P}_r(T)}(\mathcal{R}_r V_r, \mathcal{R}_r W_r).$$

Both limits are limits of spectra, including their derived information.

Proof. On a finite torsor the associated-sheaf functor is an exact strong tensor inverse image. Hence its derived reduction square commutes canonically, including iterated reductions. Finite-action descent extends these squares to the selected continuous source. At each precision a finite presentation is a bounded complex of finite modules with action through a finite quotient; its underlying perfectness makes its realized complex locally perfect on that torsor. Thus it has the same local Tor bounds and lies in $\mathcal{P}_r(T)$.

For perfect objects, derived scalar extension preserves duals and identifies internal Hom with dual tensor. Tensor adds Tor intervals, and duality sends $[a, b]$ to $[-b, -a]$; these bounds are uniform. This proves closure of both selected categories under these coefficient operations

and proves cartesianness of their results. It does not assert closure under arbitrary geometric pushforwards.

The universal property of a limit of stable categories computes mapping spectra by the displayed limits for cartesian sections. Passing to a full selected subcategory leaves those maps unchanged. The finite-precision assertion of Lemma 4.3 identifies each map by realization, and its torsor/reduction naturality identifies their transition squares. Taking the limit proves full faithfulness. No interchange of homotopy groups with limits, and hence no Mittag-Leffler assertion, is used here. Perfect recovery for point outputs is the separate result of Lemma A.9. Rationalization is performed on these integral categories and mapping spectra afterwards. $\square$

Corollary 4.5 (Local essential image and the scope of closure). *The fixed-source realization of Theorem 4.2 is an equivalence onto its full analytic essential image. That image is closed under finite cones of arbitrary analytic morphisms between its objects and under retracts. Thus there is no further mapping or attaching-map existence problem inside this one local Kummer image.*

Proof. The preceding lemma lifts every analytic mapping spectrum between two realized objects to the source. Exactness then realizes its cones. The selected underlying-perfect source is idempotent complete, so full faithfulness also lifts and splits retracts. This argument is confined to a fixed geometric face and source; it neither glues different characteristic strata nor identifies the distinct proper-image realizations in Proposition F.1. $\square$

5 Compact support and Verdier duality

We compare analytic compact support with nearby costalks by their Verdier pairings with ordinary cohomology. For a smooth geometric source of dimension N , the coefficient dual $F^{\vee_0}$ gives the source dual $F^{\vee_0}(N)[2N]$. The same factor occurs in the finite-carrier trace. The general-base conventions are collected in Definition A.11.

Lemma 5.1 (Open-neighbourhood locality of the nearby costalk). *Let $k : W \hookrightarrow \overline{X}$ be an open neighbourhood of Y , with $i_Y = k_* i_{Y,W}$. Canonical open base change gives*

$$k_s^* R\Psi_{\overline{X}} F \simeq R\Psi_W k_{\eta}^* F, \quad i_Y^! R\Psi_{\overline{X}} F \simeq i_{Y,W}^! R\Psi_W k_{\eta}^* F.$$

These also hold for tame nearby cycles and retain coefficient maps.

Proof. Open restriction is localization of the ordinary etale site. Apply Lemma E.16 to the generic direct image in nearby cycles and commute the special inverse images. This proves the first identity with its canonical map. The second uses transitivity of exceptional pullback and $k_s^! = k_s^*$. Wild invariants commute with restriction, giving the tame assertion. For W obtained by deleting the generic boundary closure, $W_{\overline{\eta}} \subset U_{\overline{\eta}}$; both restrictions of $Rj_{\eta,*} E$ and $Rj_{\eta,!} E$ equal $E|_{W_{\overline{\eta}}}$. Thus duality is applied on a separated finite-type neighbourhood to the bounded finite-Tor lisse source and its smooth source dual. The displayed identities return this local costalk to the global notation, without a global wild-triviality hypothesis. $\square$

Proposition 5.2 (Nearby-cycle duality). *Let $\overline{X}/R$ be separated of finite type and use geometric nearby cycles with R_r -coefficients. With dualizing complexes relative to the geometric generic and special points, the canonical comparison is*

$$R\Psi_{\overline{X}} \mathbb{D}_{\overline{X}_{\overline{\eta},r}} F \xrightarrow{\sim} \mathbb{D}_{\overline{X}_s,r} R\Psi_{\overline{X}} F$$

for the bounded constructible finite-Tor coefficient class used here. The duality map is [10, Corollary 3.8]; Corollary 3.9 supplies bounded constructibility and finite-Tor preservation.

These are assertions for finite-type schemes. The duality comparison preserves its evaluation pairing; see also [10, §3.1].

Suppose that, near a selected face Y , the model is vertical log smooth, its generic fibre is smooth of pure dimension N , the generic boundary misses Y , and F is the generic restriction of a lisse Kummer complex with the tame descent convention of Definition E.2. Then both $R\Psi F$ and $R\Psi \mathbb{D}F$ have trivial wild action on their cohomology in an open neighbourhood of Y . Consequently the ordinary duality identity supplies the tame identity needed for the costalk formula on that neighbourhood.

Proof. In [10, §3.1], the base is a Henselian valuation trait and the coefficient ring is Noetherian torsion, annihilated by an integer invertible on the base. Here it is $\mathrm{Spec} R$ and R_r , and our bounded finite-Tor constructible input belongs to their D_c^- and D_{cft} . Their target is $\overline{X}_s \times_s \eta$; restriction to the chosen geometric generic point gives ordinary geometric nearby cycles. Under this restriction their relative dualizing objects become those over $\bar{\eta}$ and $\bar{s}$ (Remark 3.6). Thus a smooth generic N -fold has dualizing object $R_r(N)[2N]$; there is no additional trait shift. Pairing compatibility is the one following their Corollary 3.9. We use no Artin-stack extension of that result; the passage to tame cycles and costalks is below.

For the second assertion use Lemma 5.1 and remove the closure of the generic boundary. All ensuing tame identities, $j_*/j_!$ identifications and their Verdier mates are taken on this restricted open neighbourhood; only afterwards is $i_Y^!$ applied. They are not assertions of wild triviality for the unlocalized global Rj_*F . On the remaining neighbourhood both j_* and $j_!$ of the coefficient restrict to that generic coefficient. Since the generic fibre is smooth, its absolute dual is $F^{\vee_0}(N)[2N]$; dualizing the log-lisse extension and adding this single twist and shift gives a log-lisse extension of the dual. Log local acyclicity and the log/ordinary exchange of Lemma E.5 apply to both extensions. Write W for this one neighbourhood, $G = \mathrm{Gal}(\bar{K}/K)$ and $P \subset G$ for wild inertia. The action of G on the geometric log special fibre factors through G/P [13, §8.1]. The chosen log-lisse extension $\mathcal{L}$ is pulled back from the specified Kummer coefficient; its transport is the induced geometric transport, with no additional wild representation on its coefficients. Consequently P acts trivially on that log special fibre and its restricted coefficient. The local-acyclicity unit and the log/ordinary exchange give

$$C_F := R\Psi_W F \simeq R\varepsilon_{s,*}(\mathcal{L}|_{W_s^{\mathrm{log}}}).$$

Here the comparison is the exchange map in the category with smooth G -action, as in [13, §8.2, (8.2.3)], not an isomorphism after forgetting the action. Functoriality of $R\varepsilon_{s,*}$ therefore makes P act trivially on every cohomology sheaf of C_F . This is the argument of [13, Corollary 8.4, (8.4.1)–(8.4.2)]; the extension from constant to the selected log-lisse coefficients is supplied by Lemma E.5. It applies also to the dual extension on the same W : dualization preserves the trivial wild transport, and the prime-to- p Tate factor has trivial wild action.

For either nearby complex C , the canonical map from its smooth wild invariants to C has cohomology maps $H^j(C)^P \rightarrow H^j(C)$. Exactness of these invariants, as proved in Lemma E.15, makes all these maps isomorphisms. Thus tame and ordinary nearby cycles agree on W , before applying $i_Y^!$. This order is essential: trivial wild action on i_Y^*C alone would not identify $i_Y^!C$ with the costalk of its tame part. The actual algebraic identities $\mathbb{D}i_Y^* = i_Y^!\mathbb{D}$ and $\mathbb{D}Rj_* = Rj_!\mathbb{D}$ now give the asserted tame costalk duality on this domain. No invariants under the remaining tame base inertia are taken; its action remains external. $\square$

Definition 5.3 (The local supported domain). Use the actual source of Theorem 4.2, with Y a closed geometric point. Let $N = d + e$ be the pure generic dimension, and choose the supplied proper algebraic compactification $\overline{X}/R$, with Y disjoint from the generic-boundary

closure. Set $T =]Y[_{\bar{\eta}}$. Require the selected coefficients to have the uniform perfectness conditions already used in Theorem 4.2. Write $j_{\eta} : U_{\bar{\eta}} \hookrightarrow \bar{X}_{\bar{\eta}}$ for the given generic open and $i_Y : Y \hookrightarrow \bar{X}_s$. The two actual supported objects are

$$A_r^c(E) = R\Gamma_c(T, \mathcal{E}_r^{\text{an}}), \quad N_r^c(E) = i_Y^! R\Psi_{\bar{X}}^t(Rj_{\eta,*}\mathcal{E}_r).$$

The first uses compact support on the ordinary analytic etale site. The second uses the algebraic costalk; its outer face direct image is the identity because the face base is the geometric point. Neither object is defined here as the dual of a star object.

Proposition 5.4 (Analytic compact-support duality). *In Definition 5.3, T is a separated smooth boundaryless analytic space of pure dimension N . For a selected lisse perfect F on T , its actual compact-support trace gives*

$$R\Gamma_c(T, F) \xrightarrow{\sim} D_r R\Gamma(T, F^{\vee 0}(N)[2N]), \quad D_r = R\text{Hom}_{R_r}(-, R_r).$$

For the realized Kummer coefficients the two sides are perfect.

Proof. The closed-point tube T is an open analytic domain in $\bar{X}_{\bar{\eta}}^{\text{an}}$. The boundary-disjoint hypothesis places it in the smooth generic neighbourhood, so its structural map is separated, smooth and boundaryless of dimension N . With $j_T : T \hookrightarrow \bar{X}_{\bar{\eta}}^{\text{an}}$, $R\Gamma_c(T, F) = R\Gamma(\bar{X}_{\bar{\eta}}^{\text{an}}, j_{T,!}F)$.

Apply [11, Theorem 7.3.1] to this structural map, the source coefficient F , and the target coefficient R_r . Its hypotheses are $G \in D^-$, target in D^+ and $\ell \neq p$; no quasi-compactness of the source is required. The actual trace gives

$$D_r R\Gamma_c(T, F) \simeq R\Gamma(T, F^{\vee 0}(N)[2N]).$$

For realized coefficients the right side is perfect by Theorem 4.2. The ring R_r is an injective cogenerator over itself: duality is exact and faithful, and a module whose dual is finite embeds in its finite double dual. The actual compact-support complex therefore has bounded finite cohomology. Finite R_r -modules are reflexive, so, writing $C = R\Gamma_c(T, F)$, biduality gives $C \simeq D_r D_r C$. Since $D_r C$ is already perfect by the displayed star comparison, and D_r takes bounded finite projective complexes to such complexes, C is perfect. This proves the asserted trace-induced equivalence without presupposing compact-support finiteness. $\square$

Theorem 5.5 (Compact support and nearby costalks). *In Definition 5.3, there is a canonical comparison on the realized coefficient source*

$$\theta_r^! : A_r^c(E) \xrightarrow{\sim} N_r^c(E),$$

and both objects are naturally identified with

$$H_r^!(E) = D_r C_{\text{cts}}^*(\Gamma, E_r^{\vee 0})(-N)[-2N].$$

The comparison respects the common star-algebra action on coefficients. It is compatible with coefficient reduction and has uniformly bounded, derived-cartesian adic recovery on the actual coefficient source of Theorem 4.2.

Proof. Let $\mathcal{DE}_r = \mathcal{E}_r^{\vee 0}(N)[2N]$. Its actual log/generic realization is obtained by dualizing the source construction of Theorem 4.2, so the already proved star comparison applies to this coefficient and to its maps. Denote that comparison by

$$\beta_{\mathcal{DE}} : i_Y^* R\Psi^t Rj_{\eta,*} \mathcal{DE}_r \xrightarrow{\sim} R\Gamma(T, \mathcal{DE}_r^{\text{an}}).$$

On the left, Proposition 5.2 and the actual finite algebraic duality identities give

$$\begin{aligned} D_r i_Y^* R\Psi^t Rj_{\eta,*} \mathcal{DE}_r &\simeq i_Y^! \mathbb{D}_{\overline{X}_{s,r}} R\Psi^t Rj_{\eta,*} \mathcal{DE}_r \\ &\simeq i_Y^! R\Psi^t Rj_{\eta,!} \mathcal{E}_r = N_r^c(E). \end{aligned}$$

The last step uses source biduality. Proposition 5.2 justifies the tame exchange on a neighbourhood before the costalk is taken. Proposition 5.4 identifies the dual of the right side of $\beta_{\mathcal{DE}}$ with $A_r^c(E)$. Define $\theta_r^!$ to be $D_r(\beta_{\mathcal{DE}})$ under these two geometrically proved identifications. This constructs a map between the two independently defined supported objects and proves it invertible.

Applying Theorem 4.2 to $\mathcal{DE}_r$ gives (5.5). The factor $(N)[2N]$ belongs to the source dual and becomes $(-N)[-2N]$ exactly once under D_r . The ordinary star comparison preserves algebra/module structures. Its dual therefore preserves their transpose actions. Analytic Poincare duality and nearby duality identify those actions with the actual cup-product/evaluation actions on compact support and costalks. This proves the module assertion for the constructed map, with its coherent cells. For coefficient reduction, use the canonical coefficient-change maps of the actual operations and their adjunction mates. The comparison maps above are constructed from pullback, evaluation and trace, so these maps form the corresponding natural coefficient-change diagram. On the star side Theorem 4.2 has already proved it cartesian. On perfect point-valued outputs, the canonical base-change map for D_r is an isomorphism. Dualizing that diagram consequently proves cartesianness for the actual supported outputs and compatibility for $\theta_r^!$. If the input has a common perfect amplitude $[a, b]$, the formula gives supported amplitude in $[a + 2N - d, b + 2N]$, uniformly in r . The same finite construction gives the finite-constructibility bounds. The proved cartesian system therefore admits the adic recovery used in this paper. Take its coefficient inverse limit only after this verification, and rationalize afterwards. $\square$

Proposition 5.6 (Refinement traces and finite descent). *The comparison of Theorem 5.5 is natural for coefficient maps and for compatible algebraized finite Kummer covers whose selected geometric tubes are finite etale over the original tube and whose chosen compactified models admit compatible finite morphisms after the specified tame base change. Under the supported/star duality pairings, actual finite-cover trace is the adjoint of actual star pullback. These adjunctions respect composition. If an actual finite semilinear action and its star-source comparison are supplied, the supported comparison is equivariant for that action; prime-to- ℓ finite descent preserves its recovery bounds.*

Proof. For a finite etale cover $q : T' \rightarrow T$, the counit $q_* q^* \rightarrow \text{id}$ is the trace, locally the sum over its finite sheets. The equality between pairing after trace and pairing after pullback follows from the projection formula and transitivity of the structural trace. The generic algebraic cover has the same finite-etale adjunction; nearby-duality and the completion comparison carry its natural maps to the corresponding analytic diagram. Dualizing this star diagram is exactly the supported diagram defining $\theta^!$. Composition follows from composition of the counits, including their existing adjunction coherence. No claim is made that a finite root Artin-stack bar equals the finite analytic covering complex.

An actual automorphism preserving the model, face and coefficient data preserves these trace and evaluation maps. Thus the same construction is equivariant on the supplied finite action diagram. Exactness of prime-to- ℓ group invariants gives the last assertion. This does not manufacture equivariant source realizations for arbitrary choices of a root neutralization, nor does it establish chart-change incidence. $\square$

Corollary 5.7 (Geometric realization of the continuous root trace). *On the domain of Theorem 5.5, with the fixed rank- N root carrier and the finite duality formalism of Theorem 6.7, there is an integral adic comparison*

$$A_{\text{ad}}^c(E) \simeq N_{\text{ad}}^c(E) \simeq G_{!,\text{ad}}^{\text{tr}}(E).$$

It identifies the unit with the base-linear dualizing module of the continuous star algebra. Together with Theorem 4.2 this realizes the local star/trace datum with its finite-carrier supported functor.

Proof. Theorem 6.7 computes the continuous finite-root trace by the right side of (5.5), with the same contragredient coefficient and the same single rank- N source twist. Theorem 5.5 identifies both actual supported objects with that formula, through the already identified continuous star functor. Perfect duality fixes the associated unit module and its action. The comparison is made after continuous trace assembly. All adic transitions have been verified before taking their inverse limit. $\square$

Corollary 5.8 (The supported degree and twist on the unit). *In Theorem 5.5, for the unit coefficient and $L = M^\vee$,*

$$H_c^{2N-j}(T, R_r) \simeq \bigwedge^j (M \otimes R_r)(j - N), \quad 0 \leq j \leq d,$$

and the other cohomology groups vanish. The same formula holds for the nearby costalk. In particular an open smooth disc contributes only $R_r(-1)$ in degree 2, whereas a semistable node annulus contributes R_r in degree 1 and $R_r(-1)$ in degree 2.

Proof. Theorem 4.2 identifies the ordinary unit cohomology with $\bigwedge^j (L \otimes R_r)(-j)$ in degree j . Apply the single $D_r(-)(-N)[-2N]$ in Theorem 5.5 and dualize the finite free exterior factors. The two examples have respectively $(d, N) = (0, 1)$ and $(1, 1)$. This computes the supported cohomology, not a formality statement for the cochain algebra or a normal-pair incidence map. $\square$

6 The finite-root trace

We now identify the supported root term geometrically. At fixed r the star maps are inflation; their duals must be the counits of the genuine finite-stack shriek adjunctions. This map-level identification, followed by the continuous limit, relates the root trace to the supported comparison of the preceding section. The targets in this section are point-valued complexes and their star-algebra modules, with external tame transport forgotten. The base throughout this section is a geometric point, with $\Gamma = \mathbf{Z}_\ell^d$, $K_n = \Gamma/\ell^n\Gamma$ and fixed rank- N vector-bundle carriers. There is no additional descent group.

Proposition 6.1 (The enhanced finite-quotient comparison). *Let $A = R_r$ and let G be a finite constant etale group over the algebraically closed field k , with ℓ invertible in k . Use the unbounded stable etale category $\mathcal{D}(BG, A)$ of [19, §6.2]. There is a natural equivalence*

$$\Phi_{G,A} : \mathcal{D}(BG, A) \simeq D(A[G]).$$

It identifies $D_c^b(BG, A)$ with the full subcategory of bounded finite-cohomology modules inside $D(A[G])$. For $h : BG \rightarrow \text{Spec } k$ and a surjection $G \rightarrow K$ inducing $q : BG \rightarrow BK$, the functors and their adjunctions are identified as follows:

f	$f^* = f^!$	f_*	$f_!$
h	trivial action	$R\text{Hom}_{A[G]}(A, -)$	$A \otimes_{A[G]}^L -$
q	$\text{Res}_{A[G]}^{A[K]}$	$R\text{Hom}_{A[G]}(A[K], -)$	$A[K] \otimes_{A[G]}^L -$

The restriction notation in the second row means restriction along $A[G] \rightarrow A[K]$. These are comparisons of enhanced adjunctions, including the geometric units, counits and their composite cells. They commute with coefficient restriction and identify the smooth shriek scalar-change maps with derived tensor associativity. Under these comparisons, the geometric cup and supported actions are the cochain and coinvariant actions described below. Their evaluation pairing is balanced, and quotient traces are linear over the source cochain algebra via inflation.

Proof. The category before the operations. The finite etale atlas $\mathrm{Spec} k \rightarrow BG$ has Čech terms G^s . Enhanced descent [19, Theorem 6.2.13(1)] gives

$$\mathcal{D}(BG, A) \simeq \varprojlim_{[s] \in \Delta} \mathcal{D}(G^s, A) = \varprojlim_{[s] \in \Delta} \prod_{G^s} \mathcal{D}(A).$$

The theorem applies to this smooth surjection with torsion coefficients in the unbounded enhancement. For a Deligne–Mumford stack this is its etale derived category; its lisse-etale description uses cartesian cohomology, not the whole lisse-etale category. The displayed limit is $\mathrm{Fun}(BG, \mathcal{D}(A))$, the category of homotopy-coherent G -actions. Evaluation at the unique object is conservative and preserves colimits. Its left adjoint is free action $M \mapsto \bigoplus_{g \in G} M$, with monad $A[G] \otimes_A -$; composition of group elements is the monad multiplication. The free-action bar resolution is augmented to each action object. After evaluation its augmentation is split by the identity element, so its realization is that object. The same bar resolves a module for the monad. These resolutions give inverse comparison functors between actions and $A[G]$ -modules, together with their units and counits. This proves the equivalence in the derived enhancement. Finite cohomology is detected at the atlas, which proves the constructible restriction. The atlas description also identifies tensor with diagonal tensor of actions and commutes with restriction of coefficient rings.

Which adjunction is transported. The map of nerves for $G \rightarrow K$ restricts an action, and the nerve map to the point gives trivial action. Thus these are the actual f^* , with their structural comparison cells. The maps h, q are smooth of dimension zero: after a torsor base change the fibres of q are classifying stacks of its finite etale kernel. Enhanced smooth purity [19, Theorem 6.2.9(2)] identifies $f^!$ with this same f^* through the geometric trace counit. This is an assertion about nonrepresentable smooth maps in that theorem, not representable proper base change.

For $B = A[G]$, $C = A[K]$ (or $C = A$ for h), restriction has adjoints $C \otimes_B^L -$ and $R\mathrm{Hom}_B(C, -)$ on unbounded module categories. The comparison with geometric $f_!$ is the Yoneda comparison induced by the following specified chain:

$$\begin{aligned} \mathrm{Map}_C(\Phi f_! E, N) &\simeq \mathrm{Map}_{\mathcal{D}(BG, A)}(E, f^! \Phi^{-1} N) \\ &\simeq \mathrm{Map}_B(\Phi E, \mathrm{Res} N) \\ &\simeq \mathrm{Map}_C(C \otimes_B^L \Phi E, N). \end{aligned}$$

The first equivalence is the geometric adjunction, the second uses its fixed purity map and the atlas equivalence. Consequently the geometric counit maps to the tensor adjunction counit, namely $C \otimes_B^L \mathrm{Res} N \rightarrow N$ induced by the action. The unit is $M \rightarrow \mathrm{Res}(C \otimes_B^L M)$. Using $f^* \dashv f_*$ in the same way identifies its unit and counit with those of Hom . These mapping-space equivalences are natural before Yoneda is applied. For composable quotients their pasting is tensor associativity (and right-adjoint composition). The composite-adjunction identities therefore supply the same composition cells, including on transformations.

Cup actions and the balanced pairing. Write $T = h^*$, $C = Rh_*$, $Q = Rh_!$, and $\mathcal{A}_G = C(A)$ for the cochain algebra. The actual supported action is the composite

$$\lambda : \mathcal{A}_G \otimes QW \xrightarrow{\sim} Q(T\mathcal{A}_G \otimes W) \xrightarrow{Q(\epsilon_A \otimes 1)} QW, \quad \epsilon_A : TC(A) \rightarrow A. \quad (2)$$

Here and below tensors are derived over A . The projection isomorphism holds for the full cochain complex: in the module model it is tensor associativity for $Q = A \otimes_{A[G]}^L -$. Since T is strong symmetric monoidal, its right adjoint C is lax symmetric monoidal, and ϵ_A is a map of algebra objects. Thus this construction gives a coherent unital $\mathcal{A}_G$ -action. On $X = C(W^\vee)$, write $\mu : \mathcal{A}_G \otimes X \rightarrow X$ for cup. The invariants counit $TX \rightarrow W^\vee$ is $T\mathcal{A}_G$ -linear, where $T\mathcal{A}_G$ acts on $W^\vee$ through ϵ_A .

For $Y = QW$, the actual pairing is

$$e : Y \otimes X \simeq Q(W \otimes TX) \longrightarrow Q(W \otimes W^\vee) \longrightarrow Q(TA) \longrightarrow A.$$

It uses the invariants counit, coefficient evaluation and the coinvariants counit. Give Y the right action $\rho = \lambda \circ \tau_{Y, \mathcal{A}_G}$, with τ the symmetric braiding. Substitution of (2) into this composite gives

$$e \circ (\rho \otimes 1_X) = e \circ (1_Y \otimes \mu) \quad \text{on } Y \otimes \mathcal{A}_G \otimes X. \quad (3)$$

Indeed, after the projection isomorphism, both composites apply ϵ_A to $T\mathcal{A}_G$, $\epsilon_{W^\vee}$ to TX , and then coefficient evaluation and the same trace counit. This formulation includes the Koszul sign when rewritten using left actions on homogeneous elements.

For $q : BG \rightarrow BK$, inflation $\mathcal{A}_K \rightarrow \mathcal{A}_G$ is the right-adjoint comparison. The composite-adjunction identity is

$$\epsilon_{G,A} \circ T_G(\text{inf}) = q^* \epsilon_{K,A} : T_G \mathcal{A}_K \longrightarrow A.$$

The projection formula for $q_!$ therefore makes its counit, and hence $t : Q_G q^* W \rightarrow Q_K W$ for W on BK , $\mathcal{A}_K$ -linear:

$$t \lambda_G(\text{inf} \otimes 1) = \lambda_K(1 \otimes t).$$

These are identities of the indicated enhanced transformations. Their unit, multiplication and iterated composition cells are those of the monoidal counit and the projection formula: on the action atlas these are the full-bar diagonals, evaluation and action augmentations. Thus the balancing and linearity include the coherent module structures, not only their induced cohomology maps.

Coefficient maps and the carrier. For $A \rightarrow A'$ write $S = - \otimes_A^L A'$ and U for restriction. For these smooth maps, purity and inverse image give the actual coefficient-restriction square $f_A^! U \simeq U f_{A'}^!$. Its left mate is the geometric map $S f_{A,!} \rightarrow f_{A',!} S$. Under the just constructed comparisons this is precisely

$$S(C \otimes_B^L M) \simeq C' \otimes_{B'}^L (SM).$$

For example the whole counit square, with N a C -module, becomes

$$\begin{array}{ccc} S\Phi_K q_{A,!} q_A^* \Phi_K^{-1} N & \xrightarrow{S\Phi(\epsilon_q)} & SN \\ \downarrow \wr & & \parallel \\ C' \otimes_{B'}^L \text{Res}(SN) & \xrightarrow{\text{act}} & SN. \end{array}$$

Both horizontal arrows are the action augmentation; the vertical arrow is the geometric scalar mate followed by Φ . The same argument applies to h . It does not invoke the perfect-coefficient-map hypothesis of general coefficient-change theorems: $R_{r+1} \rightarrow R_r$ need not be perfect.

For the fixed carrier $p : \mathbf{A}_{BG}^N \rightarrow BG$, smooth purity has its single factor $A(N)[2N]$. The projection formula and the representable Thom trace identify $p! p^* W = W(-N)[-2N]$. Its trace and purity commute with coefficient restriction and with pullback of this bundle along q . Consequently inserting p in the preceding square tensors it with $A'(-N)[-2N]$; the carrier refinement is a bundle pullback with the identity on the affine-space direction. Transitivity of

these actual traces and the composite-adjunction identities give the joint root/scalar/Thom cells used below. This verifies the specified arrows throughout the construction. The same scalar mate respects (2) along the multiplicative map $SA_G \rightarrow \mathcal{A}'_G$: its defining counit square is precisely scalar extension of ϵ_A . Applying it also to $\epsilon_{W^\vee}$ and evaluation preserves (3). This uses that map, without requiring it to be an equivalence at finite root level. The Thom trace tensors these identities with $A(-N)[-2N]$ and respects the same projection and counit diagrams. $\square$

Proposition 6.2 (Finite geometric bars and their duals over a point). *Let $h_n : BK_n \rightarrow \text{Spec } k$, let $p_n : X_n \rightarrow BK_n$ be a rank- N vector bundle, and put $g_n = h_n p_n$. For an underlying- R_r -perfect coefficient V_n on BK_n , write $E_n = p_n^* V_n$ and $V_n^\vee = R\text{Hom}_{R_r}(V_n, R_r)$ with contragredient action. In the finite six-functor theory with smooth purity, composition, the projection formula, and their adjunctions, there are natural identities*

$$Rg_{n,*} E_n \simeq R\text{Hom}_{R_r[K_n]}(R_r, V_n), \quad Rg_{n,!} E_n \simeq (R_r \otimes_{R_r[K_n]}^L V_n)(-N)[-2N],$$

where R_r is the trivial right module in the second formula. In $D(R_r)$ the second term is canonically

$$R\text{Hom}_{R_r}(C^*(K_n, V_n^\vee), R_r)(-N)[-2N].$$

For compatible vector bundles under $K_m \rightarrow K_n$, the geometric trace is the dual of full-bar inflation. The same conclusions transport to finite nilpotent thickenings with these reductions.

Proof. Proposition 6.1 identifies the actual enhanced adjunctions for h_n with trivial action, derived invariants and derived coinvariants. In particular the latter computes the given geometric $h_{n,!}$ with its counit. Homotopy invariance and the compactly supported Thom isomorphism for the vector bundle give $Rp_{n,*} p_n^* V_n = V_n$ and $Rp_{n,!} p_n^* V_n = V_n(-N)[-2N]$. Composition proves the two formulas.

Use the full free group-bar resolution. Its duality pairing with the opposite bar identifies derived invariants of $V_n^\vee$, dualized over R_r , with derived coinvariants of V_n . This can be checked with a bounded finite-module representative of V_n : R_r is self-injective, so ordinary R_r -linear dual is exact on finite modules, and each bar diagonal against that bounded representative is finite. Biduality is termwise finite-module biduality. Equivalently the calculation uses the Frobenius trace on $R_r[K_n]$ and the group inversion on the opposite bar. Thus the internal Hom here is literally the one in $D(R_r)$; no identification of two different ambient dualities is being assumed. The positive unbounded bar becomes a negative unbounded chain complex, not another cochain complex.

For a quotient of groups, inflation is the adjunction unit for pullback and derived invariants. Dualizing its bar pairing gives the counit for derived coinvariants and trivial relative action. Since these are the geometric adjunctions just identified, this is the geometric trace. Adjunction composition supplies its coherent composition law. Lemma A.4 gives transport through nilpotent reductions. This proof neither replaces $h_!$ by h_* nor asserts that every unbounded finite star bar commutes with derived coefficient reduction. $\square$

Definition 6.3 (Finite presentations at a fixed precision). Let $\mathcal{C}_{n,r}$ be the full stable subcategory of $D(R_r[K_n]\text{-Mod})$ on underlying- R_r -perfect objects. Use the stable localization of complexes, with inflation along $K_m \rightarrow K_n$. Write $\mathcal{C}_r = \mathcal{S}_r(\Gamma)$. Proposition 3.1 identifies $\text{colim}_n \mathcal{C}_{n,r}$ with $\mathcal{C}_r$. A finite presentation of $V \in \mathcal{C}_r$ is $W \in \mathcal{C}_{n,r}$ with an equivalence $\text{Inf}_n^\Gamma W \simeq V$. Its tail uses $W_m = \text{Inf}_n^m W$ and $E_m = p_m^* W_m$. Root composition cells are those of inflation itself. When computing a coefficient map, its finite source, target and comparison can be placed at a common root level. This convention does not choose a lift of the entire adic diagram to finite levels.

Lemma 6.4 (Fixed-precision trace functors and coefficient maps). *The geometric root counits define an exact functor $H_{!,r} : \mathcal{C}_r \rightarrow D(R_r)$. For a finite presentation W at level n , its value is naturally*

$$\hat{H}_{n,r}(W) = R \varprojlim_{m \geq n} Rg_{m,!} p_m^* \text{Inf}_n^m W.$$

For each coefficient reduction $S = - \otimes_{R_{r+1}}^L R_r$ there is a natural transformation

$$b_r : SH_{!,r+1} \longrightarrow H_{!,r} S.$$

These transformations have the composition cells for successive reductions. Their construction precedes the proof that they are isomorphisms.

Proof. At fixed (n, r) , the functor into $\text{Fun}(\{m \geq n\}^{\text{op}}, D(R_r))$ sends W to the displayed geometric complexes. The arrow $m \rightarrow l$, $m \geq l$, is the transformation

$$Rg_{m,!} q_{m,l}^* E_l \simeq Rg_{l,!} Rq_{m,l,!} q_{m,l}^! E_l \xrightarrow{Rg_{l,!} \epsilon_{m,l}} Rg_{l,!} E_l.$$

Here $q^* = q^!$ and the geometric/module counit comparison are those of Proposition 6.1. Inflation and these derived adjunctions are functors and transformations on stable module localizations. Their composite-counit law defines the diagram on all simplices. Composing with the limit functor therefore defines $\hat{H}_{n,r}$ on mapping spectra as well.

Restriction to a cofinal tail gives a natural equivalence $\hat{H}_{n,r} \simeq \hat{H}_{l,r} \text{Inf}_n^l$. The comparison for $n \leq l \leq m$ is the composition of restriction maps of diagrams, so these equivalences form a coherent cocone. The universal property

$$\text{Fun}(\text{colim}_n \mathcal{C}_{n,r}, D(R_r)) \simeq \varprojlim_n \text{Fun}(\mathcal{C}_{n,r}, D(R_r))$$

constructs $H_{!,r}$, with presentation comparisons and higher maps. Limits in a stable category are exact, so these are exact functors.

Scalar reduction $S : \mathcal{C}_{n,r+1} \rightarrow \mathcal{C}_{n,r}$ commutes with inflation. The left-adjoint coinvariant calculation and the carrier Thom trace give finite-level natural maps

$$S(Rg_{m,!} p_m^* W_m) \longrightarrow Rg_{m,!} p_m^* (SW_m).$$

The counit square in Proposition 6.1 identifies these particular geometric maps with scalar extension of the coinvariant augmentation and the same Thom trace. They therefore commute with the root counits. Compose them with the canonical comparison $S \lim Y_m \rightarrow \lim SY_m$ to obtain a transformation on $\mathcal{C}_{n,r+1}$. Tail restriction commutes with this construction. The preceding equivalence of functor categories, now also on natural transformations, descends it to b_r .

For three coefficient rings, the direct and iterated maps from scalar extension of a limit are the same map of limit cones; finite scalar extension and the coinvariant augmentations have their usual associativity cells. This proves the composition cells before any presentation is chosen for an adic object. Applying these functors and transformations to its original cartesian section produces the output diagram. Thus no inverse system of spaces of choices of finite presenting diagrams is needed. $\square$

Lemma 6.5 (The trace pairing and the specified coefficient arrow). *The fixed-precision geometric functor has a natural equivalence*

$$\phi_r : H_{!,r}(V)(N)[2N] \xrightarrow{\sim} D_r X_r(V), \quad X_r(V) = C_{\text{cts}}^*(\Gamma, V^\vee).$$

Put $A = R_{r+1}$, $B = R_r$, $S = - \otimes_A^L B$ and omit the common Thom factor. Let $\alpha : SX_{r+1}(V) \rightarrow X_r(SV)$ be the actual star coefficient map, and let $\delta : SD_A X \rightarrow D_B(SX)$ be the map adjoint to scalar extension of evaluation. Then the previously constructed b_r satisfies the following square, whose lower arrow points to the left:

$$\begin{array}{ccc} SH_{l,r+1}(V)(N)[2N] & \xrightarrow{b_r} & H_{l,r}(SV)(N)[2N] \\ \delta S\phi_{r+1} \downarrow & & \downarrow \phi_r \\ D_B(SX_{r+1}(V)) & \xleftarrow{D_B\alpha} & D_B X_r(SV). \end{array} \quad (4)$$

In particular it identifies the actual b_r with standard perfect-dual reduction once continuous star perfectness and base change are used. The equivalence ϕ_r identifies the geometric supported action with the transpose cup action of $\mathcal{A}_r = C_{\text{cts}}^*(\Gamma, R_r)$ on $D_r X_r(V)$. The square is a module square along the actual coefficient map $S\mathcal{A}_{r+1} \rightarrow \mathcal{A}_r$.

Proof. The fixed-precision pairing. For a presentation W put

$$X_m = C^*(K_m, W_m^\vee), \quad Y_m = (Rg_{m,!} p_m^* W_m)(N)[2N].$$

Proposition 6.2 supplies the actual pairings $e_m : Y_m \otimes X_m \rightarrow R_r$ and equivalences $Y_m \simeq D_r X_m$. For $m \geq l$, write $t_{m,l} : Y_m \rightarrow Y_l$ for the geometric counit and $a_{l,m} : X_l \rightarrow X_m$ for inflation. Their paired identity is

$$e_l(t_{m,l} \otimes 1) = e_m(1 \otimes a_{l,m}).$$

It follows from the invariants-unit/coinvariants-counit adjunction on the opposite full bars; Frobenius evaluation takes the identity coefficient, with group inversion, and never divides by $|K_m|$. The identity is compatible with composition by the counit law. Hence $Y = \lim_m Y_m$ pairs with $X = \text{colim}_m X_m$. Internal Hom sends this colimit to the dual limit, proving $Y \simeq D_r X$. Finite-action descent, applied to maps from the trivial object to $V^\vee$, identifies X with $X_r(V)$. The functor construction in the preceding lemma descends the pairing itself, so ϕ_r is a natural transformation in the enhanced functor category.

The continuous module action. Put $\mathcal{A}_{n,r} = C^*(K_n, R_r)$, with inflation transitions. Fix n and restrict every Y_m for $m \geq n$ in the presenting tail along $\mathcal{A}_{n,r} \rightarrow \mathcal{A}_{m,r}$. The finite trace linearity just proved makes this a diagram in $\text{Mod}_{\mathcal{A}_{n,r}}$. Its limit has underlying object Y , since the forgetful functor creates limits. Equivalently, the maps

$$\mathcal{A}_{n,r} \otimes Y \longrightarrow \mathcal{A}_{n,r} \otimes Y_m \longrightarrow Y_m$$

form the cone defining its action. Cofinal restriction of tails and the quotient composition cells make these module structures compatible as n increases. They therefore give an action of the algebra

$$\text{colim}_n \mathcal{A}_{n,r} \simeq \mathcal{A}_r = C_{\text{cts}}^*(\Gamma, R_r) :$$

indeed, compatible algebra maps $\mathcal{A}_{n,r} \rightarrow \underline{\text{End}}(Y)$ extend uniquely across this filtered colimit in associative algebras. The cochain action on $X = \text{colim}_m X_m$ is the corresponding continuous cup product. Restrict balancing to $\mathcal{A}_{n,r}$ and X_l and choose a common level $m \geq n, l$ in the presenting tail. Equation (3) at m , together with the trace/inflation identities, proves balancing for Y and X . The coherent finite diagrams pass to these limits and colimits. Consequently $Y \simeq D_r X$ identifies the constructed action with the transpose cup action. This construction does not exchange a tensor product with an inverse limit.

Coefficient change before taking either limit. Choose a finite presentation over A and reduce it to B ; this computes the natural transformation on the finite source category. Use primes for the B -complexes. Let $t_m : SY_m \rightarrow Y'_m$ be the finite shriek coefficient map and let

$\alpha_m : SX_m \rightarrow X'_m$ be the usual derived-invariants coefficient map, using $S(W_m^\vee) \simeq (SW_m)^\vee$. In $D(B)$ one has

$$e'_m(t_m \otimes \alpha_m) = Se_m : SY_m \otimes_B SX_m \longrightarrow B. \quad (*)$$

For a map-level verification, denote trivial action by T_A , its left adjoint (coinvariants) by Q_A , and its right adjoint (invariants) by C_A . The finite pairing is the composite

$$\begin{aligned} Q_A(W_m) \otimes C_A(W_m^\vee) &\simeq Q_A(W_m \otimes T_A C_A(W_m^\vee)) \\ &\longrightarrow Q_A(W_m \otimes W_m^\vee) \longrightarrow Q_A(T_A A) \longrightarrow A. \end{aligned}$$

The first arrow uses the invariants counit, the second coefficient evaluation, and the last the coinvariants counit. The projection identity follows from tensor associativity for derived coinvariants. This is the full-bar contraction of Proposition 6.2, hence the geometric pairing already used in ϕ_r .

By definition, α_m is adjoint to scalar extension of the invariants counit $T_A C_A(W_m^\vee) \rightarrow W_m^\vee$, using $S(W_m^\vee) \simeq (SW_m)^\vee$. Scalar change commutes with T_A and the left-adjoint coinvariant tensor Q_A and carries coefficient evaluation to evaluation over B . Applying it to the displayed composite therefore gives exactly $e'_m(t_m \otimes \alpha_m)$. This proves (*) for the specified adjunction maps. The Thom factor is obtained from the same scalar-compatible vector-bundle trace. No finite star base-change isomorphism or truncation of a full bar is used.

Passage to the continuous pairing. Let $\pi_m : Y \rightarrow Y_m$ and $\pi'_m : Y' \rightarrow Y'_m$ be projections. The construction of b_r in the preceding lemma says precisely

$$\pi'_m b_r = t_m S \pi_m.$$

Since S preserves colimits, $SX = \operatorname{colim}_m SX_m$. Restrict the two pairings on $SY \otimes SX$ to each SX_m . The displayed projection identity and (*) show that both restrictions are $S(e_m(\pi_m \otimes 1))$. Their compatible equality on this colimit diagram proves

$$e'(b_r \otimes \alpha) = Se.$$

Tensor–Hom adjunction turns this identity into exactly the displayed square. All equalities here are the indicated natural homotopies of bar, evaluation and limit-cone transformations; they descend through the functor-category equivalence. No tensor/inverse-limit interchange has been used.

Finally the length- d finite free augmentation Koszul resolution over $R_r[[\Gamma]]$ computes the continuous invariants. The identification with continuous invariants preserves the augmentation and its right-adjoint counit: it is the comparison of maps from the trivial object under finite-action/torsion descent. Therefore α is carried to the canonical derived-Hom scalar map for this resolution. Finite-projective base change makes that map invertible; the underlying-perfect coefficient makes X perfect. Thus δ is invertible too, and the square gives

$$\phi_r b_r = (D_B \alpha)^{-1} \delta S \phi_{r+1}.$$

This is the standard reduction of a perfect dual. It proves the claim for the original geometric coefficient arrow. Successive reductions retain the composition cells already constructed for b_r and for evaluation. At finite level the coefficient mates are semilinear over $S\mathcal{A}_{m,r+1} \rightarrow \mathcal{A}_{m,r}$ by Proposition 6.1. The limit projections and the preceding tail construction therefore make the same b_r semilinear over $S\mathcal{A}_{r+1} \rightarrow \mathcal{A}_r$. Together with balancing, the identity $e'(b_r \otimes \alpha) = Se$ proves the asserted module square, with its inherited unit and multiplication cells. $\square$

Proposition 6.6 (Six operations on the point carriers). *Let k be algebraically closed of characteristic different from ℓ , let $K_n = (\mu_{\ell^n})^d$, and let $p_n : Z_n = \mathbf{A}_{BK_n}^N \rightarrow BK_n$ be the fixed rank- N carrier. For $g_n : Z_n \rightarrow \mathrm{Spec} k$, the published finite-coefficient Artin-stack operations supply the point finite trace domain used in Theorem 6.7. Its ambient dual is literally $R\mathrm{Hom}_{R_r}(-, R_r)$ in $D(R_r)$. Thus Assumption A.3 is not an additional hypothesis for this sequence of split point carriers. This assertion does not verify that assumption on a general base.*

Proof. The base field is excellent, and its finite-type schemes have finite ℓ -cohomological dimension in each fixed dimension. The ring R_r is an Artin local Gorenstein ring of characteristic invertible on the base. Thus these finite-type Deligne–Mumford stacks and coefficients are in the domain of [17]. The smooth-purity identification for $BK_n \rightarrow \mathrm{Spec} k$ is $h_n^! = h_n^*$: its stack dimension is zero [17, Corollary 4.6.2]. For the enhancement we use [19, §6.2, Theorems 6.2.9 and 6.2.13], as instantiated in Proposition 6.1. That proposition constructs the category comparison from atlas descent and gives the mapping-space chain transporting each geometric adjunction. The classical operations are the ones recovered on homotopy categories; the enhanced transformations used here are fixed by this construction. The published classifying-stack calculation [17, Example 4.9.2] in particular retains the negative unbounded shriek complex; properness of BK_n does not identify it with the positive star complex.

Smooth purity and the usual compactly supported Thom trace for the representable vector-bundle map supply the single factor $(-N)[-2N]$. The finite free group bar, paired with its opposite bar over the self-injective ring R_r , identifies its linear dual with the chain complex for coinvariants on bounded finite-module coefficients. This is the actual point Verdier dual in $D(R_r)$, including on the one-sided unbounded bar outputs. Inflation and its paired map are precisely the already identified geometric adjunction maps; their composition cells are those of these adjunctions. The coefficient/counit square and its carrier extension are those of Proposition 6.1; no unspecified enhancement of a triangulated equivalence is used.

Coefficient extension commutes with the left-adjoint coinvariant and Thom constructions. No corresponding unconditional finite-bar star reduction is asserted. Internal Hom in $D(R_r)$ sends the star colimit to the trace limit, and only the resulting finite continuous Koszul complex is used for bounded reduction and adic recovery. This verifies the ambient requirements actually used by Theorem 6.7. If the fixed carriers are replaced by the stipulated finite nilpotent thickenings, the representable nil-transport lemma applies; no nonrepresentable proper star/shriek equality is used. $\square$

Theorem 6.7 (The continuous trace of finite root carriers). *Use the point carriers of Proposition 6.2; their six-functor domain is verified in Proposition 6.6. Let $V_\bullet$ be a cartesian source system of Definition 2.1, now with $\Gamma = \mathbf{Z}_\ell^d$, and write $V = R\varprojlim_r V_r$ for its underlying perfect point complex. Put $\Lambda = \mathbf{Z}_\ell[[\Gamma]]$. Finite presentations of V_r exist by Proposition 3.1. Use the fixed-precision functor and coefficient transformations of Lemma 6.4; each value may be computed on any inflated finite presenting tail with its geometric counit transitions. No compatible choice across all r is required. Then*

$$H_{!,r}(V_r) := R\varprojlim_{n, \mathrm{Tr}^!} Rg_{n,!}E_{n,r} \simeq R\mathrm{Hom}_{R_r}(C_{\mathrm{cts}}^*(\Gamma, V_r^\vee), R_r)(-N)[-2N].$$

The inverse-limit output $H_{!,r}$ is presentation-independent and perfect over R_r ; the individual $Rg_{n,!}E_{n,r}$ may remain unbounded. The coefficient maps b_r constructed above are equivalences after derived reduction, and

$$R\varprojlim_r H_{!,r}(V_r) \simeq R\mathrm{Hom}_{\mathbf{Z}_\ell}(K_\Gamma(V_\bullet^\vee), \mathbf{Z}_\ell)(-N)[-2N].$$

Write

$$\mathbf{G}_{!,\text{ad}}^{\text{tr}}(V) := R\varprojlim_r H_{!,r}(V_r), \quad \mathbf{G}_!^{\text{tr}}(V) := \mathbf{G}_{!,\text{ad}}^{\text{tr}}(V)[1/\ell].$$

Thus the root supported term is defined by the actual finite-carrier trace system before it is identified with the continuous dual. Here $V^\vee$ denotes the coefficient dual system and its recovered point complex. The Iwasawa derived Hom in this formula is continuous in the sense of Definition 2.1.

Proof. Lemma 6.4 constructs the geometric functors and their coefficient transformations independently of the dual formula. Lemma 6.5 then proves that formula as an equivalence of functors, and its square (4) identifies exactly their previously constructed coefficient maps with perfect-dual reduction. Thus the given output diagram is cartesian, with uniform perfect bounds from the finite continuous Koszul complex. The finite root bars themselves need not be bounded.

Apply Lemma A.9 to this output system and to the continuous star system on $V_\bullet^\vee$. Perfect $\mathbf{Z}_\ell$ -duality has the standard finite reductions; the proved square identifies their entire diagram with the geometric one. The reduction/limit adjunction therefore gives the stated integral comparison with its original maps. Rationalize only after this recovery. The root limit has been taken first throughout. $\square$

7 Proof of the theorem and examples

Proof of Theorem 1.1. Compactify X in $(\mathbf{P}_R^1)^2 \times (\mathbf{P}_R^1)^e$ by the equation

$$X_0 Y_0 = \pi X_1 Y_1, \quad x = X_0/X_1, \quad y = Y_0/Y_1.$$

Take the standard affine charts for the z_i . This is proper, and Y lies in its displayed affine open; the closure of the generic boundary is in the complementary infinity divisors and misses Y . Near Y the chart is $\mathbf{N} \rightarrow \mathbf{N}^2$, $1 \mapsto (1, 1)$, with smooth disc directions. It is fs, saturated, vertical and log smooth, with generic dimension N and relative rank one. The tube is the actual open annulus times open discs, so the analytic supported domain of Theorem 5.5 applies.

The source construction of Corollary 4.1 uses the actual Kummer torsors $a^{\ell^n} = x$. Point finite-action descent handles all bounded derived coefficients and maps at each r , and Theorem B.13 identifies the chosen adic coefficient systems. Proposition B.14 proves that the Iwasawa computations apply to the entire point source quantified in the theorem. Corollary 3.2 identifies actual structural star with continuous invariants. On the actual Kummer torsors, Proposition E.4 localizes the specialization unit, and Lemma E.5 proves its invertibility from the constant-coefficient theorem. The prime-to- p log-Kummer source supplies the required tame descent. Theorem E.6 then gives actual nearby and tube star by its specialization/completion maps. Lemma 4.3 proves that the induced mapping map, including the ordinary-etale/quasi-etale bridge, is an equivalence at finite precision; Proposition 4.4 passes that same map to the specified adic categories.

The geometric nearby and analytic trace dualities of Proposition 5.2 and Proposition 5.4 apply to this compactification and this actual lisse realization. Dualizing the same star map therefore constructs the actual compact-support/costalk comparison in Theorem 5.5. Proposition 6.6 supplies the published finite stack domain and its ambient module duality. Theorem 6.7 consequently identifies the genuine inverse trace tower with the dual of the same continuous star. This proves the second row, including its single source twist and shift. Its module actions are those induced by geometric evaluation and trace.

At each fixed r use the length-one continuous Koszul construction, only after the root-bar passage. It and its perfect linear dual have the displayed uniform Tor-amplitudes, and the previously constructed geometric coefficient-change maps are cartesian by Theorem 4.2, Theorem 5.5 and Theorem 6.7. Lemma A.9 now recovers actual perfect $\mathbf{Z}_\ell$ -outputs and their canonical reductions. Lemma A.8 identifies the canonical reductions through its pro-zero Tor tower. Completing the comparison maps and then rationalizing proves the remaining assertions. $\square$

Corollary 7.1 (The geometric unipotent interface for refinement). *Set $e = 0$ in Theorem 1.1, put $Q = \mathbf{Q}_\ell$, and retain its coordinate and compatible base roots. Let $\mathcal{U}_\Gamma$ be the full subcategory of the rational coefficient source with bounded finite-dimensional unipotent cohomology representations (successive quotients have trivial Γ -action). Let $\mathcal{U}^b(\mathbf{G}_{m,\bar{K}})$ denote the actual geometric constructible complexes with unipotent lisse cohomology. The associated Kummer sheaf functor and continuous cochains give equivalences*

$$\mathcal{U}^b(\mathbf{G}_{m,\bar{K}}) \simeq \mathcal{U}_\Gamma \xrightarrow[\sim]{K_\Gamma} \text{Perf}(A_\Gamma), \quad A_\Gamma = K_\Gamma(Q).$$

Here A_Γ has its continuous-cochain E_∞ structure; modules use the right-module convention. The tensor on the last category is $\otimes_{A_\Gamma}$, corresponding to the diagonal coefficient tensor over Q .

For the actual associated coefficient F , the star comparison is an A_Γ -module comparison, and its supported companion is

$$R\Gamma(Y, i^! R\Psi_{\bar{X}}^t Rj_{\eta,!} F) \simeq D_Q K_\Gamma(V^\vee)(-1)[-2], \quad D_Q = R\text{Hom}_Q(-, Q).$$

The action on the right is dual to the cochain action on $K_\Gamma(V^\vee)$; the identification uses the actual evaluation and trace pairing of the theorem.

Proof. In either coefficient category, unipotent sheaves are obtained from constant finite-dimensional sheaves by their actual extension triangles. Finite Postnikov towers then show that every displayed object is in the thick closure of the unit. Conversely kernels, cokernels, extensions and summands of unipotent representations remain unipotent, so this thick closure has exactly the stated cohomology condition. The associated-sheaf functor on the torsors $z_n^{\ell^n} = x$ is exact and tensor. Its map on unit endomorphisms is the actual Kummer map

$$C_{\text{cts}}(\Gamma, Q) \longrightarrow R\Gamma(\mathbf{G}_{m,\bar{K}}, Q).$$

It is an equivalence: both sides have Q in degree zero, $Q(-1)$ in degree one and no other cohomology, and the map sends the universal Kummer class to that of x . The geometric torus calculation is the rank-one Kummer calculation; extension of the algebraically closed field to C preserves it. Restriction to the annulus gives the same class and the actual star map already proved in the theorem. Exactness and retracts extend the unit mapping comparison in each variable to full faithfulness; unit generation then gives essential surjectivity onto the actual geometric category. The construction includes the extension maps in the Postnikov towers.

On $\mathcal{U}_\Gamma = \text{thick}(Q)$, cochains are $R\text{Hom}(Q, -)$. The unit mapping comparison and finite cones and retracts identify this functor with the one-object Morita equivalence onto $\text{Perf}(A_\Gamma)$. The lax tensor structure supplies the balanced natural map

$$K_\Gamma(V) \otimes_{A_\Gamma} K_\Gamma(W) \longrightarrow K_\Gamma(V \otimes_Q W).$$

It is the unit identification on (Q, Q) and hence an equivalence on both thick closures, by exactness in each variable. Its symmetry and associativity are inherited from this same cochain transformation. The supported assertion and its module action are the theorem's trace comparison, restricted to this source and rationalized. $\square$

Remark 7.2 (Coordinate changes and the refinement interface). Write $t = \pi$, and fix $t_n \in \bar{K}$ with $t_n^{\ell^n} = t$ and $t_{n+1}^\ell = t_n$. On any generic overlap with $x = t^a u$, the maps of actual torsors are

$$z_n^{\ell^n} = x \mapsto w_n = z_n/t_n^a, \quad w_n^{\ell^n} = u.$$

They commute with root transition and deck action. For $v = u^{-1}$ the torsor map is $w_n \mapsto w_n^{-1}$ and the deck map is $\zeta \mapsto \zeta^{-1}$. Thus translation by t^a identifies the relative action, while inversion pulls it back along $\gamma \mapsto \gamma^{-1}$. Applying the associated-sheaf functor and the bar cochain functor gives the specified maps of cochain algebras and their coefficient modules, including all composition maps: division by t_n^{a+b} is successive division by t_n^a, t_n^b , and inversion squares to the identity. The familiar signs on H^1 are consequences of these maps.

Evaluation commutes with these torsor identifications and group automorphisms. The induced finite quotient equivalences preserve adjunction counts; hence the finite-carrier trace pairings and their coefficient reductions commute as well. On a geometric overlap, analytic trace has the same compatibility by functoriality under isomorphism. Dualizing the same star map therefore transports the supported comparison together with its Tate twist. All these assertions use the fixed compatible base roots; no base-Galois-equivariant choice is asserted.

This is the coefficient interface with the refinement paper: on a thickness-one point chart its actual star algebra and coefficient module are A_Γ and $K_\Gamma(V)$, and its supported direct image is the displayed costalk. For a general stratum S , identifying $H_S, N_S(F)$ and the direct image of $\kappa_S^! R\Psi F$ with these local data still uses that paper's actual specialization, restriction and gluing maps. In particular, the coordinate isomorphisms above compare coefficients on generic overlaps; cross-stratum links and Gysin maps must come from the actual open/closed geometry. They are not supplied by root-chart functoriality or lattice projections. Neither a thickness- m node nor a positive-dimensional exceptional support is covered by this corollary. The Morita assertion concerns only the geometric unipotent subsource, not the entire source of the node theorem.

Remark 7.3 (A nontrivial character and its integral torsion). In the node theorem choose a generator γ of relative inertia and a rank-one coefficient $V = \mathbf{Z}_\ell$ on which γ acts by $u = 1 + \ell$ for odd ℓ , or $u = 1 + 4$ for $\ell = 2$. These give continuous characters of $\mathbf{Z}_\ell$; their reductions factor through finite root quotients. Put $a = u - 1$. The star complex is $[\mathbf{Z}_\ell \xrightarrow{a} \mathbf{Z}_\ell]$ in degrees 0, 1. It has only $H^1 = \mathbf{Z}_\ell/a$. The dual character has differential $u^{-1} - 1 = -a/u$, so the supported formula has only

$$H_c^{2N}(T, \mathcal{R}(V)) = (\mathbf{Z}_\ell/a)(-N).$$

Its perfect representative nevertheless occupies degrees $2N - 1, 2N$. After derived reduction modulo ℓ , both of those degrees have nonzero cohomology, agreeing with the constant mod- ℓ coefficient. The extra degree comes from Tor, not an ordinary reduction of the integral cohomology group. Rationalization kills both star and supported complexes for this nontrivial character.

Appendix F gives examples of the obstructions to extending the coefficient source beyond the stated Kummer domain.

A General charts and coefficient conventions

This appendix fixes the general-base conventions used by the descent arguments. The node theorem uses their geometric-point cases. The final duality lemma separates the constructible star domain from the lisse-perfect duality domain.

A.1 Adic coefficients and finite operations

Definition A.1 (Integral Kummer coefficients). Put $R_r = \mathbf{Z}/\ell^r$. On a coefficient topos X let $\mathcal{D}(X, R_r)$ be the stable localization of module-sheaf complexes at quasi-isomorphisms. Define

$$\mathcal{D}_{\text{ad}}(X, \mathbf{Z}_\ell) = \varprojlim_{r, -\otimes_{R_{r+1}}^L R_r} \mathcal{D}(X, R_r), \quad \mathcal{D}_{c,\text{ad}}^b(X, \mathbf{Z}_\ell) \subset \mathcal{D}_{\text{ad}}(X, \mathbf{Z}_\ell). \quad (5)$$

The full subcategory consists of derived cartesian systems with uniform bounded amplitude and the stated constructible perfect restrictions on a finite stratification. Mapping spectra are those of this limit category. Reduction is taken in the ambient category: $\mathbf{F}_\ell \otimes_{\mathbf{Z}/\ell^2}^L \mathbf{F}_\ell$ shows why it is not a functor on all bounded constructible complexes. Rational coefficients mean inversion of ℓ in this integral category.

On finite Artin stacks we use the enhanced six operations [17, 18, 19, 20]. Individual star and shriek images lie in their respective $+$ and $-$ constructible domains. All root (co)limits are formed in the ambient unbounded category: finite classifying-stack cohomology is unbounded [17, Example 4.9.2]. Boundedness of the continuous outputs is proved by the Iwasawa calculation below.

For $\Gamma_M = M \otimes \mathbf{Z}_\ell(1)$ put $\Lambda_M^{\text{Iw}} = \mathbf{Z}_\ell[[\Gamma_M]]$ and $\Lambda_{M,r}^{\text{Iw}} = \Lambda_M^{\text{Iw}}/\ell^r$; in particular

$$\Lambda_{M,r}^{\text{Iw}} \neq R_r[\Gamma_M/\ell^r \Gamma_M]. \quad (6)$$

At a point, $\text{Mod}_{\Lambda_M^{\text{Iw}}}^{\text{pc}}$ denotes pseudocompact modules, derived from inverse systems of finite-length modules for (ℓ^r, I_M^a) , with completed tensor and continuous Hom. An *Iwasawa-admissible coefficient* is a bounded, locally pseudo-coherent object of finite Tor-amplitude over the Iwasawa algebra whose underlying $\mathbf{Z}_\ell$ -complex is perfect. This is the classical point terminology. Over a stratified base, “selected” or “admissible” means the uniformly underlying-perfect cartesian class in Definition B.12, without an implicit global pseudocompact sheaf condition. Torsion cohomology is allowed; the regular Iwasawa module is excluded when $M \neq 0$.

For sheaf coefficients the independent adic module model is Definition B.12. Theorem B.13 identifies it with the selected continuous-action systems. A separate global pseudocompact-sheaf interpretation is not used. At precision r , finite-action descent supplies a root level $n(r)$, including derived maps; these levels need not be bounded as r varies. Ambient coefficient recovery is Lemma A.8, and perfect recovery at a point is Lemma A.9.

We write $V = V_{\text{cont}}[1/\ell]$ and $\Lambda_{M, \mathbf{Q}_\ell}^{\text{Iw}} = \Lambda_M^{\text{Iw}}[1/\ell]$. This is the bounded-denominator algebra, rather than $\mathbf{Q}_\ell[[T_1, \dots, T_d]]$; coefficients throughout admit the specified perfect integral model.

Assumption A.2 (Ambient six-functor domain). All algebraic models, their finite root stacks, and their quotient stacks are locally of finite type over one fixed affine excellent noetherian scheme B of finite Krull dimension. The prime ℓ is invertible on B , and finite-type B -schemes have uniformly finite ℓ -cohomological dimension in each dimension occurring below. A dualizing object on B and the constructible enhanced finite and adic six-functor formalisms are fixed once and for all. The strictly local face S , although used to split the chart, is obtained by strict localization inside this domain; all statements over S and $\mathcal{S} = [S/\Delta]$ mean the corresponding base changes of the fixed formalism. In particular, every use of Rf_* , $Rf_!$, f^* , $f^!$, Verdier duality, and a Beck–Chevalley mate below takes place in its stated D_c^+/D_c^- constructible domain.

Assumption A.3 (Finite trace domain and ambient duality compatibility). For each finite enhanced root map g_n and each coefficient precision r , use the genuine finite Artin-stack

operations in the chosen enhanced formalism. The coefficient class and its source dual lie in the domain of

$$Rg_{n,!}E_{n,r} \simeq \mathbb{D}_{\mathcal{S},r} Rg_{n,*} \mathbb{D}_{\text{root},n,r} E_{n,r}.$$

The formalism is required to have smooth purity, proper pushforward, and nil-invariant coefficient pullback, with their usual composition and coefficient-change compatibilities. Lemma A.4 deduces the central-root identity $q_{m,n}^! \simeq q_{m,n}^*$ from these axioms. This identity concerns central faces; ramified normal-pair maps use their own purity formulas.

At each r there is a fixed closed presentable stable category of sheaf complexes on $\mathcal{S}$ in which the full-bar colimits and their dual limits are formed. Its internal Hom into $\omega_{\mathcal{S},r}$ restricts to the finite Verdier duals in the displayed formula. No constructible bounded or one-sided bounded category is assumed to contain arbitrary such limits. The boundedness and adic recovery of the continuous outputs are proved from the Iwasawa model. For the split carriers over a geometric point used in the local trace theorem, Proposition 6.6 verifies these requirements. On a general base they remain explicit additional hypotheses, and no general-base trace theorem is asserted here.

Lemma A.4 (Nil transport and central gerbe purity). *Within the finite six-functor domain, let $i : X_0 \hookrightarrow X$ be a nilpotent closed immersion for which i^* is a coefficient equivalence. Then $i^! \simeq i^*$ canonically. For the central refinements, with the same carrier on both sides, $q_{m,n}^! \simeq q_{m,n}^*$; these identities are compatible with composition and coefficient change wherever the specified finite operations have coefficient-change isomorphisms.*

Proof. Nil-invariance makes Ri_* the inverse of i^* ; representable properness gives $Ri_! = Ri_*$. Taking right adjoints therefore gives $i^! = i^*$, compatibly with structural duality. On reductions a central refinement is locally $BH \rightarrow S$ for a finite etale group H . This map is smooth of stack dimension zero, so smooth purity gives $q_0^! = q_0^*$. For reduction immersions i_m, i_n , transitivity gives $i_m^! q^! = q_0^! i_n^!$. Replacing the two $i^!$ by i^* and using their inverse equivalences yields $q^! = q^*$. The carrier map is the identity. Transitivity of purity and the uniqueness of adjunction transport provide the composition cells; coefficient compatibility holds wherever the finite operations have coefficient-change isomorphisms. $\square$

These hypotheses specify the base range for the Artin-stack six operations. Continuous Iwasawa cohomology, root-site descent, and geometric comparisons are established in the following appendices.

A.2 Charts and relative inertia

Definition A.5 (Standard split tame product chart over a strictly local base). Let R be a complete strictly Henselian discrete valuation ring with fraction field K , residue characteristic p , and fix $\ell \neq p$. Let $M \simeq \mathbf{Z}^d$ be the free cocharacter (relative characteristic) lattice, write $M^\vee := \text{Hom}_{\mathbf{Z}}(M, \mathbf{Z})$ for its character lattice, and put

$$\Gamma_M := M \otimes_{\mathbf{Z}} \mathbf{Z}_\ell(1), \quad \Lambda_M^{\text{Iw}} := \mathbf{Z}_\ell[[\Gamma_M]] = \varprojlim_{U \subset \Gamma_M} \mathbf{Z}_\ell[\Gamma_M/U],$$

A *standard split tame product chart* of type (M, e) is a split log-smooth formal chart over the displayed strictly Henselian base which is fs, saturated, algebraizable in a neighbourhood of the chosen relative face, and topologically of finite type over $\text{Spf } R$, together with the following data. The selected tube is required to be separated and compactifiable over the face base: it admits an open immersion into a proper finite-type chart over that base.

- (a) The algebraization consists of a finite-type log-smooth morphism $f : X^{\text{alg}} \rightarrow \text{Spec } R$ in a neighbourhood of a specified locally closed stratum $i : Y \hookrightarrow X_s^{\text{alg}}$, an open $j : U \hookrightarrow X_\eta^{\text{alg}}$ whose specialization meets precisely that neighbourhood, and a proper log compactification $X^{\text{alg}} \hookrightarrow \overline{X}^{\text{alg}}$ over the face base. Completion along Y is the given formal chart, and the tube of Y has a fixed product

$$]S[_{\text{an}} \simeq \mathbb{T}_{M,K}^{\text{an}} \times \mathbb{D}^e, \quad \mathbb{T}_{M,K} = \text{Spec } K[M^\vee],$$

compatible with specialization and with the face projection. The symbols $i, j, U, Y, X^{\text{alg}}$ and their Čech pullbacks always refer to these supplied objects; in particular the nearby open is not silently replaced by the whole generic fibre. Here $\mathbb{D}^e$ is a relatively open polydisc and the displayed product is a specified chart isomorphism. The chart also supplies the rank- e smooth tangent bundle $\mathcal{E}_{\text{sm}}$ on S obtained by restricting the relative tangent bundle of the polydisc factor to its zero section. On the split chart it is $\mathcal{O}_S^e$; its descent datum is part of the product chart.

- (b) We use the chosen pro- ℓ Kummer subsite: it is generated by the finite covers

$$[\ell^n] : \mathbb{T}_{M,K} \longrightarrow \mathbb{T}_{M,K}, \quad m \longmapsto \ell^n m, \quad n \geq 0,$$

where the displayed multiplication is on the cocharacter lattice M (equivalently, characters pull back by $\lambda \mapsto \ell^n \lambda$), and by their pullbacks to the log chart. Coefficients are required to be trivial on the prime-to- ℓ tame quotient; the discarded finite prime-to- ℓ part may be retained separately as finite descent data. The resulting deck group is $\Gamma_M = M \otimes \mathbf{Z}_\ell(1)$. This is an auxiliary pro- ℓ Kummer face site, not the ordinary étale site of the unrigidified quotient stack and not, by notation alone, a geometric Artin fan.

- (c) Coefficients are taken in the constructible adic Kummer category of Definition A.1; in particular, every coefficient used in a theorem below is represented by a compatible finite-level system in the selected cartesian action/module-sheaf model. Over a geometric point Proposition B.14 identifies the entire selected class with the classical perfect Iwasawa model.
- (d) The chosen compactification restricts to the fixed toric/polydisc compactification of the displayed tube. Analytic compact support uses extension by zero followed by proper direct image. Its nearby counterpart is the costalk construction of (43); their comparison on the supported domain is Theorem 5.5.

We denote the pro- ℓ site in (b), together with its finite descent groupoid, by $\mathcal{F}_M^{\text{rel, ket}}$. Throughout the paper, I_t^{base} means the selected pro- ℓ quotient $I_{t,\ell}^{\text{base}}$; any retained prime-to- ℓ finite tame factor is incorporated into the finite descent groupoid. The finite tame descent datum may act on M through a finite group of lattice automorphisms. The root coefficient constructions respect that action, with quotient base specified in Definition A.7. Geometric analytic equivariance additionally uses the actual chart automorphisms and a compatible compactification.

Remark A.6 (Actual tubes and the separate log-smooth chart domain). The entire-torus product in Definition A.5 is an additional restriction, not a consequence of log smoothness. For example, in $X = \text{Spec } R[x, y]/(xy - \pi)$, let Y be $x = y = 0$ in the special fibre. A point specializes to Y precisely when $|x| < 1$ and $|y| < 1$. Since $xy = \pi$, its geometric tube is

$$]Y[_{\overline{\eta}} = \{ |\pi| < |x| < 1 \},$$

an open annulus, not the entire torus in the supplied monomial coordinate. Adding smooth coordinates vanishing on Y adds open discs. Also, completion along Y uses the ideal (x, y) ; modulo π its completed ring contains $k[[x, y]]/(xy)$, rather than a finite-type k -algebra. Such completions belong to the special formal-scheme domain and need not be topologically of finite type for the π -adic topology.

Theorem E.6 computes star cohomology on these completions in the domain of Definition E.2. Entire-torus product charts retain their separate compactification and support hypotheses.

Definition A.7 (Relative root chart). On a strictly local face base S , fix a torsion-free fs sharp monoid P and a primitive $\rho \in P$ with $L_{\text{rel}} = P^{\text{gp}}/\mathbf{Z}\rho$ free. Put $M = \text{Hom}(L_{\text{rel}}, \mathbf{Z})$. For a nonclosed face use its supplied pair (P_τ, ρ_τ) . This root datum is independent of the entire-torus condition of Definition A.5. A finite group Δ acts on (P, ρ) and on S , with

$$|\Delta| \in \mathbf{Z}_\ell^\times. \quad (7)$$

The quotient base and its Iwasawa algebra are

$$\mathcal{S} = [S/\Delta]. \quad (8)$$

On S the algebra is Λ_M^{Iw} with its semilinear Δ -action; when the base action is trivial one may use $\Lambda_M^{\text{Iw}} \rtimes \Delta$.

A base-preserving saturated chart map $\alpha : (P_{\tau'}, \rho_{\tau'}) \rightarrow (P_\tau, \rho_\tau)$ preserves ρ exactly and induces a relative lattice map with torsion-free cokernel. It must also define a morphism of the supplied log points [13, §1.2]: writing $\epsilon_Q : \mathcal{O}_S[Q] \rightarrow \mathcal{O}_S$ for the augmentation, we require $\epsilon_{P_\tau} \circ \mathcal{O}_S[\alpha] = \epsilon_{P_{\tau'}}$. For the sharp zero-augmentation charts used here this is precisely $\alpha^{-1}(0) = \{0\}$. Preserving ρ and saturation alone do not imply this condition. The map, including its log-point structure, is Δ -equivariant when descent is present. Its root maps and the 2-cells of the trivial-root sections come from the root fibre's universal property, hence compose coherently. Such a map need not be strict. Root refinement $P \rightarrow \ell^{-n}P$ is a separate Kummer operation, with torsion cokernel. Writing $\bar{\alpha} : L_{\text{rel}, \tau'} \rightarrow L_{\text{rel}, \tau}$, precomposition gives

$$\alpha_M : M_\tau \rightarrow M_{\tau'}, \quad \alpha_{\mathcal{K}, n} : \mathcal{K}_{\tau, n} \rightarrow \mathcal{K}_{\tau', n}, \quad \alpha_\Lambda : \Lambda_{M_\tau}^{\text{Iw}} \rightarrow \Lambda_{M_{\tau'}}^{\text{Iw}}. \quad (9)$$

For rank-changing maps preservation of the selected perfect coefficient class must be checked for the operation in question.

The root datum above defines the central root fibre. To equip it with a carrier, additionally supply an algebraized log-smooth model $\mathfrak{U}$, its chosen face S , and, when descent is present, a compatible lift of the Δ -action to the model. Assume its relative logarithmic dimension is $N = d + e$. Set

$$\mathcal{V}_{M, e} = \mathcal{T}_{\mathfrak{U}/\mathfrak{s}_{\mathbf{N}_\rho}}^{\log}|_S. \quad (10)$$

This rank- N bundle retains the geometric dimension absent from the bare root fibre. Its Thom trace supplies $(-N)[-2N]$, so the supported construction belongs to the root datum equipped with this model and carrier. If compatible product coordinates splitting the tangent sequence are supplied, we may write

$$\mathcal{V}_{M, e} \simeq (M \otimes \mathcal{O}_S) \oplus \mathcal{E}_{\text{sm}}.$$

The induced Δ -action is diagonal in this decomposition when the splitting is Δ -equivariant. The intrinsic bundle and its action require no such splitting. This tangent decomposition is separate from an entire-torus product description of the tube; for the node, the z_i supply the smooth directions. Strict log-etale changes of the fixed model preserve the bundle.

Use $\mathbf{V}(\mathcal{V}) = \mathrm{Spec}_S \mathrm{Sym}(\mathcal{V}^\vee)$ with compactification $\mathbf{P}(\mathcal{V} \oplus \mathcal{O}_S)$. Refinement traces over a geometric point are constructed in Theorem 6.7.

Write $\mathfrak{s}_{P/S}$ and $\mathfrak{s}_{\mathbf{N}\rho/S}$ for S with the indicated constant log charts, nonzero characteristic elements mapping to zero. The relative root construction uses their fibre; it does not require $P/\mathbf{N}\rho$ to be fs or sharp.

A.3 Recovery of adic systems

Lemma A.8 (Derived coefficient recovery). *Put $A = \mathbf{Z}_\ell$ and $R_s = A/\ell^s$. Fix an ambient presentable stable A -linear category of module complexes, with its module categories for R_s and their derived scalar changes. For a derived-cartesian system*

$$F_{s+1} \otimes_{R_{s+1}}^L R_s \simeq F_s,$$

form $\hat{F} = \varprojlim_s F_s$ after restriction to that specified ambient category. Then the canonical map

$$\hat{F} \otimes_A^L R_r \xrightarrow{\sim} F_r$$

is an equivalence for every r , and $\hat{F}$ is derived ℓ -complete. No Mittag-Leffler hypothesis or exactness of products of cohomology sheaves is needed.

For the selected adic sheaf categories in this paper, the adic object is the cartesian system itself. The displayed ambient realization recovers its reductions; it is not asserted to be a bounded locally free ordinary A -sheaf on the small étale site. Integral bounded perfectness at a geometric point is proved in Lemma A.9. Rationalization is performed on the specified integral/admic realization, never termwise on the torsion coefficients.

Proof. Fix r and discard the finitely many levels $s < r$. In $D(R_s)$ the multiplication map has the canonical fibre sequence

$$R_r[1] \longrightarrow R_s \otimes_A^L R_r \longrightarrow R_r.$$

Indeed, the middle term is $[R_s \xrightarrow{\ell^r} R_s]$ in degrees $-1, 0$; its kernel is $\ell^{s-r} R_s \simeq R_r$ and its cokernel is R_r . Under $s+1 \rightarrow s$ the map on that kernel is multiplication by ℓ , since $\ell^{s+1-r} = \ell \ell^{s-r}$. The cokernel map is the identity. The calculation also identifies the module structures and the canonical maps to R_r .

Tensor this fibre sequence over R_s with F_s and use cartesianness. It gives a fibre sequence of towers

$$(F_r[1], \ell) \longrightarrow (F_s \otimes_A^L R_r)_s \longrightarrow (F_r, \mathrm{id}).$$

Compatibility follows by first restricting the ring-level sequence at s to R_{s+1} , and then tensoring with F_{s+1} ; the prescribed cartesian cells identify the result with the sequence at s . The left-hand tower is pro-zero: r successive maps multiply by $\ell^r = 0$ on the R_r -module $F_r[1]$. More explicitly, in its product formula for homotopy limit the shift operator T satisfies $T^r = 0$, so $1 - T$ is invertible with inverse $1 + T + \cdots + T^{r-1}$. Its homotopy limit is zero.

The object R_r is perfect over A : it is the cofiber of $A \xrightarrow{\ell^r} A$. Tensoring by it therefore commutes with all limits in the stable ambient category. Taking homotopy limits of the preceding fibre sequence proves the asserted canonical recovery. The middle tower is not eventually constant; only its extra Tor fibre is pro-zero.

For any object K , the compatible triangles $K \xrightarrow{\ell^s} K \rightarrow K \otimes_A^L R_s$ give

$$R\mathrm{Hom}_A(A[1/\ell], K) \longrightarrow K \longrightarrow \varprojlim_s (K \otimes_A^L R_s).$$

Here the first term is the limit of the multiplication-by- ℓ tower of K . It vanishes for every R_s -module by the same nilpotent-shift argument. Internal Hom in its second variable preserves limits, so it also vanishes for $\hat{F}$. This proves derived completeness in the specified ambient category. Nothing here exchanges geometric stalks with infinite products or asserts that ordinary sheaf cohomology of $\hat{F}$ is finite. $\square$

Lemma A.9 (Perfect cartesian systems over a geometric point). *Let $(F_s)_s$ be a cartesian system in $D(R_s)$ with each F_s perfect of a common Tor-amplitude $[a, b]$. Then $R\varprojlim_s F_s$ is perfect over $\mathbf{Z}_\ell$, with the same Tor-amplitude, and has the given canonical reductions. Its cohomology is recovered by the ordinary limits of $H^i(F_s)$. These cohomology towers are Mittag-Leffler because their terms are finite modules.*

Proof. Form $F = R\varprojlim_s F_s$ in $D(A)$, $A = \mathbf{Z}_\ell$, using the given enhanced diagram. Lemma A.8 already supplies its canonical reductions. Since each $H^i(F_s)$ is finite, its tower is Mittag-Leffler. The module Milnor sequence therefore gives $H^i(F) = \varprojlim_s H^i(F_s)$ and $F \in D^{[a, b]}(A)$.

We prove perfectness by induction on the common Tor interval. If $F_1 = 0$, the minimal-complex test below gives $F_s = 0$ for every s , and $F = 0$. Otherwise one may decrease b to the actual top degree of F_1 . Equivalently, when $H^b(F_1) = 0$ the following step uses the empty basis, $m = 0$, and decreases the interval without attaching a cell. Choose a basis of $H^b(F_1)$. The triangle $F \xrightarrow{\ell} F \rightarrow F_1$ and $H^{b+1}(F) = 0$ lift this basis to a map $u : A^{\oplus m}[-b] \rightarrow F$. Projecting u to the given diagram and using scalar adjunction gives a morphism of cartesian systems $(R_s^{\oplus m}[-b])_s \rightarrow (F_s)_s$, with all its compatibility cells. Let $(G_s)_s$ be its cofiber system.

Each G_s is perfect. Its reduction to R_1 is the cone of a map which is an isomorphism in degree b and has no other source cohomology, so it has amplitude $[a, b - 1]$. Over the local Artin ring R_s , a minimal finite free representative has term ranks given by the cohomology dimensions of its derived residue-field reduction. Consequently every G_s has Tor-amplitude $[a, b - 1]$. This finite-ring test applies separately at each s . By induction $G = R\varprojlim_s G_s$ is perfect in that interval. Exactness of the stable limit and $R\varprojlim_s R_s = A$ give a triangle $A^{\oplus m}[-b] \rightarrow F \rightarrow G$. Thus F is perfect in $[a, b]$. The empty-interval case is zero: a perfect R_s -complex with zero residue-field reduction is zero by the same minimal-complex test.

The recovered object F , its maps and its reductions are canonical; the cells above establish perfectness. Indeed scalar reduction $c : D(A) \rightarrow \varprojlim_s D(R_s)$ is left adjoint to $L(F_\bullet) = R\varprojlim_s F_s$, since

$$\mathrm{Map}(cM, F_\bullet) = \varprojlim_s \mathrm{Map}_{R_s}(M \otimes_A^L R_s, F_s) = \mathrm{Map}_A(M, LF_\bullet).$$

The preceding recovery lemma identifies the counit $cLF_\bullet \rightarrow F_\bullet$ with an equivalence. On derived-complete M the unit is an equivalence as well. These adjunction maps identify the *given* coherent systems, all morphisms and their higher cells. $\square$

A.4 Coefficient duality and Verdier duality

Definition A.10 (Duality conventions). At a geometric point use $\omega_r = R_r$. For the general-base adic duality assertions fix a coherent derived-cartesian system $\omega_{\mathcal{J}, \bullet}$ of dualizing complexes:

$$\omega_{\mathcal{J}, r+1} \otimes_{R_{r+1}}^L R_r \xrightarrow{\sim} \omega_{\mathcal{J}, r}.$$

Require uniform bounded constructibility and, when a selected perfect output is asserted, a common finite underlying- R_r Tor interval $[c_0, c_1]$ on a fixed finite stratification. These are system hypotheses on the chosen dualizing system. Verdier biduality is used only on the

reflexive domain of these complexes. For $f : X \rightarrow \mathcal{S}$ set $\omega_{X,r} = f^! \omega_{\mathcal{S},r}$, $\omega_{X/\mathcal{S},r} = f^! R_r$ and $\mathbb{D}_{X,r} = R\mathcal{H}om_{R_r}(-, \omega_{X,r})$. Use $\mathbb{D}_{\mathcal{S},r}$ for the base dual. The finite six-functor domain gives

$$\mathbb{D}_{Y,r} Rf_! K \simeq Rf_* \mathbb{D}_{X,r} K, \quad K \in \mathcal{D}_c^-(X, R_r), \quad (11)$$

on the stated reflexive domains. In particular a proper nonrepresentable classifying-stack map still has distinct star and shriek images.

The notation $\mathbb{D}_{X,\text{cont}}$ denotes the system of finite Verdier duals, with a derived-limit realization only after its coefficient reductions have been verified. At a point this verification for the continuous outputs follows from perfectness. Over a general base, Lemma A.12 proves it for the tensor-dualizable inputs specified there; it is not asserted for every stratified-perfect input. Whenever the finite six-functor comparisons form cartesian systems on the stated domain, their actual mates assemble to

$$\mathbb{D}_{\mathcal{S},\text{cont}} Rf_! \simeq Rf_* \mathbb{D}_{X,\text{cont}}. \quad (12)$$

This notation asserts neither a general-base supported realization nor a new exceptional functor on the bare infinite-root fibre.

The coefficient dual has contragredient action:

$$V_{\text{cont}}^\vee = R\mathcal{H}om_{\mathbf{Z}_\ell}^{\text{cts}}(V_{\text{cont}}, \mathbf{Z}_\ell), \quad (\gamma\phi)(v) = \phi(\gamma^{-1}v). \quad (13)$$

At a point this is classical continuous $\mathbf{Z}_\ell$ -linear Hom. Over a base, for a uniformly lisse-perfect system, it denotes the cartesian system of finite R_r -coefficient duals proved below, with its ambient limit used only for recovery; it is not ordinary Hom from an unspecified integral etale sheaf. The Δ -action is likewise semilinear and contragredient. Rational duals are obtained after inverting ℓ . A coefficient is *lisse-perfect* here if it is tensor-dualizable on the whole base topos: locally a bounded complex of finite free constant-coefficient sheaves. This differs from being perfect only on each constructible stratum. For a uniformly lisse-perfect adic base coefficient, with Hom interpreted in the adic system category and computed levelwise, put

$$C^{\vee_0} = R\mathcal{H}om_{\mathbf{Z}_\ell}(C, \mathbf{Z}_\ell). \quad (14)$$

For $\mathcal{L}$ in the Verdier-reflexive domain, at each precision tensor–Hom adjunction and the tensor dual of C give the following identity. Its adic form uses compatible systems for $\mathcal{L}$ and its Verdier dual; reflexivity alone does not prove coefficient change for an arbitrary constructible $\mathcal{L}$:

$$\mathbb{D}_{\mathcal{S},\text{cont}}(\mathbb{D}_{\mathcal{S},\text{cont}} \mathcal{L} \otimes C) \simeq \mathcal{L} \otimes C^{\vee_0}. \quad (15)$$

Thus coefficient duality and source Verdier duality are distinguished by their dualizing objects.

Definition A.11 (Absolute source duals and coefficient duals). Fix the compatible base dualizing objects $\omega_{\mathcal{S},r}$. For a geometric source X with structural map p in the finite formalism, write $\omega_{X,r} = p^! \omega_{\mathcal{S},r}$ and

$$\mathbb{D}_{X,r} F := R\mathcal{H}om_{R_r}(F, \omega_{X,r}).$$

In particular $\mathbb{D}_{\text{an},r}$ is the absolute dual on the geometric analytic source, and $\mathbb{D}_{\text{near},r}$ denotes the absolute dual on the generic algebraic source of E_r ; it is not an undefined dual on the value of $R\Psi^t$. For the finite enhanced root face put

$$\mathbb{D}_{\text{root},n,r} E_{n,r} := R\mathcal{H}om_{R_r}(E_{n,r}, g_n^! \omega_{\mathcal{S},r}).$$

The notation $\mathbb{D}_{\text{root}}E$ denotes its compatible continuous system only under Assumption A.3 and the adic recovery statement. On a product coefficient, with $N = d + e$,

$$\mathbb{D}_{\text{root},n,r}(\mathcal{L}_r \boxtimes V_{n,r}) \simeq (\mathbb{D}_{\mathcal{J},r}\mathcal{L}_r \boxtimes V_{n,r}^{\vee_0})(N)[2N], \quad V_{n,r}^{\vee_0} = R\text{Hom}_{R_r}(V_{n,r}, R_r),$$

with contragredient action. We abbreviate $\mathcal{L}^\dagger = \mathbb{D}_{\mathcal{J}}\mathcal{L}$. The factor $(N)[2N]$ is already inside the source dual; it is inserted only once. A_M^* and $A_{M,e}^!$ below use base-linear coefficient duality, not a second copy of base Verdier duality.

Fix a smooth projective toric compactification $j_M : \mathbb{T}_{M,K} \hookrightarrow \overline{\mathbb{T}}_{M,K}$ and a compactification of the polydisc factor for which the extension-by-zero definition gives

$$R\Gamma_c(\mathbb{D}^e, \mathbf{Q}_\ell) \simeq \mathbf{Q}_\ell(-e)[-2e].$$

These product calculations fix the normalization of the continuous dual. The actual local compact-support domain and its analytic duality are specified in Section 5. Finite-root traces use the genuine adjunction counits in the finite-stack domain; their point implementation is verified in Proposition 6.6.

Choose a basis $\gamma_1, \dots, \gamma_d$ of Γ_M and put $T_i = \gamma_i - 1 \in \Lambda_M^{\text{Iw}}$. The elements $T_1, \dots, T_d$ form a regular sequence. Consequently the Iwasawa–Koszul complex

$$\mathbf{K}_M^{\text{Iw}} := \bigotimes_{i=1}^d [(\Lambda_M^{\text{Iw}})^{-1} \xrightarrow{T_i} (\Lambda_M^{\text{Iw}})^0] \longrightarrow \mathbf{Z}_\ell \quad (16)$$

is a finite free resolution of the augmentation module. This is the only Koszul resolution used in the paper.

For comparison, let $\mathbf{B}_M^{\text{cts}} \rightarrow \mathbf{Z}_\ell$ be the completed normalized bar resolution of the profinite group Γ_M . At a finite root level and finite coefficient precision we write $\mathbf{B}_{M;n,r}$ for the full normalized bar resolution of the finite deck group $\Gamma_M/\ell^n\Gamma_M$ over $\mathbf{Z}/\ell^r$. In general it is unbounded. In particular,

$$[(\mathbf{Z}/\ell^r)[C_{\ell^n}] \xrightarrow{\gamma-1} (\mathbf{Z}/\ell^r)[C_{\ell^n}]]$$

is not a resolution of the trivial module, and no finite-level bar–Koszul quasi-isomorphism is asserted. Indeed, for $n = r = 1$ and one generator, $(\mathbf{Z}/\ell)[C_\ell] \simeq (\mathbf{Z}/\ell)[u]/(u^\ell)$ with $u = \gamma - 1$, so $u^{\ell-1}$ is a nonzero element of the kernel of multiplication by u . This also agrees with the nonvanishing of $H^q(C_\ell, \mathbf{Z}/\ell)$ in arbitrarily high degrees.

Lemma A.12 (Iwasawa base change and the coefficient-duality domain). *Put $\Lambda = \Lambda_M^{\text{Iw}}$, $\Lambda_r = \Lambda/\ell^r$, and $\mathbf{K}_r = \mathbf{K}_M^{\text{Iw}} \otimes^L R_r$. Let $V_\bullet$ be any selected system of Definition B.12 over a noetherian split base, with common underlying- R_r Tor interval $[a, b]$. At a point this is the entire source of Proposition B.14. Write*

$$H_{*,r}(V) = R\mathcal{H}om_{\Lambda_r}(\mathbf{K}_r, V_r).$$

The following domains are distinct.

- (i) *For every such selected system, $\mathbf{K}_r$ is a length- d finite free resolution of R_r . The actual coefficient maps give*

$$H_{*,r+1}(V) \otimes_{R_{r+1}}^L R_r \simeq H_{*,r}(V).$$

The outputs are constructible and stratified-perfect with common Tor interval $[a, b + d]$. Finite-projective Hom also commutes with reduction of the ambient derived-limit realization. Thus the full general-base star and coefficient-comparison domain is unchanged.

- (ii) Suppose additionally that each V_r is lisse-perfect on the whole base, uniformly in r . Then the coefficient dual $V_r^\vee = R\mathrm{Hom}_{R_r}(V_r, R_r)$, with contragredient inertia, is a selected cartesian system with Tor interval $[-b, -a]$. Its coefficient biduality map is an equivalence. All these maps respect the supplied finite semilinear descent and group inversion. At a point no additional restriction is imposed; classical Iwasawa-admissibility of the dual follows from the point-source theorem.
- (iii) For inputs in (ii), use the coherent dualizing system of Definition A.10, and put $N = d + e$ and

$$H_{!,r}(V) = \mathbb{D}_{\mathcal{S},r} H_{*,r}(V^\vee)(-N)[-2N].$$

These are cartesian, uniformly bounded constructible outputs. With the stated Tor bound $[c_0, c_1]$ on the dualizing system they are stratified-perfect with interval $[a - d + c_0 + 2N, b + c_1 + 2N]$. They recover by reduction from their ambient derived inverse limit. At a geometric point $\omega_r = R_r$, so the bound is $[a + 2N - d, b + 2N]$ and perfect integral recovery holds for the full point source. The finite cohomology towers there are Mittag-Leffler.

Part (i) remains valid after tensoring with any selected stratified-perfect base system. Parts (ii)–(iii) remain valid after tensoring with a uniformly lisse-perfect base system. No common trivializing cover or finite root level for all precisions is required.

Proof. The regular sequence $T_1, \dots, T_d$ in $R_r[[T_1, \dots, T_d]]$ gives the finite free augmentation resolution. Applying Hom is a finite complex of finite sums of V_r , with degrees from 0 to d . Its terms and differentials commute with scalar reduction. This proves (i), including its Tor bound, without exchanging a finite-group bar with reduction. Recovery of the original coefficient system in the ambient category is Lemma A.8; the finite free resolution then gives the asserted integral Hom formula.

For (ii), tensor duality on the whole base gives the actual maps

$$V_r \longrightarrow (V_r^\vee)^\vee, \quad V_{r+1}^\vee \otimes_{R_{r+1}}^L R_r \longrightarrow (V_{r+1} \otimes_{R_{r+1}}^L R_r)^\vee.$$

Both are equivalences by a local finite free computation; it also reverses the Tor interval. Dualizing the action uses group inversion, so evaluation and coevaluation are equivariant and the maps retain the given descent cells. Their cohomology is again finite and locally constant. On each geometric stalk its continuous finite inertia representation factors through a finite ℓ -group; over R_r its augmentation ideal acts nilpotently. Hence the dual belongs to the selected action category and, by the already proved relative action/Iwasawa equivalence, to the selected module model. Tensor dualizability on the base supplies the internal- Hom identities used here.

Put $H_r = H_{*,r}(V^\vee)$. The same finite Koszul construction makes H_r lisse-perfect with Tor interval $[-b, d - a]$. Consequently

$$H_{!,r}(V) = H_r^\vee \otimes_{R_r}^L \omega_{\mathcal{S},r}(-N)[-2N].$$

The reduction map is the tensor product of the dual reduction map for H and the specified reduction map of $\omega_\bullet$. It is therefore an equivalence with its inherited composition cells. The displayed formula proves the stated bounds and constructibility; with finite Tor for ω it proves stratified-perfectness. Lemma A.8 gives ambient recovery. At a point use Lemma A.9; in each finite target cohomology module, the descending images stabilize, which proves the Mittag-Leffler assertion. Tensoring preserves the respective domains, giving the final clauses. $\square$

Remark A.13 (Coefficient duality at a singular base point). For a closed point $i : s \hookrightarrow S$ on a strictly henselian nodal curve over an algebraically closed field, $i_* R_r$ is a selected stratified-perfect cartesian system. At $r = 1$, set $F = \mathbf{F}_\ell$ and $D_0 = R\mathcal{H}om_S(-, F_S)$. The normalization exact sequence and purity on its two regular branches [17, Proposition 4.10.1] give

$$i^! F_S \simeq F[-1] \oplus F(-1)^{\oplus 2}[-2] =: C.$$

Here the degree-one summand is the connecting term of the normalization sequence. Closed-immersion adjunction gives $D_0(i_* F) = i_* C$ and $D_0^2(i_* F) = i_* R\mathcal{H}om_F(C, C)$. The latter has degree-zero dimension $1 + 2^2 = 5$. Thus (ii) cannot hold on all stratified-perfect coefficients even though (i) and the general-base category equivalences still do. The distinction is between coefficient duality and Verdier duality; replacing $V^\vee$ by a Verdier dual would change the stated formulas.

B Descent over a noetherian base

The root tower uses finite quotient actions, whereas its continuous limit uses an adic Iwasawa module. We compare these coefficient categories before discussing geometric direct images. The finite quotient may depend on the coefficient precision and on the finite diagram being represented.

B.1 Relative finite-action descent on a noetherian base

This subsection treats the ordinary etale action-sheaf model on a noetherian scheme S . It is independent of the actual infinite-root site of Theorem C.20. The distinction between this derived action model and an independently defined sheaf of pseudocompact Iwasawa modules is kept until the latter comparison has been established.

Definition B.1 (The relative continuous action-sheaf category). Fix r and let $\mathcal{A}_r = \text{Mod}(S_{\text{ét}}, R_r)$. Let $\Gamma = \mathbf{Z}_\ell^d$, $H_n = \ell^n \Gamma$, and $K_n = \Gamma/H_n$. Write $\mathcal{A}_{n,r}$ for sheaves in $\mathcal{A}_r$ with a K_n -action, and $\mathcal{A}_{\Gamma,r}$ for sheaves with a smooth Γ -action:

$$\text{colim}_n F^{H_n} \xrightarrow{\sim} F \quad \text{as sheaves in } \mathcal{A}_r.$$

Thus every local section is locally fixed by an open subgroup. This defines continuity at the sheaf level. These are Grothendieck abelian categories, with exact forgetful functors. Subscripts c denote the full subcategories whose underlying sheaves on S are constructible. The derived categories $D_c^b(\mathcal{A}_{n,r})$ and $D_c^b(\mathcal{A}_{\Gamma,r})$ mean the full subcategories of the ambient derived categories with bounded constructible cohomology in the ambient continuous-action category.

Lemma B.2 (Noetherian action sheaves and a global finite quotient). *Every object of $\mathcal{A}_r$ is a union of constructible subobjects, and its noetherian objects are exactly its constructible objects. The same assertions hold in $\mathcal{A}_{n,r}$ and $\mathcal{A}_{\Gamma,r}$, with constructibility measured after forgetting the action. In particular every object of $\mathcal{A}_{\Gamma,r,c}$ has action through a single K_n on all of S .*

Proof. For a noetherian scheme, constructible R_r -sheaves form a strong Serre subcategory and satisfy the ascending chain condition for arbitrary subsheaves; every sheaf is a filtered colimit of constructible sheaves [6, Sections 59.73–59.74]. Taking images gives the asserted union. Conversely, a noetherian object in such a union equals one of its constructible subobjects. This step uses the noetherian hypothesis on S .

For a smooth action, F^{H_n} is stable under Γ since H_n is normal. A constructible subobject $C \subset F^{H_n}$ need not be stable, but the sum of its translates by the finite group K_n is a constructible, Γ -stable subobject of F . Its action factors through K_n . Such subobjects cover F : first use smoothness, then the constructible-subobject union inside F^{H_n} . Their finite sums form a directed union. The analogous finite-group argument applies to $\mathcal{A}_{n,r}$.

An equivariant object with constructible underlying sheaf is noetherian, because equivariant subobjects are underlying subobjects. Conversely a noetherian equivariant object must equal one member of the union just constructed, and is therefore underlying-constructible. Finally, for constructible F , the increasing chain of subsheaves F^{H_n} is stationary. Its union is F , so $F^{H_n} = F$ at some level. This proves factorization on the entire base; no intersection of infinitely many stalkwise stabilizers is used. $\square$

Lemma B.3 (Filtered localization and finite diagram descent). *Suppose exact full inclusions of small abelian categories $\mathcal{B}_n$ have filtered union $\mathcal{B}$, with a common exact faithful underlying functor detecting cohomology. Then*

$$\operatorname{colim}_n D^b(\mathcal{B}_n) \simeq D^b(\mathcal{B})$$

in stable infinity-categories with exact functors. Mapping spectra are filtered colimits over common stages, and finite simplicial diagrams, including finitely specified homotopies, descend up to equivalence to a common stage. If an object condition is preserved and detected by the unchanged underlying complex, the same result holds for the associated full stable subcategories.

Proof. Use $N \operatorname{Ch}^b(\mathcal{B}_n)$ marked by all quasi-isomorphisms W_n , and its infinity-categorical localization $N \operatorname{Ch}^b(\mathcal{B}_n)[W_n^{-1}]$ as $D^b(\mathcal{B}_n)$. Inflation here is the exact functor on the actual complexes before localization. A bounded complex, a finite string of chain maps, and a chain homotopy involve finitely many objects and equations. They all belong to a common stage. Quasi-isomorphisms are detected by the underlying functor, so the marked nerve for $\mathcal{B}$ is the filtered colimit of these marked nerves.

For every infinity-category $\mathcal{E}$, the universal property is an equivalence of functor categories, including transformations:

$$\operatorname{Fun}(D^b(\mathcal{B}), \mathcal{E}) \simeq \operatorname{Fun}_W(N \operatorname{Ch}^b(\mathcal{B}), \mathcal{E}) \simeq \varprojlim_n \operatorname{Fun}_{W_n}(N \operatorname{Ch}^b(\mathcal{B}_n), \mathcal{E}).$$

The subscript means the full subcategory sending the marked arrows to equivalences. Saturating those arrows does not alter this condition: every such functor already inverts their saturation. This proves localization commutes with this colimit, with no extension-closedness assertion for a finite stage in the union. The functors preserve shifts and cones, so the equivalence is stable.

For a filtered categorical colimit, objects come from finite stages and mapping spaces are colimits of the mapping spaces at common stages [3, Lemma 0.2.1]. For the finite diagram assertion one can use the explicit model in [3, §0.5]: a filtered diagram has a quasicategory model whose simplicial colimit represents the categorical colimit. A map from a finite simplicial set factors through a stage, since it has finitely many nondegenerate simplices and relations. The same applies to a specified homotopy (use its product with Δ^1). This proves descent for the whole finite homotopy-coherent diagram. Stability and compatibility with shifts identify the corresponding mapping spectra as well.

Finally, represent an object of the selected target at a finite stage. Its underlying complex has not changed, so that representative already satisfies the object condition. Fullness means its mapping spectra are the ambient ones just compared. One takes a full subcategory of the derived category, not complexes all of whose terms individually satisfy perfectness. Intermediate resolutions and Postnikov pieces may therefore have nonperfect terms. $\square$

Lemma B.4 (Bounded noetherian derived replacement). *Let $\mathcal{A}$ be any of the abelian categories in Lemma B.2, and let $\mathcal{B}$ be its subcategory of noetherian objects. The canonical exact functor is an equivalence of stable enhanced categories*

$$D^b(\mathcal{B}) \xrightarrow{\sim} D_{\mathcal{B}}^b(\mathcal{A}).$$

Proof. The noetherian objects form a Serre subcategory. If $X \twoheadrightarrow Y$ with Y noetherian, the directed union of noetherian subobjects of X has images exhausting Y , so one already surjects. This verifies [7, Lemma 13.17.4]. Good truncation gives bounded representatives. The exact functor on derived enhancements identifies Hom in every shift, hence mapping spectra; bounded replacement gives essential surjectivity. $\square$

Theorem B.5 (Relative finite derived-action descent). *On a noetherian scheme S , inflation induces an equivalence*

$$\operatorname{colim}_n D_c^b(\mathcal{A}_{n,r}) \xrightarrow{\sim} D_c^b(\mathcal{A}_{\Gamma,r}). \quad (17)$$

This equivalence restricts to the full subcategories with stratified underlying- R_r -perfect complexes. Its mapping spectra are the filtered colimits of the finite-level mapping spectra. Every finite simplicial diagram with its specified higher cells has a finite-level presentation. The construction includes attaching maps across strata and is natural for pullback between noetherian bases and for compatible homomorphisms of the split inertia towers.

Proof. Write $\mathcal{B}_{n,r}$ and $\mathcal{B}_{\Gamma,r}$ for the noetherian subcategories. Lemma B.2 proves that each object of $\mathcal{B}_{\Gamma,r}$ is in some $\mathcal{B}_{n,r}$. A morphism between two such sheaves is equivariant at their common level if and only if it is Γ -equivariant. Consequently the continuous abelian category is the filtered union of these finite-action abelian categories. The finite categories are not asserted to be extension-closed inside that union.

Apply Lemma B.3 to this filtered union, using the common underlying sheaf functor to detect quasi-isomorphisms. Lemma B.4 then identifies each localization with the corresponding full constructible subcategory of the ambient action derived category. This gives (17), its mapping spectra and finite higher-diagram descent. The argument uses sheaf complexes on the entire base, not bounded representatives chosen only at stalks. Inflation need not be fully faithful after deriving at a fixed stage.

The underlying complex is unchanged by inflation. Stratified perfectness can therefore be imposed after the equivalence. The intermediate noetherian terms need not be R_r -perfect, so this restriction does not discard torsion Postnikov pieces. Because all complexes and roofs above are sheaves on the entire base, the argument already includes cross-stratum attaching morphisms. Finally pullback is exact, preserves constructible sheaves under the stated noetherian base changes, and commutes with inflation and quasi-isomorphisms. Its canonical localization squares give naturality; no particular choice of a bounded representative is made into a functor. $\square$

Lemma B.6 (Derived rectification of finite semilinear descent). *For the Grothendieck abelian action-sheaf categories $\mathcal{A}$ used here, finite semilinear descent satisfies $D(\mathcal{A}^\Delta) \simeq D(\mathcal{A})^{h\Delta}$. On the selected bounded perfect subcategories, finite-action descent can consequently be performed in $\mathcal{A}^\Delta$ before deriving.*

Proof. Forgetful evaluation has the exact left adjoint $M \mapsto \bigoplus_{\delta \in \Delta} \delta^* M$. On derived categories evaluation is conservative and preserves colimits: it is exact and coproducts and filtered colimits are computed on the underlying complexes. Its monad is the same finite sum with multiplication given by the group composition maps. The derived monadic comparison therefore identifies $D(\mathcal{A}^\Delta)$ with modules for this monad on $D(\mathcal{A})$. Evaluation from the homotopy fixed-point category has the same left adjoint and monad, is conservative,

and computes colimits on the underlying objects. Its monadic comparison gives the same category. This proves rectification; exact averaging alone would not prove it.

For the finite root levels, work first with semilinear sheaves. A finite union or sum of the Δ -translates of a constructible subobject is still constructible; Δ normalizes H_n . The noetherian finite-action replacement and finite Postnikov arguments therefore apply inside this abelian descent category. They descend objects, attaching maps and finite diagrams to a common root level. The just proved rectification identifies this with the displayed homotopy fixed-point notation. The equivalence preserves the underlying complex, so restricts to the selected perfect class and commutes with its scalar reductions. No exchange of a filtered colimit with the infinite nerve $B\Delta$ is used. $\square$

Corollary B.7 (finite descent and adic action systems). *Theorem B.5 also holds for finite semilinear descent by a group Δ acting on S and normalizing the subgroups H_n . On the stratified underlying-perfect subcategory its equivalences commute with derived coefficient reduction. In particular they induce an equivalence on the uniformly bounded constructible cartesian adic systems:*

$$\varprojlim_r \left(\operatorname{colim}_n D_{c,\text{perf}}^b(\mathcal{A}_{n,r}) \right) \simeq \varprojlim_r D_{c,\text{perf}}^b(\mathcal{A}_{\Gamma,r}),$$

with the same restrictions on the two limit categories. For two such systems represented at finite levels at each precision, the mapping spectrum is

$$\varprojlim_r \operatorname{colim}_{n \geq n_0(r)} \operatorname{Map}_{D(\mathcal{A}_{n,r})}(E_{n,r}, F_{n,r}).$$

No finite integral action level, and no reversed order of these limits, is asserted.

Proof. Lemma B.6 identifies the resulting derived descent category with homotopy-coherent finite descent. Work in the abelian category of descent sheaves from the start. Saturating a constructible subobject by the finitely many Δ -translates preserves constructibility. The H_n are normalized by Δ , so the invariant-subobject and finite-saturation arguments still apply. The resulting abelian categories satisfy the same noetherian lifting property. Apply the proof of Theorem B.5 there. This does not commute a colimit with the infinite nerve $B\Delta$; it proves the equivariant assertion before deriving.

Inflation commutes with derived extension of coefficient scalars: both retain the same underlying complex and action, and the canonical extension-of-scalars comparison follows from tensor associativity. On the stratified underlying-perfect class the reductions remain bounded and constructible; no such claim is made for an arbitrary bounded complex over the nonregular ring R_r . The levelwise comparison squares therefore form an equivalence of coefficient systems of categories. Taking their categorical limits proves the adic statement. The conditions on cohomological amplitude, constructibility, and the prescribed adic recovery class are the same after forgetting the action, so the equivalence restricts to them. Mapping spectra in a limit category are limits of mapping spectra; apply the fixed- r equivalence to obtain the displayed formula. Finite presentations and reduction cells may be chosen successively with an increasing bound $\nu(r)$, using descent of each finite diagram. This proves compatibility without positing one bound for the entire adic system. $\square$

B.2 Torsion and completed module sheaves

Proposition B.8 (The two finite coefficient topologies). *Put $\Lambda = \mathbf{Z}_\ell[[T_1, \dots, T_d]]$, $I = (T_1, \dots, T_d)$, and identify $1 + T_i$ with the chosen generators of Γ . The ideals*

$$J_{r,n} = (\ell^r, (1 + T_1)^{\ell^n} - 1, \dots, (1 + T_d)^{\ell^n} - 1) \quad \text{and} \quad Q_{r,a} = (\ell^r, I^a)$$

define cofinal systems, already at each fixed r . Consequently smooth Γ -action sheaves of R_r -modules are exactly sheaves of Λ/ℓ^r -modules which are locally I -power torsion. This is an exact equivalence of abelian categories, and hence of their derived categories and bounded constructible-cohomology subcategories. It concerns the derived category of torsion modules; comparison with the unrestricted or completed module derived category is not asserted.

Proof. In $\Lambda/J_{r,n} = R_r[K_n]$, reduction modulo ℓ identifies the augmentation algebra with $\mathbf{F}_\ell[T_1, \dots, T_d]/(T_1^{\ell^n}, \dots, T_d^{\ell^n})$. Thus for $L = d(\ell^n - 1) + 1$ one has $I^L \subset \ell R_r[K_n]$, and hence $I^{rL} = 0$. This proves $Q_{r,rL} \subset J_{r,n}$. Conversely, for $1 \leq k < a$ the identity

$$k \binom{\ell^n}{k} = \ell^n \binom{\ell^n - 1}{k - 1}$$

gives $v_\ell \binom{\ell^n}{k} \geq n - v_\ell(k)$. Choose n with $\ell^n \geq a$ and $n \geq r + \lceil \log_\ell(\max\{1, a\}) \rceil$. All coefficients in degrees $1, \dots, a - 1$ are divisible by ℓ^r ; the higher terms lie in I^a . Therefore $J_{r,n} \subset Q_{r,a}$. The case $d = 0$ is the trivial augmentation case.

A section killed by $J_{r,n}$ has a finite K_n -action, and the first inclusion makes it I -power torsion. A section killed by I^a has an action through K_n by the second inclusion. These constructions are compatible on overlapping annihilator bounds and with restriction of sheaves. They give inverse exact functors, without choosing a uniform bound for all sections of an arbitrary sheaf. The global bound for constructible sheaves is supplied separately by Lemma B.2. Deriving an exact abelian equivalence proves the last assertion. $\square$

Remark B.9 (Completion on the small etale site). A cartesian adic system is not in general a locally free ordinary $\mathbf{Z}_\ell$ -sheaf on the small etale site. For example let $\eta = \text{Spec } k((t))$, with k algebraically closed and $\text{char } k \neq \ell$. Choose a generator of the tame $\mathbf{Z}_\ell$ -quotient and let it act on $V = \mathbf{Z}_\ell$ by $u = 1 + \ell^2$. The reductions V_r define nonzero rank-one lisse R_r -sheaves $\mathcal{L}_r$ on η , with cartesian reduction maps. Every finite extension has inertia image of finite index in this quotient. Its integral invariant module is zero since $u^m - 1 \neq 0$ in $\mathbf{Z}_\ell$ for every positive integer m . Limits of sections commute with invariants, so the ordinary inverse limit sheaf $\varprojlim_r \mathcal{L}_r$ is zero on the small etale site. In particular $H^0(R\varprojlim_r \mathcal{L}_r) = 0$ there. This does not assert that the derived inverse limit itself is zero; it rules out interpreting it as the expected degree-zero locally free sheaf. The example also occurs on the generic stratum of a strictly henselian trait. It does not obstruct an adic-system, pro-sheaf, or pro-etale realization with a correctly specified completion functor.

Lemma B.10 (Sections and torsion on the etale basis). *Let S be noetherian, A a noetherian ring, and $\mathcal{M} = \text{Mod}(S_{\text{ét}}, \underline{A})$. On the basis of quasi-compact etale $U \rightarrow S$ the following hold: sections commute with filtered colimits; evaluation $J \mapsto J(U)$ has an exact left adjoint; and, for injective $J \in \mathcal{M}$ and a finitely generated ideal $I = (t_1, \dots, t_d)$, the augmented stable Čech map*

$$\Gamma_I J \longrightarrow \bigotimes_{i=1}^d [\underline{A} \rightarrow \underline{A}[1/t_i]] \otimes_A J$$

is a quasi-isomorphism. Here Γ_I means locally I -power torsion, and the Čech complex starts in degree zero.

Proof. A quasi-compact etale scheme over S is noetherian. Its covering families have finite subcovers, and the fibre products in their matching conditions are again quasi-compact. The finite-cover sheaf condition is an equalizer of finite products. Filtered colimits of modules commute with these equalizers, so the presheaf colimit restricted to this basis is already a sheaf. Its value at U is therefore the colimit of the values at U .

For the étale localization $j_U : U_{\text{ét}} \rightarrow S_{\text{ét}}$, put $L_U(M) = j_{U,!}\underline{M}_U$. The constant-sheaf adjunction and the localization adjunction give

$$\text{Hom}_{\mathcal{M}}(L_U M, J) = \text{Hom}_A(M, J(U)).$$

At a geometric point $\bar{s}$ its stalk is $\bigoplus_{\bar{u} \in U_{\bar{s}}} M$; this is exact in M . Enough geometric points detect exactness, so L_U is exact. The left adjoint is extension along an étale localization, not a structural proper or analytic shriek. Applying this adjunction to an injective J shows that $J(U)$ is an injective A -module.

The annihilator of I^a is a finite kernel, since I^a is finitely generated. The first part gives $(\Gamma_I J)(U) = \Gamma_I(J(U))$. Localization by t_i is a filtered colimit of multiplication maps; hence $J[1/t_i](U) = J(U)[1/t_i]$, also for multiple localizations. The displayed sheaf Čech complex therefore has, on every basis object, exactly the usual module Čech complex of $J(U)$. For a noetherian ring its only cohomology on an injective module is $\Gamma_I(J(U))$ in degree zero [5, Lemma 47.10.1]. The augmented cone is consequently exact on every basis object. Taking the filtered colimit over étale neighbourhoods makes its geometric stalks acyclic, so the sheaf cone is acyclic. This last step does not require sections to be exact on arbitrary sheaves. $\square$

Proposition B.11 (Derived torsion embedding over the base). *Let S be noetherian, $A = R_r[[T_1, \dots, T_d]]$, and $I = (T_1, \dots, T_d)$. Let $\mathcal{M} = \text{Mod}(S_{\text{ét}}, \underline{A})$ and let $\mathcal{T} \subset \mathcal{M}$ be the sheaves locally killed by powers of I . Inclusion induces an equivalence*

$$D^+(\mathcal{T}) \xrightarrow{\sim} D_{I\text{-tor}}^+(\mathcal{M}),$$

and hence is fully faithful on bounded constructible objects, with their mapping spectra. The target is a full subcategory of the unrestricted module-sheaf derived category, not the derived category of a separately chosen finite quotient ring.

Proof. Write $\Gamma_I F = \text{colim}_a \underline{\text{Hom}}_A(A/I^a, F)$. It is the right adjoint to the exact inclusion of torsion sheaves. We verify the derived comparison in that category.

Lemma B.10 supplies the sheaf-level calculation on injectives. More explicitly, Γ_I sends injectives of $\mathcal{M}$ to injectives of $\mathcal{T}$, since its left adjoint (inclusion) is exact. On a bounded-below injective resolution the finite-length flat Čech complex from that lemma computes the underlying object of $R\Gamma_I$. Its augmentation is the inclusion counit. Bounded-below totalization and the finite Čech length suffice here; no unbounded torsion-completion equivalence is needed.

For a torsion sheaf all the displayed localizations vanish. More generally, for a bounded-below complex with torsion cohomology they are acyclic, by flatness. Thus the counit $R\Gamma_I K \rightarrow K$ is an equivalence on precisely this part of the argument. For K in $D^+(\mathcal{T})$, apply this assertion to its included complex. The triangle identity identifies the included unit with the inverse counit; exact faithful inclusion detects its being an equivalence. Thus both unit and counit of the derived adjunction are equivalences on the stated subcategories, proving the claim. Its canonical enhanced functor identifies Hom in every shift, hence mapping spectra. No unbounded sheaf injective-completion assertion is needed here. $\square$

Definition B.12 (Independent adic Iwasawa module-sheaf model). At precision r let $\mathcal{M}_r^{\text{sel}}$ be the full subcategory of $D(\text{Mod}(S_{\text{ét}}, \underline{\Lambda}_r))$ with bounded I -power-torsion cohomology, constructible underlying R_r -cohomology, and stratified underlying R_r -perfectness. Here $\Lambda_r = R_r[[T_1, \dots, T_d]]$; modules and derived morphisms are defined in the unrestricted module-sheaf category before selecting this subcategory. Define

$$\mathcal{M}_{\text{Iw,ad}}^{\text{sel}}(S) = \varprojlim_{r, -\otimes_{R_{r+1}}^L R_r} \mathcal{M}_r^{\text{sel}},$$

restricted to a common cohomological and underlying- R_r Tor interval and a finite constructible stratification for the system. These are all the selection conditions; no further finite-Tor condition over an integral global Iwasawa sheaf is implicit. Its mapping spectrum, denoted $\mathrm{Map}_{\mathrm{Iw},\mathrm{ad}}$, is the mapping spectrum in this limit category. This is an independent algebraic definition: it uses module sheaves over completed coefficient rings, with no group action or root-site category in its construction.

This is the explicit relative completed coefficient model used in the proved comparison below. It does not define a global topologically projective resolution theory on the small étale site. The stronger identification with any such separately proposed pseudocompact sheaf category is a separate comparison problem and is not assumed implicitly.

Theorem B.13 (Adic Iwasawa realization of continuous actions). *On the noetherian split bases of Theorem B.5, including the specified finite semilinear descent, there is a natural equivalence*

$$\varprojlim_r D_{c,\mathrm{perf}}^b(\mathcal{A}_{\Gamma,r}) \xrightarrow{\sim} \mathcal{M}_{\mathrm{Iw},\mathrm{ad}}^{\mathrm{sel}}(S).$$

The same uniform conditions are imposed on both sides. This equivalence, together with relative finite-action descent, proves the coefficient equivalence for this explicit target model. At fixed precision it identifies relative continuous invariants with

$$R\mathrm{Hom}_{\Lambda_r}(R_r, E_r),$$

computed by the length- d augmentation Koszul resolution. This invariants complex commutes with derived coefficient reduction on the selected class. Integral mapping spectra are the coefficient inverse limits of the unrestricted Λ_r -module-sheaf mapping spectra.

Proof. Proposition B.8 identifies the smooth action abelian category with $\mathcal{T}$. Proposition B.11 identifies its bounded derived objects and their mapping spectra with the full torsion-cohomology subcategory of unrestricted Λ_r module sheaves. Restrict to the underlying-perfect constructible objects to obtain the equivalence at each r . The functors preserve the underlying R_r -complex and are compatible with scalar change; on this selected class reduction remains bounded. The canonical squares for reduction consequently give an equivalence of the limit categories, including their stated uniform restrictions. Finite semilinear descent is handled in the same abelian sheaf categories; all the constructions, including torsion and scalar change, respect it. This identifies the independently defined action and module models.

Trivial relative action is restriction along $\Lambda_r \rightarrow R_r$. Full faithfulness of the torsion embedding identifies its right adjoint with the displayed internal derived Hom: test the adjunction against arbitrary base complexes with trivial relative action in the bounded-below domain. The sequence $T_1, \dots, T_d$ is regular in Λ_r , even when $r > 1$, so its finite free Koszul complex resolves R_r . Applying internal Hom constructs the sheaf-level comparison and proves coefficient reduction and boundedness for these invariants. Coordinates are used for the computation only; the right adjoint is intrinsic. Mapping spectra in the adic category are limits by its universal property. This last assertion is categorical and does not interchange sheaf stalks with inverse limits. $\square$

B.3 Pseudocompact models at a point

Proposition B.14 (The point source and Iwasawa perfectness). *Let $A = \mathbf{Z}_\ell$, $\Gamma = \mathbf{Z}_\ell^d$ and $\Lambda = A[[\Gamma]]$. The entire category $\mathcal{S}_{\mathrm{ad}}(\Gamma)$ of Definition 2.1 is equivalent to*

$$\{M \in \mathrm{Perf}(\Lambda) : M|_A \in \mathrm{Perf}(A)\}.$$

The equivalence identifies the specified reductions with $M \otimes_{\Lambda}^L \Lambda_r = M \otimes_A^L R_r$ and their mapping spectra. These objects have classical pseudocompact representatives and are derived complete for $(\ell, T_1, \dots, T_d)$. Underlying- A -perfectness in the displayed class is essential: it makes every reduction have finite, I -power-torsion cohomology. For $d > 0$, the free module Λ alone is not in this class. In particular no extra Iwasawa finite-Tor or pseudo-coherence condition is needed in the definition of the point source.

Proof. Recovery with the action and the specified arrows. Write $\Lambda_r = \Lambda \otimes_A^L R_r$ and use the stable localizations of module complexes. Ideal cofinality and the derived torsion embedding (Propositions B.8 and B.11, over a point) identify bounded finite-cohomology continuous coefficients with the corresponding full subcategory of $D(\Lambda_r)$. These enhanced inclusions commute with derived coefficient change, as in Theorem B.13. Thus the input, including its specified scalar-change cells, gives an object $V_{\bullet}$ of $\mathcal{L} = \varprojlim_r D(\Lambda_r)$. Here the limit uses derived scalar extension, not restriction.

There is an adjunction

$$c : D(\Lambda) \rightleftarrows \mathcal{L} : L, \quad cM = (M \otimes_{\Lambda}^L \Lambda_r)_r, \quad L(V_{\bullet}) = R \varprojlim_r \text{Res}_{\Lambda} V_r.$$

Indeed the mapping spectrum into a cartesian section is

$$\text{Map}_{\mathcal{L}}(cM, V_{\bullet}) = \varprojlim_r \text{Map}_{\Lambda_r}(M \otimes_{\Lambda}^L \Lambda_r, V_r) = \text{Map}_{\Lambda}(M, LV_{\bullet}).$$

Set $M = LV_{\bullet}$. Its Λ -action is therefore present before any finiteness argument. The counit has components $\rho_r : M \otimes_{\Lambda}^L \Lambda_r \rightarrow V_r$, adjoint to the limit projections. The forgetful functor $D(\Lambda) \rightarrow D(A)$ preserves limits, and scalar extension here is tensor by the central A -perfect object R_r . After forgetting, ρ_r is exactly the recovery map of Lemma A.8 applied to the given cartesian system. That lemma applies directly in $D(\Lambda)$ as well. The forgetful functors are conservative, so ρ_r is an equivalence of Λ_r -objects. Moreover

$$\rho_r = \theta_r \circ (\rho_{r+1} \otimes_{R_{r+1}}^L R_r)$$

under scalar associativity, with the homotopy supplied by the counit in $\mathcal{L}$; θ_r is the original reduction. The same counit supplies the higher cells. This proves recovery of the given action diagram, including its reduction maps. Lemma A.9 now makes $M|_A$ perfect, and the recovery lemma makes M derived ℓ -complete.

Perfectness over the Iwasawa ring. Each $H^i(M)$ is finite over A . Its finite A -generating set also generates it over Λ , since $A \subset \Lambda$ and the action has just been constructed. A basis of Γ identifies Λ with the regular local ring $A[[T_1, \dots, T_d]]$ of dimension $d+1$. The finite-global-dimension theorem for regular local rings makes this bounded finite-cohomology complex perfect. A bounded finite free Λ -representative gives its classical pseudocompact model and derived maximal-ideal completeness. Its reductions have uniform Λ_r -Tor bounds.

Return to continuous and finite actions. Conversely, let M belong to the displayed category. The reductions $N_r = M \otimes_A^L R_r$ are underlying- R_r -perfect with one Tor interval, and their cohomology modules are finite sets. On each nonzero cohomology module the image of the local ring Λ_r in its endomorphism ring is a finite local quotient. Its maximal ideal is nilpotent; thus the cohomology is I -power torsion. This does not assert that N_r itself is a complex over one finite quotient merely because its cohomology has finite action. Instead Proposition B.11 identifies N_r , through the counit $R\Gamma_I N_r \rightarrow N_r$, with a bounded object of the derived torsion category. Ideal cofinality identifies that abelian category with discrete continuous Γ -modules. Proposition 3.1 then descends the whole derived object: successive Postnikov attaching maps, not only its cohomology modules, descend to common

finite quotients. Full faithfulness of the enhanced torsion inclusion transports the specified scalar arrows and cells of $N_\bullet$ back to the source. Their boundedness and Tor bounds follow from the unchanged underlying complexes. This produces the required source system.

Maps and independence of coordinates. Every M under consideration is derived ℓ -complete, since its underlying A -perfect complex is complete and forgetting is conservative. Thus both the unit of $c \dashv L$ on these objects and its counit on the selected systems are equivalences. In particular

$$\mathrm{Map}_\Lambda(M, N) = \varprojlim_r \mathrm{Map}_{\Lambda_r}(M \otimes_\Lambda^L \Lambda_r, N_r).$$

This follows from completeness in the second variable and scalar adjunction; no compactness of M is required for this identity. The finite-precision torsion equivalences identify it with the source mapping spectrum. All functors used in recovery are intrinsic to $A[[\Gamma]]$. A basis is used only to prove regularity; changing it acts by an automorphism of this same ring. Supplied semilinear actions therefore preserve the recovery maps and their cells. The statement asserts no global pseudocompact sheaf comparison over a general base. $\square$

Proposition B.15 (Agreement with classical pseudocompact Hom at a point). *Over a geometric point, let V, W be classical pseudocompact Λ -complexes in the stated admissible class, in particular V is perfect over Λ and W is derived ℓ -complete. Their image in Definition B.12 satisfies*

$$R\mathrm{Hom}_\Lambda^{\mathrm{cts}}(V, W) \simeq R\varprojlim_r R\mathrm{Hom}_{\Lambda_r}(V \otimes_\Lambda^L \Lambda_r, W \otimes_\Lambda^L \Lambda_r).$$

Proof. Choose a bounded finite-projective Λ -resolution of V . Finite projective modules are topologically projective and every Λ -linear map from them to a pseudocompact module is continuous. Thus the same finite Hom complex computes classical continuous and algebraic derived Hom. Its reduction computes the right-hand term at each r . Finite projectivity commutes with the derived inverse limit recovering W , proving the formula. This identifies the new relative model with the classical meaning on points; it does not infer a global topologically projective sheaf resolution from that point calculation. $\square$

Remark B.16 (Proved adic model and the stronger pseudocompact interpretation). The adic model on a noetherian split chart retains finite-action descent, attaching maps and finite semilinear descent. Its derived Hom is computed in unrestricted Λ_r module sheaves before coefficient completion. Over a point this agrees with the classical pseudocompact calculation on the admissible class.

The comparison with classical pseudocompact Hom is the point result of Proposition B.15. Over a general base we use the adic module-sheaf category just defined; an identification with a global topologically-projective pseudocompact sheaf category is a separate question. The next section identifies the actual root topos.

C The relative root site over a base

For the geometric root statements, a *split* base includes a chosen compatible identification of the relative finite etale inertia tower with the constant groups $K_n = (\mathbf{Z}/\ell^n)^d$. On the strictly local base of Definition A.7 this follows by choosing compatible primitive roots of unity, since ℓ is invertible. The semilinear descent action records its action on these choices. On another noetherian base such a compatible splitting must be supplied; it is not inferred for an arbitrary finite etale group scheme. The preceding abstract action-sheaf results use a constant profinite group on any noetherian base. The finite-quotient lemma below itself needs only a finite etale group scheme and does not assume it constant.

We now identify the geometric source of continuous inertia. The relative root fibre is first constructed as an actual fpqc stack. Finite-presentation descent then compares its representable étale site with finite root levels; the resulting topos identifies the structural star functor and its natural transformations.

C.1 The infinite-root central fibre and its relative pro- ℓ category

Talpo–Vistoli introduced the pullback of an infinite root stack along the lifted closed point as its central fibre; it is a pro-algebraic stack and need not itself be an infinite root stack [14]. The same construction is used in Scherotzke–Sibilla–Talpo [16]. The object below is its chartwise relative form followed by the prescribed finite tame descent. Its construction is prior art. What we isolate on its intrinsic full pro-root site is a relative-inertia-selected stratified-perfect constructible category and the removal of the base inertia direction. It is not the closed stratum of the ordinary Artin fan, which retains a diagonalizable stabilizer in that direction. Whenever a face τ is fixed in this subsection, we write $P = P_\tau$ and $\rho = \rho_\tau$ (and likewise for the associated relative lattice). Thus all root-stack constructions below are understood facewise; the suppressed face index may be restored along the supplied augmented log-point maps of Lemma C.13. This convention constructs no specialization or incidence map between strata.

For a fine saturated monoid Q and an integer a invertible on S , write $\sqrt[a]{\mathfrak{s}_{Q/S}}$ for the a th root stack. In the chart above it has the standard presentation

$$\sqrt[a]{\mathfrak{s}_{Q/S}} = \left[(\mathrm{Spec} S[\tfrac{1}{a}Q] \times_{\mathrm{Spec} S[Q]} S) / D\left((\tfrac{1}{a}Q)^{\mathrm{gp}}/Q^{\mathrm{gp}}\right) \right],$$

where $D(A) = \underline{\mathrm{Hom}}(A, \mathbf{G}_m)$ and the map $\mathrm{Spec} S \rightarrow \mathrm{Spec} S[Q]$ is the log-point augmentation. Thus every finite level is a genuine finite-type geometric Artin stack; when Q^{gp} is free and S is strictly local its reduced geometric fibre is $B\mu_a^{\mathrm{rk} Q}$. For example, for $Q = \mathbf{N}$ over a field k one obtains

$$\sqrt[a]{\mathfrak{s}_{\mathbf{N}/k}} \simeq [\mathrm{Spec} k[x]/(x^a) / \mu_a, \quad (\sqrt[a]{\mathfrak{s}_{\mathbf{N}/k}})_{\mathrm{red}} \simeq B\mu_a.$$

This elementary example is why all classifying-stack statements below are made for the reduction or for the nil-invariant representable étale topos, never for the unreduced finite root stack itself. For a possibly nonreduced base S , the notation $(-)_{\mathrm{red}}$ below means the *relative monomial reduction*: quotient by the nilpotent ideal generated by positive fractional monomials while leaving S unchanged. Its classifying stack at order ℓ^n is therefore $B_S \mathcal{K}_n$.

Let $\mathbf{St}_{\mathrm{fpqc}}(S)$ denote the infinity-topos of fpqc sheaves of spaces on S -schemes (in a fixed universe). The root stacks are its 1-truncated objects, equivalently stacks in groupoids. Limits of these remain 1-truncated and compute their 2-categorical limits. In the following display every inverse limit is formed in $\mathbf{St}_{\mathrm{fpqc}}(S)$:

$$\sqrt[\infty]{\mathfrak{s}_{Q/S}} := \varprojlim_{(a,p)=1} \sqrt[a]{\mathfrak{s}_{Q/S}}, \quad \sqrt[\ell^\infty]{\mathfrak{s}_{Q/S}} := \varprojlim_{n \geq 0} \sqrt[\ell^n]{\mathfrak{s}_{Q/S}}.$$

The first is the prime-to- p infinite root stack and the second is its actual pro- ℓ limit stack. We reserve $\{\sqrt[\ell^n]{\mathfrak{s}_{Q/S}}\}_n$ for the associated pro-object. The distinction matters: the representable finitely presented étale site of the actual limit will be compared with the filtered finite-level site below, by the finite-presentation descent theorem below. The map $\mathbf{N}\rho \rightarrow P$ induces maps at every root level. The zero (trivial-root) section, induced at each finite level by the trivial line bundle with its zero section,

$$e_{\rho,S} : S \longrightarrow \sqrt[\infty]{\mathfrak{s}_{\mathbf{N}\rho/S}}$$

is compatible with the finite-level sections $e_{\rho,S,n}$. Define first the split relative infinite-root central fibre over S and then its descended quotient over $\mathcal{S} = [S/\Delta]$:

$$\tilde{\mathcal{A}}_M^{\text{rel}} := e_{\rho,S} \times_{\sqrt[a]{\mathfrak{s}_{\mathbf{N}\rho/S}}}^{(2)} \sqrt[a]{\mathfrak{s}_{P/S}}, \quad \boxed{\mathcal{A}_M^{\text{rel}} := [\tilde{\mathcal{A}}_M^{\text{rel}}/\Delta] \longrightarrow \mathcal{S}}. \quad (18)$$

For an arbitrary root order a prime to p , reserve bracketed notation

$$\tilde{\mathcal{A}}_{M,[a]}^{\text{rel}} := e_{\rho,S,[a]} \times_{\sqrt[a]{\mathfrak{s}_{\mathbf{N}\rho/S}}}^{(2)} \sqrt[a]{\mathfrak{s}_{P/S}}. \quad (19)$$

Unbracketed n below always denotes the exponent in the pro- ℓ root order ℓ^n ; coefficient precision is denoted by r . Thus the three indices a, n, r are never interchangeable. Its pro- ℓ refinement is obtained by replacing every infinite root stack in (18) with its ℓ^∞ -part; denote this restricted geometric object by

$$\tilde{\mathcal{A}}_{M,\ell}^{\text{rel}} := e_{\rho,S} \times_{\ell^\infty \sqrt[\mathfrak{s}_{\mathbf{N}\rho/S}]}^{(2)} \ell^\infty \sqrt[\mathfrak{s}_{P/S}], \quad \mathcal{A}_{M,\ell}^{\text{rel}} := [\tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}/\Delta] \longrightarrow \mathcal{S}. \quad (20)$$

The finite-level relative faces are

$$\tilde{\mathcal{A}}_{M,n}^{\text{rel}} := e_{\rho,S,n} \times_{\ell^n \sqrt[\mathfrak{s}_{\mathbf{N}\rho/S}]}^{(2)} \ell^n \sqrt[\mathfrak{s}_{P/S}], \quad \mathcal{A}_{M,n}^{\text{rel}} := [\tilde{\mathcal{A}}_{M,n}^{\text{rel}}/\Delta] \longrightarrow \mathcal{S}.$$

The bare central fibre records relative characteristic inertia but not the ordinary smooth directions of the product chart. It is therefore the correct coefficient space for the root comparison, but it is not by itself the third geometric object for compact supports. From now on, supply the additional log-smooth model and compatible action of Definition A.7, and let

$$v : \mathbf{V}_{M,e} := \mathbf{V}(\mathcal{V}_{M,e}) \longrightarrow S, \quad N := d + e,$$

and define the *smooth-enhanced root faces*

$$\tilde{\mathcal{R}}_{M,e,n}^{\text{rel}} := \tilde{\mathcal{A}}_{M,n}^{\text{rel}} \times_S \mathbf{V}_{M,e}, \quad \mathcal{R}_{M,e,n}^{\text{rel}} := [\tilde{\mathcal{R}}_{M,e,n}^{\text{rel}}/\Delta], \quad g_{M,e,n} : \mathcal{R}_{M,e,n}^{\text{rel}} \rightarrow \mathcal{S}. \quad (21)$$

In addition to the inverse system of these finite levels, define its actual fpqc limit stack and structural map by

$$\tilde{\mathcal{R}}_{M,e,\ell}^{\text{rel}} := \tilde{\mathcal{A}}_{M,\ell}^{\text{rel}} \times_S \mathbf{V}_{M,e}, \quad \mathcal{R}_{M,e,\ell}^{\text{rel}} := [\tilde{\mathcal{R}}_{M,e,\ell}^{\text{rel}}/\Delta], \quad g_{M,e,\ell}^{\text{act}} : \mathcal{R}_{M,e,\ell}^{\text{rel}} \rightarrow \mathcal{S}. \quad (22)$$

The notation $\{\mathcal{R}_{M,e,n}^{\text{rel}}\}_n$ is reserved for its finite-level pro-object. Coefficients on the actual limit are pulled back from the central-root factor using Corollary C.24. Homotopy invariance gives $Rv_* v^* \simeq \text{id}$ on these coefficients, whereas smooth purity gives $Rv_! v^* \simeq (-N)[-2N]$. Thus the carrier leaves the $*$ -realization unchanged and realizes the full logarithmic-smooth compact-support shift. In the stress-test case $d = 0, e = 1$ and $P = \mathbf{N}\rho$, the object is $\mathbf{A}_S^1 \rightarrow S$; its constant-coefficient shriek image is correctly $\mathbf{Z}/\ell^r(-1)[-2]$.

Proposition C.1 (Intrinsic tangent carrier on the fixed log model). *The vector bundle $\mathcal{V}_{M,e}$ and its zero section are independent of any product-coordinate splitting of the bundle in (10). Two strict log-etale product charts on the same log model induce canonically identified carriers, projective-bundle compactifications, Thom orientations, and finite-level $Rg_*/Rg_!$ operations. Under the product decomposition, the determinant line is*

$$\det(\mathcal{V}_{M,e}) \simeq \det(M \otimes \mathcal{O}_S) \otimes \det(\mathcal{E}_{\text{sm}}).$$

Proof. For a log-smooth morphism the logarithmic cotangent complex is a locally free sheaf in degree zero; its dual commutes with strict log-etale pullback. The relative characteristic exact sequence removes the base summand $\mathbf{Z}\rho$ and identifies the remaining logarithmic tangent directions with $M \otimes \mathcal{O}_S$. The ordinary smooth tangent directions form the quotient $\mathcal{E}_{\text{sm}}$. A product coordinate splits this exact sequence but does not change its middle term. Vector-bundle formation, the projective completion $\mathbf{P}(\mathcal{V}_{M,e} \oplus \mathcal{O}_S)$, Thom purity, and the six operations are functorial under the resulting bundle isomorphism. Taking determinants gives the displayed identity. $\square$

Remark C.2 (Canonicity and locality of the carrier). Proposition C.1 makes the carrier intrinsic to the fixed log-smooth model and independent of a product splitting. It is not asserted to be intrinsic to the bare infinite-root central fibre, nor is invariance under a logarithmic modification proved. Comparison between genuinely different log models still requires an isomorphism or a future modification-descent theorem. Within the fixed six-functor formalism, $Rg!$ is independent of the chosen compactification, but this standard independence does not imply log-model independence. Theorem 1.1 uses this fixed-model, local, canonicity.

C.2 The relative limit and its atlas

Lemma C.3 (Relative 2-fibres are the finite-level inverse system). *The compatible finite-level root maps induce canonical equivalences of groupoids, functorial in the test scheme over S before descent,*

$$\tilde{\mathcal{A}}_{M,\ell}^{\text{rel}} \simeq \varprojlim_{n \geq 0} \tilde{\mathcal{A}}_{M,n}^{\text{rel}}, \quad \tilde{\mathcal{A}}_M^{\text{rel}} \simeq \varprojlim_{(a,p)=1} \tilde{\mathcal{A}}_{M,[a]}^{\text{rel}}.$$

In the first display, n is the exponent in the root order ℓ^n ; in the second, a runs through all integers prime to p . Taking the quotient by Δ gives the corresponding inverse systems over $\mathcal{S}$. Here the limits are limits of 2-groupoids (equivalently, of the corresponding fibred groupoids). In particular, the relative 2-fibre is not being defined by first choosing the auxiliary Kummer site.

Proof. For a test scheme $S' \rightarrow S$, an object of the left-hand side is a compatible system of roots on the P -chart over S' , together with a compatible 2-isomorphism between its restriction along $\mathbf{N}P \rightarrow P$ and the trivial root section. Giving this datum at the infinite root level is, by definition of the inverse-limit root stack, the same as giving it at every finite level with compatibility under divisibility of the root index. The same assertion holds for morphisms: a morphism and its 2-isomorphism are determined levelwise, and the compatibility equations are exactly those in the inverse-limit groupoid. Thus the two sides have equivalent mapping groupoids for every S' , and the displayed equivalence follows from the 2-Yoneda lemma.

All transition maps are Δ -equivariant over the same S . For the quotient assertion take the limit in stacks over the fixed $\mathcal{S} = [S/\Delta]$. A test map $T \rightarrow \mathcal{S}$ fixes one Δ -torsor $P \rightarrow T$ and its equivariant map $P \rightarrow S$. A lift to the quotient at level n is an equivariant lift of this same P to the split root fibre at level n . A compatible family of lifts, including their equivariance cells, is exactly a lift to the split inverse limit: mapping into a limit preserves its limit and the descent equations. The same holds for morphisms of lifts. Descent on the fixed torsor therefore proves the assertion over $\mathcal{S}$. It is not a general claim that taking orbit quotients commutes with inverse limits merely because Δ is finite. $\square$

Lemma C.4 (Torsor transitions in the relative quotient atlas). *In the split chart, put $L = P^{\text{gp}}/\mathbf{Z}\rho$ and $H_n = \text{Hom}(L, \mu_{\ell^n})$. The relative finite root fibre has a compatible presentation $[Z_n/H_n]$ with Z_n affine. For an affine test object T carrying compatible maps to these fibres,*

let $P_n = T \times_{[Z_n/H_n]} Z_n$. Then P_n is an H_n -torsor, and $P_m \rightarrow P_n$ is a torsor under $\ker(H_m \rightarrow H_n)$ for $m \geq n$. In particular these test-torsor transitions are finite faithfully flat. This statement does not assert flatness of $Z_m \rightarrow Z_n$.

Proof. Work over an affine open $S = \operatorname{Spec} B$ of the split base. For root order $a = \ell^n$, the absolute root atlas and the rigidified relative atlas have coordinate rings

$$B[a^{-1}P]/(X^p : p \in P \setminus \{0\}), \quad B_n = B[a^{-1}P]/(X^p : p \neq 0, X^{\rho/a}),$$

respectively: the second quotient is pullback along the zero section of the base-root coordinate. The base log point sends every positive monomial to zero. Choose nonzero generators $p_1, \dots, p_t$ of P . The ideal $J_n = (X^{p_1/a}, \dots, X^{p_t/a})$ in B_n satisfies $J_n^{t(a-1)+1} = 0$, since each generator has a th power zero. Sharpness gives $B_n/J_n = B$, even when B itself is nonreduced. This proves the required relative nilpotence with an explicit bound; no uniform bound as n varies is used.

The residual action is by the finite etale group H_n below. Its nerve has coordinate schemes $U_{n,q} = \operatorname{Spec} B_n \times_S H_n^q$, affine and qcqs. For $m \geq n$ the monoid map sends X^{p/ℓ^m} to $(X^{p/\ell^n})^{\ell^{m-n}}$; together with $H_m \rightarrow H_n$ it makes all nerve transitions affine. These formulas commute with localization on S , so the presentations and their properties descend from affine opens. They are the geometric input to the finite-presentation limit theorem.

Write the absolute root chart as $[V_n/G_{P,n}]$ and the base root chart as $[W_n/G_{\rho,n}]$, where $G_{P,n} = \operatorname{Hom}(P^{\text{gp}}, \mu_{\ell^n})$ and $G_{\rho,n} = \mu_{\ell^n}$. Primitivity of ρ and freeness of L imply that $G_{P,n} \rightarrow G_{\rho,n}$ is a finite faithfully flat surjection with kernel H_n . Pulling back along the trivial base-root object is equivalent to choosing a trivialization of its associated $G_{\rho,n}$ -torsor. The remaining structure group is H_n , and the coordinate scheme is $Z_n = V_n \times_{W_n} S$, where $S \rightarrow W_n$ is the zero-root section. Thus the relative two-fibre is $[Z_n/H_n]$. This argument uses the groupifications and an affine fibre product; it does not require the fs monoid P itself to be free.

Refinement gives an equivariant map $Z_m \rightarrow Z_n$ for the surjection $H_m \rightarrow H_n$. In the quotient-stack description a map from T is a torsor with an equivariant map to Z_n . Compatibility of the test objects therefore gives the specified isomorphism $P_m \times^{H_m} H_n \simeq P_n$. After a faithfully flat cover trivializing P_m , its map to P_n is exactly $H_m \rightarrow H_n$. The kernel is finite etale of rank $\ell^{(m-n)\operatorname{rk} L}$, since ℓ is invertible. The corresponding coordinate algebra is locally free of this positive rank; flatness, faithful flatness and the torsor identity descend. This proves the assertion without any flatness claim about the monoid coordinate maps. The construction uses the supplied refinement two-cells, so it respects composition. For finite Δ -descent the same assertion is checked after the fpqc cover $S \rightarrow [S/\Delta]$; torsors and finite faithfully flat maps descend, also when no Δ -equivariant lattice splitting is chosen. $\square$

Proposition C.5 (Effective atlas of the actual infinite-root fibre). *Let $U_{n,\bullet} \rightarrow \tilde{\mathcal{A}}_{M,n}^{\text{rel}}$ be the compatible affine action-groupoid presentations constructed from the two finite root charts, and put*

$$U_{\infty,q} := \varprojlim_n U_{n,q}.$$

Then $U_{\infty,0}$ is affine over S , the map

$$U_{\infty,0} \longrightarrow \tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}$$

is an effective fpqc epimorphism, and its Čech nerve is canonically $U_{\infty,\bullet}$. Consequently

$$\tilde{\mathcal{A}}_{M,\ell}^{\text{rel}} \simeq |U_{\infty,\bullet}| \quad \text{in } \mathbf{St}_{\text{fpqc}}(S), \quad (23)$$

and this is simultaneously the actual inverse-limit stack and the stack presented by the limiting root groupoid. The same statement holds after effective finite Δ -descent.

Proof. Test after an arbitrary affine map $T \rightarrow \tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}$. Its projections give compatible maps $T \rightarrow \tilde{\mathcal{A}}_{M,n}^{\text{rel}}$, and

$$P_n := T \times_{\tilde{\mathcal{A}}_{M,n}^{\text{rel}}} U_{n,0}$$

is the finite diagonalizable torsor of Lemma C.4, trivializing the level- n relative root data. For $m \geq n$, the transition $P_m \rightarrow P_n$ is the torsor under the kernel of the finite diagonalizable quotient $\mathcal{H}_m \rightarrow \mathcal{H}_n$ (with the base-root factor rigidified); in particular it is finite faithfully flat. This remains so after *any* base change. In particular, for a geometric point $t : \text{Spec } \Omega \rightarrow T$, the maps $\mathcal{O}(P_n \times_T t) \rightarrow \mathcal{O}(P_m \times_T t)$ are injective. This is faithful-flat injectivity after base change, not an assertion that tensoring an arbitrary injective ring map remains injective. The inverse limit $P_\infty = \varprojlim_n P_n$ is $\text{Spec}_T(\varprojlim_n \mathcal{O}_{P_n})$. Each $P_n \rightarrow T$ is finite locally free and faithfully flat. Its filtered-colimit algebra is flat. Tensoring with Ω commutes with the filtered colimit; the preceding fibrewise injectivity makes its nonzero unit survive. Thus every geometric fibre is nonempty. We use the elementary criterion that a flat algebra B over a ring A is faithfully flat if and only if $B/\mathfrak{m}B \neq 0$ for every maximal ideal $\mathfrak{m}$ of A . Indeed, for $0 \neq x \in M$ the cyclic submodule $Ax = A/I$ embeds in M . Choose $\mathfrak{m} \supset I$; B/IB surjects onto the nonzero $B/\mathfrak{m}B$, and flatness injects B/IB into $M \otimes_A B$. Thus tensor detects nonzero modules; the converse follows by testing residue fields. Nonzero geometric fibres imply this residue-field condition by faithfulness of field extension. Apply the criterion to the filtered-colimit algebra above. Hence $P_\infty \rightarrow T$ is affine faithfully flat and quasi-compact. Since this holds after every affine test map, $U_{\infty,0} \rightarrow \tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}$ is fpqc.

Work in the sheaves-of-spaces infinity-topos specified above; its 1-truncated root objects compute the stack limits. Limits commute with limits in this ambient category. Hence

$$\underbrace{U_{\infty,0} \times_{X_\infty} \cdots \times_{X_\infty} U_{\infty,0}}_{q+1} \simeq \varprojlim_n \underbrace{U_{n,0} \times_{X_n} \cdots \times_{X_n} U_{n,0}}_{q+1} = U_{\infty,q}.$$

Here X_n denotes the relative root fibre at level n . The equivalences preserve the projections and diagonals defining all face and degeneracy maps. They therefore identify the whole Čech nerve and all its structure maps. The representable fpqc map is locally surjective on objects, hence epimorphic on π_0 in this sheaf infinity-topos. Such a map is an effective epimorphism: by the effective-epimorphism/Čech-descent property of an infinity-topos, its Čech realization recovers the target. On the other hand finite limits commute with inverse limits in the same infinity-topos; applying this to the defining relative 2-fibre identifies that target with the actual stack $\tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}$ of (18). This proves (23) and supplies fpqc effectivity for this presentation. The comparison with the representable étale site is proved in Theorem C.9; its continuous-action interpretation and derived structural realization are proved separately in Theorem C.20 and Corollary C.21. The atlas construction is equivariant for Δ . The quotient is formed from its action groupoid and fpqc descent; it is not a finite limit merely because Δ is finite. Its étale coefficient comparison is given by finite semilinear descent and the root-topos comparison below. $\square$

C.3 Finite-presentation descent for the root site

Definition C.6 (Representable étale objects over an fpqc stack). For an fpqc stack X , an object of $\text{RepEt}_{\text{fp}}(X)$ is a morphism $Y \rightarrow X$ of fpqc stacks such that, for every scheme $T \rightarrow X$, the pullback $Y \times_X T$ is an algebraic space étale and of finite presentation over T . The source Y need not be algebraic. Morphisms are morphisms over X , with the canonical identifications in the slice; representability makes this an ordinary category up to its canonical equivalence. Coverings are jointly surjective étale families. At finite algebraic root levels this convention gives the usual representable étale objects; at the actual limit it permits their base changes, which may be non-algebraic fpqc stacks.

Lemma C.7 (The scheme-source test at infinite inertia). *Over an algebraically closed field of characteristic different from ℓ , let $G = \varprojlim_n K_n$ be an infinite proconstant group scheme. There is no nonempty algebraic space Y admitting a representable etale finitely presented morphism $Y \rightarrow BG$. Thus restricting the source in Definition C.6 to schemes or algebraic spaces does not give a covering site for BG .*

Proof. Pull back by the trivial torsor $\mathrm{Spec} k \rightarrow BG$. The resulting $P = Y \times_{BG} \mathrm{Spec} k$ is an etale finitely presented algebraic space over k , hence a finite discrete set, and $P \rightarrow Y$ is a G -torsor. Nonemptiness of Y implies nonemptiness of P . The torsor action is free because Y is an algebraic space; after taking geometric points this would be a free action of the infinite group $G(k)$ on a nonempty finite set. This is impossible. Sources such as $BH \rightarrow BG$ for an open subgroup $H \subset G$ are instead allowed by the stated fpqc-stack convention. $\square$

Lemma C.8 (Representable etale objects on a finite quotient). *Let a finite etale group scheme G act on an affine scheme U . For the fpqc quotient stack $X = [U/G]$, pullback to U identifies $\mathrm{RepEt}_{\mathrm{fp}}(X)$ with G -equivariant etale finitely presented algebraic spaces $V \rightarrow U$. Its inverse sends V to $[V/G] \rightarrow [U/G]$. This morphism is representable even when the action on V has stabilizers. Morphisms, their equalities, and covering families are detected on the atlas U .*

Proof. The action is an isomorphism between the two pullbacks of V to $G \times U$, with its unit and cocycle identities on U and $G \times G \times U$. Form $[V/G]$ as the stack of G -torsors with equivariant maps to V . A test map $T \rightarrow [U/G]$ is a torsor $P \rightarrow T$ with an equivariant map $P \rightarrow U$. Directly from these objects and arrows,

$$T \times_{[U/G]} [V/G] \simeq (P \times_U V)/G.$$

The diagonal action here is free because P is a torsor, regardless of stabilizers on V . Its quotient is the algebraic space obtained by etale descent along the finite etale cover $P \rightarrow T$. Explicitly, an etale scheme atlas of $P \times_U V$ and its translates give an etale equivalence relation in schemes; the sheaf quotient is that descended algebraic space. Its pullback to P is $P \times_U V$, etale and finitely presented over P . These properties descend along this finite etale cover. This proves representability and the required finiteness over every T . In particular its pullback to U is exactly V .

Conversely, pulling back a representable object on X gives this action datum. The stack descent equivalence reconstructs the object: locally a lift consists of a map to V , with precisely the prescribed action isomorphisms. A map of such objects is a G -equivariant map of the algebraic spaces; its compatibility is one equation on $G \times U$. Relative representability makes the mapping groupoids discrete, so there are no additional independent higher 2-cell choices to descend. A family covers if and only if its pullback to U is jointly surjective. Composition and pullback of covers are the corresponding operations on etale algebraic spaces. $\square$

The finite action-groupoid descent below uses the same fixed Δ -torsor over each test of $\mathcal{S}$ as in Lemma C.3; no new orbit-quotient/limit interchange is used.

The following result concerns the specified relative pro- ℓ root fibre over S . Talpo–Vistoli’s [14, Theorem 6.22, arXiv v2 numbering] already gives the absolute Kummer-etale/infinite-root topos equivalence; [23, §§5–6] compare the actual root stack, its pro-system and its representable etale site after profinite shape. Here we must additionally track the chosen base-root neutralization, restriction to the ℓ -power tower, and the structural coefficient functor over S . The next argument descends actual finite-presentation objects; the later coefficient theorem handles precision r and its reduction arrows. We use the effective chart

atlas above and descend its degree-0, 1, 2 étale data. The conclusion below is a site equivalence on this chart; its identification over S and its continuous derived structural functor are proved afterwards.

Theorem C.9 (Finite-presentation descent on the actual root site). *Let $X_n = \tilde{\mathcal{A}}_{M,n}^{\text{rel}}$ and $X_\infty = \tilde{\mathcal{A}}_{M,\ell}^{\text{rel}}$ on the fixed split chart. Use their compatible affine action-groupoid presentations $U_{n,\bullet}$ and the effective limit presentation of Proposition C.5. Explicitly, over each affine open of S , $U_{n,q}$ is affine (hence qcqs), its root transitions are affine, and $U_{n,\bullet}$ is the nerve of a finite étale root-group action; ℓ is invertible on the base. These properties follow from the explicit coordinate formulas in Lemma C.4 and are the hypotheses used below. Then pullback induces an equivalence of sites*

$$\text{colim}_n \text{RepEt}_{\text{fp}}(X_n) \xrightarrow{\sim} \text{RepEt}_{\text{fp}}(X_\infty),$$

with the source convention of Definition C.6 and the finite-cover topology on the left. It is natural over S , under the specified strict base changes, and under finite Δ -descent. In particular it proves the actual-site interpretation of the filtered site. Identification of this topos with continuous action sheaves and its derived structural functor is identified in Corollary C.21.

Proof. Work first over an affine open of S . There all $U_{n,q}$ are affine and their transitions are affine. The resulting site comparisons commute with restriction to affine overlaps and hence descend to S by the same étale descent of objects and maps. These conditions come from the explicit monoid-algebra and finite-group presentations. The proof uses the algebraic-space limit theorem for finite-presentation objects and morphisms, and descent of étaleness and surjectivity after increasing the index [4]. It does not use a limit theorem for stacks with nonrepresentable root transition maps.

The descent input in addition to the algebraic-space limit theorem is the following. Relative representability makes an étale object over X_∞ a 0-truncated object in the fpqc slice topos, although its total stack may be 1-truncated. On the effective atlas it is therefore determined by an algebraic space V over $U_{\infty,0}$, an isomorphism between its two pullbacks to $U_{\infty,1}$, and its unit and cocycle equations on degrees zero and two. Higher simplices give the iterated composites of that isomorphism. Morphisms are maps over degree zero satisfying the degree-one equivariance equation.

Conversely these data glue as a sheaf of sets in the fpqc slice, by effectivity of the atlas. To verify relative representability, test on a scheme $T \rightarrow X_\infty$. Its pulled-back atlas $T' \rightarrow T$ is fpqc, and the data give an étale finitely presented algebraic space over T' with descent datum. Effective fpqc descent of étale algebraic spaces yields an étale finitely presented space over T . Thus the glued slice object has exactly the source type specified in Definition C.6. This establishes the degree-0, 1, 2 description on which the following finite-stage descent operates. At a finite stage the quotient construction is verified separately below.

Let $Y_\infty \rightarrow X_\infty$ be an object of the right-hand site. Its pullback $V_\infty \rightarrow U_{\infty,0}$ is an étale finitely presented algebraic space, with the descent isomorphism between its two pullbacks to $U_{\infty,1}$. First descend V_∞ to an étale finitely presented $V_n \rightarrow U_{n,0}$. The two pullbacks of V_n to $U_{n,1}$ are finitely presented; their isomorphism and its inverse descend to a common later stage by the morphism part of the same limit theorem. The inverse identities and the unit identity descend as equalities of morphisms. The cocycle identity descends on $U_{n,2}$ after one further refinement. This is a finite amount of ordinary groupoid descent data.

At this stage the unit and degree-two cocycle are exactly a G_n -action on V_n over $U_{n,0}$, where G_n is the finite étale group of the root quotient chart. Lemma C.8 constructs $Y_n = [V_n/G_n] \rightarrow [U_{n,0}/G_n] = X_n$ and proves it representable étale of finite presentation. In particular the quotient's representability is not supplied by the algebraic-space limit theorem.

Pulling it back to X_∞ gives exactly the descent datum of Y_∞ . Both pullbacks carry the same degree-zero object and degree-one isomorphism with the degree-two cocycle. Descent for objects and arrows of the fpqc stacks along the effective atlas identifies $Y_n \times_{X_n} X_\infty$ with Y_∞ . This proves essential surjectivity.

A morphism between two such objects pulls back to a morphism of finitely presented algebraic spaces over $U_{\infty,0}$, with its compatibility equation over $U_{\infty,1}$. Descend the morphism and then this equation. Equality of two morphisms is likewise detected after increasing the stage. This proves precisely the filtered Hom formula and full faithfulness.

For a finite family of arrows, pull back to the affine atlas of the target object. Joint surjectivity is surjectivity of a single finite coproduct of finitely presented étale morphisms. If it holds at the limit, the surjectivity part of algebraic-space limit descent makes it hold at a later stage. The converse follows by base change. An arbitrary covering of a finitely presented object has a finite subcover: its pullback has open images covering a quasi-compact algebraic space. The quotient lemma detects these covers on the atlas; their pullbacks and finite composites are again detected there. Thus the finite-cover topology generates the full induced étale topology. The only limit inputs were descent of finite-presentation algebraic spaces on degrees 0, 1, 2, followed by the finite quotient construction.

All comparisons are induced by pullback; the noncanonical finite stage choices do not enter the resulting equivalence. Finite Δ -descent can be included in the finite action-groupoid presentations before the argument. Their degrees 0, 1, 2 still have the required quasi-compactness and affine transitions, giving the same proof and the asserted naturality. $\square$

Definition C.10 (Pro-root site and double-index constructible category). For a root level $n \geq 0$, let

$$\mathcal{C}_n := \text{RepEt}_{\text{fp}}(\tilde{\mathcal{A}}_{M,n}^{\text{rel}})$$

be its full representable, finitely presented étale site. If $m \geq n$, root pullback gives $q_{m,n}^* : \mathcal{C}_n \rightarrow \mathcal{C}_m$. Define $\mathcal{C}_\infty$ as follows. An object is a pair (n, Y_n) ; a morphism is

$$\text{Hom}_{\mathcal{C}_\infty}((n, Y_n), (n', Y_{n'})) := \varinjlim_{m \geq n, n'} \text{Hom}_{\mathcal{C}_m}(q_{m,n}^* Y_n, q_{m,n'}^* Y_{n'}).$$

A finite family is covering if, after one common refinement m , it is jointly surjective in $\mathcal{C}_m$.

The action of Δ on the root tower induces an action on its sheaf category. We use equivariant descent in that category:

$$\text{Sh}(\mathcal{C}_\infty^\Delta) := \text{Sh}(\mathcal{C}_\infty)^{h\Delta} = \lim_{B\Delta} \text{Sh}(\mathcal{C}_\infty). \quad (24)$$

Here the diagram consists of the inverse-image equivalences of the action; its objects are sheaves with coherent equivariance data. The notation on the left abbreviates the action-groupoid sheaf category, not a claim that taking equivariant representable objects is already a comparison of sites. The analogous finite-level notation is used for $\mathcal{C}_n$. The nerve $B\Delta$ has arbitrarily high simplicial degrees even though Δ is finite. No commutation with a filtered colimit is inferred from calling that nerve a finite diagram. Its coefficient descent is Corollary B.7.

Let $\mathcal{X}_{\text{proot}} := \text{Sh}(\mathcal{C}_\infty^\Delta)$. The intrinsic notion of constructibility used here is the coherent-topos one. A finite stratification is a finite decomposition of the terminal topos by coherent locally closed subtopoi; a complex is constructible if its cohomology objects are locally constant with finite stalks on those subtopoi. This definition uses neither a Kummer presentation nor a finite root level. Write $\mathcal{D}_{c,\text{pf}}^b$ for the full subcategory whose restrictions to all strata are perfect over $\mathbf{Z}/\ell^r$. The site and this perfect constructible category are intrinsic. The

following Iwasawa-admissible subcategory is then selected by the relative inertia supplied by (P, ρ) :

$$\mathcal{D}_{c, \text{Iw}}^b(\mathcal{A}_{M, \ell}^{\text{rel}}/\mathcal{S}, \mathbf{Z}_\ell) := \varprojlim_r \mathcal{D}_{c, \text{pf}}^b(\mathcal{X}_{\text{proot}}, \mathbf{Z}/\ell^r)_{\text{unif, Iw}}, \quad (25)$$

where the inverse limit is formed in stable ∞ -categories. The subscript means: common finite cohomological and underlying- R_r Tor intervals, a finite stratification on which all cohomology sheaves have finite stalks, the adic reduction equivalences, and the underlying- R_r perfectness at each precision in the displayed category. The Iwasawa interpretation is Definition B.12; at points its additional classical finite-Tor properties follow from Proposition B.14. The root index does not enter this definition. These conditions specify the selected perfect adic system, rather than a locally free ordinary small-etale integral sheaf, and do not impose torsion-free cohomology; the complex $[\mathbf{Z}_\ell \xrightarrow{\ell} \mathbf{Z}_\ell]$ at $M = 0$ belongs to the category. Its finite-level derived presentation follows from Theorem B.5; its actual-root interpretation is Corollary C.21.

Lemma C.11 (Continuous mapping complexes on a pro-root stratum). *Let $\mathcal{T}$ be a coherent stratum on which the relative inertia tower is the quotient tower $\mathcal{K}_m$ of $\mathcal{K}_\ell$, and let F_n, G_n be perfect locally constant $\mathbf{Z}/\ell^n$ -complexes at one finite level. For $m \geq n$ write F_m, G_m for their inflations. Then the natural map is an equivalence of mapping spectra*

$$\text{hocolim}_{m \geq n} R\text{Hom}_{B\mathcal{K}_m}(F_m, G_m) \xrightarrow{\sim} R\text{Hom}_{B\mathcal{K}_\ell}^{\text{cts}}(F, G). \quad (26)$$

After a split geometric pullback, both sides are represented by

$$C_{\text{cts}}^*(\Gamma_M, R\text{Hom}(F, G)) = \varinjlim_m C^*(\Gamma_M/\ell^m \Gamma_M, R\text{Hom}(F, G)).$$

No finite-level Ext-amplitude is asserted. In particular, the equality allows the periodic high-degree classes at finite levels to die under inflation. The statement is compatible with finite descent groups and finite recollement diagrams.

Proof. At finite level the mapping object on the classifying stratum is the full normalized inhomogeneous cochain complex on $\mathcal{K}_m$ with coefficients in $R\text{Hom}(F, G)$. On the pro-stratum it is the analogous continuous cochain complex. A continuous cochain from $\mathcal{K}_\ell^q$ to a finite discrete coefficient factors through some $\mathcal{K}_m^q$; hence the displayed equality is degreewise before taking cohomology. Since F and G are perfect, their internal derived Hom has finite amplitude, so each total degree contains only finitely many bar/internal bidegrees. Exact filtered colimits of finite modules therefore commute with this totalization and with cohomology, proving (26). This argument deliberately uses the entire bar complex; boundedness of the objects does not truncate group cohomology. Finite recollement uses only finite limits and colimits. For Δ , (7) supplies the averaging idempotent $|\Delta|^{-1} \sum_{\delta \in \Delta} \delta$; hence invariants are exact and $H^{>0}(\Delta, -) = 0$ on the torsion coefficient categories used here. The same calculation therefore commutes with the prescribed equivariant descent without introducing another infinite group-cohomological tail. $\square$

Proposition C.12 (Actual site descent and selected derived comparison). *The finite-covering rule of Definition C.10 defines the filtered finite-level site. By Theorem C.9, with the source convention of Definition C.6, its comparison with the actual root site is*

$$\text{RepEt}_{\text{fp}}(\tilde{\mathcal{A}}_{M, \ell}^{\text{rel}}) \simeq \mathcal{C}_\infty. \quad (27)$$

After effective descent for the finite action groupoid, the corresponding statement is

$$\text{Sh}_{\text{et}}^{\text{rep}}(\mathcal{A}_{M, \ell}^{\text{rel}}) \simeq \text{Sh}(\mathcal{C}_\infty)^{h\Delta} =: \text{Sh}(\mathcal{C}_\infty^\Delta). \quad (28)$$

On the fixed noetherian split chart, the continuous-action topos identification of Theorem C.20 and Theorem B.5 then gives

$$\mathcal{D}_{c,\text{pf}}^b(\mathcal{X}_{\text{proot}}, R_r) \simeq \text{colim}_n \mathcal{D}_{c,\text{pf}}^b(\tilde{\mathcal{A}}_{M,n}^{\text{rel}}, R_r)^{h\Delta}. \quad (29)$$

These comparisons use the selected action/adic coefficient model of Remark B.16. The derived comparison follows from finite-action descent in that model.

Proof. Identity coverings, base change of a finite covering, and composition of finite coverings can each be represented at one common finite root level; this proves the pretopology assertion. The first site equivalence is Theorem C.9; finite étale descent for the quotient by Δ gives the second. Theorem C.20 identifies the actual topos and structural pullback. Relative derived-action descent then gives the coefficient equivalence; finite semilinear descent is performed in its action-sheaf category. On affine presentations the candidate transition maps are

$$U_{m,q} \longrightarrow U_{n,q}, \quad m \geq n. \quad (30)$$

Affine finite-presentation continuity supplies part of this comparison; see [21, Tags 07SK, 07SL, and 07SN]. It does not say that the non-finitely-presented atlas $U_{\infty,0}$ is a hypercover in the full representable étale site. The fpqc effectivity statement and the required étale cohomological descent statement must therefore remain separate. For the independent adic Iwasawa coefficient model, the inverse comparison and attaching mapping spectra are now proved by Theorem B.13 and relative action descent. The needed topos identification is the separate generating-site argument of Theorem C.20. $\square$

The actual pro- ℓ stack in the preceding display is the fpqc inverse limit of the finite root fibres of order ℓ^n ; its associated inverse system is a pro-DM, hence pro-geometric, object. The actual limit is not asserted to be a finite-type algebraic stack. Proposition C.5 is the bridge between that actual limit and the finite-level presentation; the transition maps are induced by divisibility of the root indices. Before finite descent by Δ , the relative root fibre over the chosen split chart is generally a nilpotent thickening of a classifying stack, not the classifying stack itself. Precisely,

$$(\tilde{\mathcal{A}}_{M,n}^{\text{rel}})_{\text{red}} \simeq B_S \mathcal{K}_n, \quad \text{Sh}_{\text{ét}}^{\text{rep}}(\tilde{\mathcal{A}}_{M,n}^{\text{rel}}) \simeq \text{Sh}_{\text{ét}}^{\text{rep}}(B_S \mathcal{K}_n) \quad (\Delta = 1), \quad (31)$$

with $\mathcal{K}_n$ as in Lemma C.14. The second equivalence is nil-invariance of the representable étale topos. After descent, the same statement is retained as a Δ -equivariant equivalence of topoi. The quotient by Δ is shorthand for the stack obtained by effective descent along the prescribed finite splitting groupoid. In particular it retains the full Δ -equivariant descent datum. The central fibre is canonical relative to the chosen chart, section, and descent groupoid: it uses the actual relative log point and does not require the quotient $P/\mathbf{N}\rho$ to be fs. It is the infinite-root enhancement of the relative central face of the local Artin fan. It should not be confused with the ordinary quotient stack $[\text{Spec}_S \mathcal{O}_S[P]/D_S(P^{\text{gp}})]$. Before descent there is a canonical characteristic map over S

$$\tilde{\mathcal{A}}_M^{\text{rel}} \longrightarrow \varprojlim \sqrt{\mathfrak{s}_{P/S}} \longrightarrow \mathcal{A}_{P/S}, \quad \mathcal{A}_{P/S} := [\text{Spec}_S \mathcal{O}_S[P]/D_S(P^{\text{gp}})],$$

which exhibits the split relative root face over the usual relative local Artin cone. These maps are Δ -equivariant. After descent the corresponding arrows are over $\mathcal{S} = [S/\Delta]$:

$$\mathcal{A}_M^{\text{rel}} \longrightarrow [\varprojlim \sqrt{\mathfrak{s}_{P/S}}/\Delta] \longrightarrow [\mathcal{A}_{P/S}/\Delta].$$

No map from the descended source to the unquotiented targets over S is asserted. The same convention applies to the pro- ℓ versions, with every root index replaced by ℓ^∞ . The absolute expression with a residue field k is used only after taking a geometric fibre in Proposition F.3.

C.4 Root-chart maps and continuous coefficients

Lemma C.13 (Functoriality of relative root faces). *Let $\alpha : (P_{\tau'}, \rho_{\tau'}) \rightarrow (P_{\tau}, \rho_{\tau})$ be one of the base-preserving saturated chart morphisms in Definition A.7, so that $\alpha(\rho_{\tau'}) = \rho_{\tau}$, the induced relative map is saturated, and α respects the log-point augmentations (equivalently $\alpha^{-1}(0) = \{0\}$ on these sharp charts). For every root order a invertible on the base, functoriality of the finite root-stack construction and the trivial-root sections induces the corresponding map of relative ath-root fibres. These maps are compatible with divisibility in a , with the finite descent groupoid Δ , and with composition of chart maps of the supplied log points. Consequently, in a selected codimension-two diagram whose two composite log-point morphisms agree, the relative root composites agree by the canonical 2-isomorphism, before passing to the pro- ℓ limit. Equality of the relative lattice maps alone is not the hypothesis of this assertion. This is functoriality of relative root charts, not an incidence theorem: it supplies no cross-stratum specialization, link or Gysin operation.*

Proof. Put $P' = P_{\tau'}$ and $P = P_{\tau}$. At finite level the algebra map $\mathcal{O}_S[\frac{1}{a}P'] \rightarrow \mathcal{O}_S[\frac{1}{a}P]$ sends $X^{p'/a}$ to $X^{\alpha(p')/a}$. The augmentation identity is exactly what allows this map to pass to the base changes along the zero sections. Thus it induces the map between the presentations (for $Q = P_{\tau'}$ and $Q = P_{\tau}$)

$$\left[(\mathrm{Spec}(S[\frac{1}{a}Q]) \times_{\mathrm{Spec}(S[Q])} \mathrm{Spec} S) / D((\frac{1}{a}Q)^{\mathrm{gp}} / Q^{\mathrm{gp}}) \right]$$

for the two log points, in the direction from the P -root stack to the P' -root stack. The map is equivariant for the induced homomorphism of the displayed diagonalizable groups. Without the augmentation identity, the projection $\mathbf{N}^2 \rightarrow \mathbf{N}$, $(a, b) \mapsto a$, would send a positive monomial to 1 and would not descend through the zero-section relation setting it to 0. This projection is therefore not an allowed log-point map here. Exact preservation of ρ makes the trivial-root sections commute with this map, so the universal property of the 2-fibre gives the relative map. The same presentation shows compatibility with refinement of a and with Δ . For a composite of monoid maps the induced maps of monoid algebras and diagonalizable groups compose strictly; the universal property supplies the canonical associativity 2-cell on the fibres. Augmentation compatibility is preserved by composition. If one also wants to use a composite as a selected chart map for coefficient operations, its other selection conditions must be checked; no closure of finite-Tor coefficient operations under an arbitrary rank change is inferred. Passing to the inverse system proves the assertion. $\square$

Lemma C.14 (Finite root reductions and the relative Kummer group). *For every root exponent $n \geq 0$, the reduction of the split relative geometric fibre is $B_S \mathcal{K}_n$ before Δ -descent, where the finite diagonalizable group scheme is*

$$\mathcal{K}_n := \underline{\mathrm{Hom}}(P^{\mathrm{gp}} / \mathbf{Z}\rho, \mu_{\ell^n}).$$

The primitivity of ρ and the torsion-freeness of $L_{\mathrm{rel}} = P^{\mathrm{gp}} / \mathbf{Z}\rho$ give an exact sequence

$$1 \longrightarrow \mathcal{K}_n \longrightarrow \underline{\mathrm{Hom}}(P^{\mathrm{gp}}, \mu_{\ell^n}) \longrightarrow \mu_{\ell^n} \longrightarrow 1.$$

Consequently

$$\mathcal{K}_{\ell} := \varprojlim_n \mathcal{K}_n \simeq \mathrm{Hom}(L_{\mathrm{rel}}, \mathbf{Z}_{\ell}(1)) \quad \text{as a pro-étale group sheaf over } S.$$

For a strict geometric point $\bar{s} \rightarrow S$ and the chosen split-lattice identification,

$$\mathcal{K}_{\ell, \bar{s}} := \varprojlim_n \mathcal{K}_n(\bar{s}) \simeq M \otimes \mathbf{Z}_{\ell}(1) = \Gamma_M.$$

After descent, the inertia-selected coefficient category carries the same continuous Γ_M -action together with the prescribed Δ -descent groupoid; no assertion is made that the absolute stabilizer of the central fibre is just Γ_M .

Proof. At level ℓ^n , the standard presentation of the root stack of the log point with chart Q has an affine atlas whose ideal generated by the positive fractional monomials is nilpotent. Because Q is sharp, its reduction is the zero section. Hence the reduced root stack is the classifying stack of $D(Q^{\text{gp}}/\ell^n Q^{\text{gp}})$, although the unreduced root stack is not. Taking the 2-fibre over the base-root section restricts characters along $\mathbf{Z}\rho \hookrightarrow P^{\text{gp}}$. The stabilizer of the reduction is therefore $D(P^{\text{gp}}/(\mathbf{Z}\rho + \ell^n P^{\text{gp}}))$, equivalently the displayed $\mathcal{K}_n$. Nil-invariance gives (31). Since ρ is primitive and the quotient is torsion-free, restriction onto the base factor is surjective. Passing to the inverse limit and using $M = \text{Hom}(L_{\text{rel}}, \mathbf{Z})$ gives the final identification. The finite group Δ acts on every term and hence on the resulting equivariant descent groupoid. $\square$

Remark C.15 (Torsion and wild boundaries). The torsion-free hypothesis on L_{rel} is substantive. If the relative quotient has torsion, the finite kernels $\mathcal{K}_n$ acquire an additional finite diagonalizable factor, and the auxiliary site must be enlarged by the corresponding finite tame characters. If $\ell = p$, the groups μ_{ℓ^n} are not étale and the present Kummer-étale argument does not apply. Neither case is included in the selected coefficient theorem.

Definition C.16 (Auxiliary relative Kummer coefficient category). At root exponent n , let $\mathcal{F}_{M,n}^{\text{rel,ket}}$ be the Kummer-étale chart site over S generated by the saturated Kummer-étale extension $P \rightarrow \frac{1}{\ell^n}P$ together with a trivialization of its restriction to $\mathbf{N}\rho$, its strict-étale base changes, and their descent data. This definition uses the logarithmic chart and not the root central fibre. The Čech action groupoid of its standard atlas has deck group scheme $\mathcal{K}_n$. Put

$$\mathcal{D}_{c,\text{ad}}^b(\mathcal{F}_M^{\text{rel,ket}}/\mathcal{S}, \mathbf{Z}_\ell)^{\text{Iw}} := \varprojlim_r \left(\left(\varinjlim_n \mathcal{D}_c^b(\mathcal{F}_{M,n}^{\text{rel,ket}}, \mathbf{Z}/\ell^r) \right)^\Delta \right)_{\text{unif,Iw}}. \quad (32)$$

Here “unif, Iw” selects derived-cartesian coefficient systems with a common finite cohomological interval, a common underlying- R_r Tor interval, and a finite constructible stratification on which the terms at every precision are underlying- R_r -perfect with finite cohomology stalks. These are exactly the adic-system conditions in (25). Reductions and their coherence maps are taken in the ambient unbounded module-sheaf categories; the displayed notation does not assert that reduction preserves every bounded complex before this selection. The superscript Δ means coherent semilinear descent along the given action groupoid, not ordinary invariants.

For clarity the Iwasawa notation in this comparison means

$$\text{Perf}_{\mathbf{Z}_\ell}^{\text{cts}}(\mathcal{S}, \Lambda_{M,\mathcal{S}}^{\text{Iw}}) := (\mathcal{M}_{\text{Iw,ad}}^{\text{sel}}(S))^{h\Delta},$$

with the semilinear action and the module-sheaf category of Definition B.12. In that category the precision- r coefficients are modules over $\Lambda_{M,r}^{\text{Iw}}$ with bounded inertia-ideal torsion cohomology; the Tor selection is over R_r , not over an integral global Iwasawa sheaf. The symbol $\text{Perf}_{\mathbf{Z}_\ell}^{\text{cts}}$ here denotes this selected adic system category. It does not impose global $\mathbf{Z}_\ell$ -perfectness or completed-inertia Tor-amplitude on an independently recovered sheaf. At a geometric point the stronger classical properties follow from Proposition B.14; that result is not used to assert global sheaf recovery. Finite ℓ -torsion cohomology remains allowed. The adjective “auxiliary” is deliberate: this site is constructed without using the root central fibre, but it is generated by the selected Kummer charts, so its identification with the action groupoid is a descent calculation. The actual-root comparison identifies this category with the specified relative pro-root coefficients and their structural functor.

Lemma C.17 (Finite-level root–Kummer action-groupoid comparison). *For every root exponent n , there are equivalences of topoi over S*

$$\mathrm{Sh}_{\acute{e}t}^{\mathrm{rep}}(\tilde{\mathcal{A}}_{M,n}^{\mathrm{rel}}) \simeq \mathrm{Sh}_{\acute{e}t}^{\mathrm{rep}}(B_S \mathcal{K}_n) \simeq \mathrm{Sh}(\mathcal{F}_{M,n}^{\mathrm{rel},\mathrm{ket}}). \quad (33)$$

They are compatible with root refinement, strict base change, the Δ -action, and the augmentation to S .

Proof. The first equivalence is nil-invariance applied to Lemma C.14. For the second, the standard relative Kummer atlas has Čech nerve $\mathcal{K}_n^{\times s^\bullet}$: trivializing the base root removes the quotient character along $\mathbf{Z}\rho$, leaving exactly the kernel $\mathcal{K}_n$. Descent for this finite étale action groupoid identifies sheaves on the Kummer chart site with sheaves on $B_S \mathcal{K}_n$. The monoid maps for $n \leq m$ give the quotient homomorphisms $\mathcal{K}_m \rightarrow \mathcal{K}_n$ on both nerves, proving refinement compatibility. Every construction is functorial in strict base change and Δ -equivariant. $\square$

Lemma C.18 (Constructible étale spaces and finite action objects). *Let S be noetherian. Étale algebraic spaces of finite presentation over S correspond to constructible sheaves of sets on $S_{\acute{e}t}$. For a finite constant group K_n (the fixed split inertia group in the root applications), representable étale finitely presented morphisms to $B_S K_n$ correspond to such sheaves with a K_n -action. Joint surjectivity corresponds to a jointly epimorphic family of sheaves. These correspondences commute with inflation and with base pullback.*

Proof. The étale-space construction represents a sheaf of sets by an étale algebraic space, under the usual universe convention [1]. Here is the needed finiteness check. A constructible sheaf F is noetherian and is generated by finitely many sections on quasi-compact étale schemes. Thus there is an epimorphism $h_U \rightarrow F$ with U a finite coproduct of such schemes. The relation $h_U \times_F h_U$ is represented by the open equality locus in $U \times_S U$: equality of two sections is étale-local, and subterminal sheaves over an étale scheme are its open subsets. This open is quasi-compact since $U \times_S U$ is noetherian. The quotient étale algebraic space is consequently quasi-compact and quasi-separated over S , hence of finite presentation.

Conversely, choose a finite quasi-compact étale scheme atlas U of an étale finitely presented space V/S . Its relation admits a finite quasi-compact étale scheme cover W . Both U and W are étale of finite presentation over S . Their representable sheaves are constructible, and their coequalizer is h_V , which is constructible by closure under finite colimits of sheaves of sets [6, Section 59.73]. This also proves full faithfulness, since the étale-space construction and restriction to the small site are inverse on these objects.

Pullback along the finite étale torsor $S \rightarrow B_S K_n$ identifies a representable étale object with an étale space V/S and a K_n -action. Conversely the quotient $[V/K_n] \rightarrow B_S K_n$ is representable étale and finitely presented. The description respects morphisms and joint surjectivity by base change along that torsor. Both inflation and base pullback are the literal pullbacks of these descent data. $\square$

Proposition C.19 (A generating site for continuous action sheaves). *Let $\mathcal{E} = \mathrm{Sh}(S_{\acute{e}t})$ and let $\mathcal{E}_\Gamma$ be the category of sheaves of sets with a smooth Γ -action, defined by $\mathrm{colim}_n F^{H_n} = F$ as in Definition B.1. This is a Grothendieck topos. Its full subcategory $\mathcal{E}_{\Gamma,c}$ of underlying-constructible objects with finite stalks on a finite stratification, equipped with the finite jointly epimorphic topology, is a generating site, and*

$$\mathrm{colim}_n \mathcal{E}_c^{K_n} \simeq \mathcal{E}_{\Gamma,c}, \quad \mathrm{Sh}(\mathcal{E}_{\Gamma,c}) \simeq \mathcal{E}_\Gamma.$$

The target topos is independently defined by smooth sheaf actions.

Proof. The constructible-subobject and ascending-chain results used in Lemma B.2 also hold for sheaves of sets. Every smooth action sheaf is the union of its H_n -fixed subsheaves. In each such subsheaf, saturate a constructible subobject by its finitely many K_n -translates. Finite unions remain constructible, so these finite-action objects cover every smooth object. If the underlying sheaf is constructible, the fixed-subobject chain stabilizes globally, proving factorization through one K_n . Equivariant morphisms are detected at a common level. This proves the first displayed equivalence without deriving anything.

For an arbitrary abstract Γ -action sheaf A , set $\mathrm{Sm}(A) = \mathrm{colim}_n A^{H_n}$, a union of subsheaves. Normality of H_n makes this a smooth invariant subsheaf. Every equivariant map from a smooth object to A lands locally in these fixed subsheaves and factors uniquely through $\mathrm{Sm}(A)$. Thus Sm is right adjoint to inclusion of smooth actions, and arbitrary limits are

$$\lim_i^{\mathrm{sm}} F_i = \mathrm{Sm}(\lim_i^{\mathrm{abs}} F_i).$$

For $F_i = \Gamma/H_i$, the tuple of identity cosets is excluded when its stabilizer $\bigcap_i H_i$ is not open. Infinite products are therefore not asserted to be the underlying products.

Finite limits and colimits of smooth actions are computed on underlying sheaves; finite limits are smooth because finitely many local stabilizers have open intersection. Arbitrary colimits are likewise smooth, since their sections are locally generated by sections of the inputs. Equivalence relations are effective and colimits are universal, as for underlying sheaves. The finite-action constructible objects just constructed form a small generating family (after fixing a universe). These facts give the topos assertion by the usual Grothendieck topos criterion.

The full generating subcategory is closed under finite limits. A jointly epimorphic family onto a constructible object has a finite subfamily: finite unions of its images form an ascending directed family of subobjects of a noetherian underlying sheaf. The topology induced by epimorphic families is therefore the stated finite topology. The generating-site comparison lemma now identifies its sheaves with $\mathcal{E}_\Gamma$. Equivalently, an object is recovered from all maps from these generators and their jointly epimorphic relations; this is the canonical presentation of an object of a topos by a full generating site. It uses objects and their relations on the entire base, using its generating site. $\square$

Theorem C.20 (The relative root topos). *On the fixed noetherian split tame chart, with the fpqc-stack source convention of Definition C.6, there is an equivalence of topoi over S*

$$\mathrm{Sh}(\mathrm{RepEt}_{\mathrm{fp}}(X_\infty)) \simeq \mathcal{E}_\Gamma, \quad X_\infty = \tilde{\mathcal{A}}_{M,\ell}^{\mathrm{rel}}.$$

The structural inverse image from $S_{\mathrm{ét}}$ is identified with trivial relative action. The equivalence is compatible with root refinement, the specified strict base changes, and finite semilinear Δ -descent.

Proof. At each finite level, étale finite-presentation objects lift uniquely across the nilpotent monomial reduction. This can be checked on the finite affine root atlas: nil-invariance of étale spaces lifts the object and every descent isomorphism uniquely, so also lifts its unit and cocycle equations. Consequently the finite site is the site of representable étale fp objects over $B_S K_n$. By Lemma C.18, it is the site $\mathcal{E}_c^{K_n}$, and the identifications commute with root pullback and covering families.

Theorem C.9 identifies the actual site with the filtered colimit of these finite sites. Proposition C.19 identifies that colimit with $\mathcal{E}_{\Gamma,c}$ and then its sheaves with $\mathcal{E}_\Gamma$. This constructs the topos equivalence, including its topology. The infinite fpqc atlas itself has not been declared an étale hypercover.

For an étale finitely presented U/S , its structural pullback to a finite root stack corresponds under nil reduction and the torsor atlas to h_U with trivial K_n -action. The same holds in the filtered site and hence in the actual topos. Such h_U generate the base topos; both inverse-image functors preserve colimits and finite limits. Thus their canonical agreement on these generators extends to the structural inverse image on all sheaves, proving the statement over S .

Every step is induced by pullback or by its canonical descent comparison. It therefore commutes with base change and with maps of the given root groupoids. Finite Δ -descent is applied to this equivalence with its supplied finite descent diagram, giving the asserted comparison over $\mathcal{S} = [S/\Delta]$. $\square$

Corollary C.21 (actual derived root star and adic coefficients). *On the chart of Theorem C.20, let $g : X_\infty \rightarrow \mathcal{S}$ be the structural morphism and let E_r be a selected bounded coefficient. Then the actual right direct image is naturally*

$$Rg_*^{\text{act}} E_r \simeq R\text{Hom}_{\Lambda_r}(R_r, E_r) \simeq C_{\text{cts}}^*(\Gamma, E_r),$$

with finite semilinear descent understood. The actual root coefficient category is equivalent to the adic Iwasawa module-sheaf category of Definition B.12, with the same selected uniform restrictions. The star outputs are uniformly bounded and derived cartesian when the input system is, and their adic value is formed by taking the coefficient inverse limit after this recovery. The assertion concerns this actual site and coefficient model; it does not construct an exceptional direct image on X_∞ .

Proof. The preceding theorem gives an exact equivalence of coefficient abelian categories, identifying structural pullback with trivial action. Its right adjoint is the invariant-sheaf functor, since a map from a trivial-action object has image in the invariant subsheaf. The exact equivalence identifies injective resolutions, so the actual derived right adjoint is derived continuous invariants. Theorem B.13 identifies this with the displayed internal Hom and its finite Koszul calculation. The comparison morphism has thus been constructed before it is checked by that calculation.

Constructibility is respected by Lemma C.22. More explicitly, subterminal objects in the smooth-action topos are exactly invariant open subobjects of the terminal sheaf, hence the corresponding base opens. On a stratum, a locally constant finite object pulls back to a locally constant finite base sheaf. Conversely a finite locally constant base sheaf with finite K_n -action becomes constant after a finite étale trivialization on the base and the regular K_n -cover in the action topos. This identifies the constructible strata and the selected coefficient subcategories; finite Tor-amplitude is tested on their local complexes. Relative derived-action descent also descends their attaching maps. Applying the exact equivalence to complexes and then to cartesian coefficient systems yields the adic coefficient comparison.

The augmentation Koszul complex has length d and finite free terms. On the selected coefficients it commutes with derived reduction and increases amplitude by at most d , uniformly in r . This proves recovery for the actual star output already identified above. The statement about its adic limit follows in the category of coefficient systems. On these bounded-cohomology coefficients, derived sheaf realization is compatible with hyperdescent through the equivalence of topoi; no unbounded hypercompletion theorem for arbitrary root coefficients is asserted. $\square$

Lemma C.22 (Constructible strata and their finite root presentations). *In the actual-root topos just identified, intrinsic finite coherent stratifications and finite locally constant coefficients agree with the underlying-constructible continuous-action model. Selected bounded derived coefficients and their attaching maps descend to finite root levels at each coefficient precision.*

Proof. A subterminal object in $\mathcal{E}_\Gamma$ is a subobject of the terminal base sheaf, with its unique action. Hence opens, and then locally closed subtopoi obtained by complements and restrictions, are exactly the base ones. On the noetherian base these opens are quasi-compact. A finite coherent stratification thus occurs already at root level zero, before any coefficient is descended. On each stratum, forgetting the action carries a finite locally constant object to a finite locally constant base sheaf. Conversely, a finite locally constant base sheaf with smooth action factors through a finite quotient by the noetherian fixed-subobject argument. An étale trivialization of that base sheaf, together with the regular finite-group torsor, trivializes it in the action topos. Theorem B.5, applied to the whole constructible sheaf category, then descends the bounded objects and attaching morphisms. Finite semilinear descent is handled by Lemma B.6. Underlying perfection is unchanged by inflation. This proves the selected constructible statement; it makes no compactness claim for infinite stalks or infinitely many strata. $\square$

Proposition C.23 (The relative log-fibre coefficient interface). *Let Y be a noetherian split log stratum with characteristic chart $\mathbf{N}\rho \rightarrow P$, with ρ primitive and $L = P^{\text{gp}}/\mathbf{Z}\rho$ free. Fix a geometric logarithmic point over the base special point, including compatible prime-to- p roots of the base parameter. Write $\varepsilon_{Y,\text{rel}} : \tilde{Y}_{\text{ket}} \rightarrow Y_{\text{ét}}$ for the projection from the resulting relative logarithmic fibre. On coefficients trivial on the non- ℓ relative tame factor, its selected bounded/adic coefficient category agrees with that of the actual pro- ℓ root fibre, and*

$$R\varepsilon_{Y,\text{rel},*}E \simeq C_{\text{cts}}^*(\Gamma, E), \quad \Gamma = \text{Hom}(L, \mathbf{Z}_\ell(1)).$$

Here the adic assertion uses the uniform restrictions of Theorem B.13. This is not an equivalence between the full prime-to- p logarithmic topos and the full pro- ℓ root topos.

Proof. On an étale split test object of Y , Kummer covers are generated by adjoining prime-to- p roots of the chart and then taking strict étale covers. The finite chart construction and its character pairing give absolute inertia $\text{Hom}(P^{\text{gp}}, \widehat{\mathbf{Z}}^{(p')}(1))$. Fixing the roots of the base parameter takes the kernel of its map to $\text{Hom}(\mathbf{Z}\rho, \widehat{\mathbf{Z}}^{(p')}(1))$. Primitivity makes this map surjective and the kernel is

$$G_{\text{rel}} = \text{Hom}(L, \widehat{\mathbf{Z}}^{(p')}(1)) = \Gamma \times H,$$

where all finite quotients of H have order prime to ℓ . There is consequently no extra finite component factor in this split relative fibre. A different chosen logarithmic lift is compared by its torsor of lifts, not by declaring a canonical splitting of absolute inertia.

The Kummer chart criterion says that every finitely presented Kummer étale object is étale locally obtained from such a finite chart cover; its strict part is ordinary étale descent. Hence the same generating-site argument used in Theorem C.20 identifies this relative log topos over Y with smooth G_{rel} -action sheaves. On the pro- ℓ selected subcategory, inflation identifies the coefficient abelian category with smooth Γ -actions. Averaging over finite quotients proves that smooth H -invariants are exact, as in Lemma E.15. Thus deriving invariants gives $R\Gamma(G_{\text{rel}}, E) = R\Gamma(\Gamma, E)$ on this subcategory. The exact equivalence, relative derived-action descent, and Theorem B.13 identify the selected derived coefficients and their adic systems with those of the actual root fibre. The structural inverse image on both sides is trivial relative action, so this comparison identifies the stated structural right adjoints as well as their sources.

The chart construction commutes with strict base pullback and with its finite Kummer refinements. Its character pairing is equivariant for the specified finite semilinear descent data. This supplies these compatibilities on a split stratum; it asserts no comparison for incidence maps between different characteristic strata. $\square$

Corollary C.24 (Intrinsic adic coefficients). *On the fixed noetherian split chart, with representable étale fpqc-stack sources as in Definition C.6, the selected adic coefficient category on the actual infinite-root central fibre is equivalent to the independently defined category $\mathcal{M}_{\text{Iw},\text{ad}}^{\text{sel}}$ of Definition B.12, with finite semilinear descent understood on both sides. This equivalence identifies mapping spectra, attaching maps and coefficient reduction. Structural star is continuous invariants, with derived adic recovery on the selected uniform class. This target does not require a global pseudocompact interpretation.*

Proof. Apply Theorem C.20 to identify the actual coefficient topos, then Theorem B.13 to identify its independent adic module-sheaf target. The structural and recovery assertions are Corollary C.21. This identifies the explicitly defined adic module-sheaf target. No geometric operation transport is inferred from this coefficient equivalence. $\square$

Proposition C.25 (Actual root star image and its coefficient scope). *Work on the fixed noetherian split chart with the selected adic module-sheaf coefficient category of Definition B.12. Then the actual structural direct image is computed by*

$$Rg_{M,e,\ell,*}^{\text{act}} E_r \simeq C_{\text{cts}}^*(\mathcal{K}_\ell, E_r). \quad (34)$$

The right side is an internal sheaf-valued continuous cochain object. For an admissible adic coefficient $E_\bullet$, write

$$\mathcal{G}_{*,\text{ad}}^{\text{cts}}(E_\bullet) := R\varprojlim_r C_{\text{cts}}^*(\mathcal{K}_\ell, E_r).$$

Its derived-cartesian recovery identifies this object with $Rg_{,\text{ad}}^{\text{act}} E_\bullet$.*

Proof. Theorem C.20 identifies structural pullback with the functor assigning trivial relative inertia action. Its right adjoint is continuous invariants; deriving that adjunction in the specified coefficient realization gives the displayed formula. The finite-projective Iwasawa calculation then proves boundedness and adic recovery for the selected coefficients. Homotopy invariance removes the affine carrier for the star image. This argument requires a comparison over the base and its derived coefficient realization. An abstract equivalence of selected categories, or an effective fpqc atlas by itself, does not identify this right adjoint. $\square$

Remark C.26 (Full site, selected coefficients). The coefficient comparison uses the full representable étale site of every finite central fibre. What is selected on that intrinsic site is the coefficient subcategory with continuous relative pro- ℓ inertia, not a chartwise-defined target site. The ordinary lisse-étale cohomology of the unrigidified Artin fan still contains the residual classifying-stack direction exposed by Proposition F.3.

D The continuous dualizing module

This appendix describes the continuous algebra and dualizing bimodule used in Section 6. The finite Iwasawa resolution computes this continuous object; the full finite-root bars and their geometric trace limit are treated in that section.

D.1 Continuous bar-Iwasawa models and the dualizing module

Lemma D.1 (Equivariant bar-Iwasawa algebra and the top orientation). *Put*

$$W_M := \text{Hom}_{\text{cts}}(\Gamma_M, \mathbf{Z}_\ell) \simeq M^\vee \otimes \mathbf{Z}_\ell(-1).$$

The basis-free object

$$C_{\text{cts}}^*(\Gamma_M, \mathbf{Z}_\ell) := R\text{Hom}_{\Lambda_M^{\text{Iw}}}^{\text{cts}}(\mathbf{Z}_\ell, \mathbf{Z}_\ell)$$

carries its canonical homotopy-coherent cup product and, for every finite group Δ acting on M , its canonical Δ -equivariant structure. The completed bar model, with its shuffle product, and every basis Koszul model represent this same multiplicative derived object. In particular,

$$H^q C_{\text{cts}}^*(\Gamma_M, \mathbf{Z}_\ell) \simeq \bigwedge^q W_M, \quad \det W_M \simeq \mathfrak{o}_M^\vee(-d), \quad \mathfrak{o}_M := \det(M \otimes \mathbf{Z}_\ell), \quad (35)$$

and all three identifications are Δ -equivariant. Thus the determinant and Tate factors used below descend without choosing a basis. What is canonical is the derived cochain algebra; no strictly functorial Koszul basis is asserted.

Proof. The augmentation ideal $I \subset \Lambda_M^{\text{Iw}}$ satisfies $I/I^2 \simeq \Gamma_M \otimes_{\mathbf{Z}_\ell} \mathbf{Z}_\ell$, functorially in M , and the associated graded algebra is the symmetric algebra on I/I^2 . The regular-sequence calculation therefore identifies the Yoneda algebra of the augmentation module with $\bigwedge^\bullet W_M$. Continuous group cochains give the intrinsic multiplicative model. The shuffle map on the completed bar resolution represents its cup product, while comparison of projective resolutions transports that product to any chosen Koszul model; the result is canonical in the chosen enhancement through comparison with the intrinsic continuous cochain algebra. No contractibility claim for an unspecified space of model choices is needed. Every construction commutes with lattice automorphisms, hence with Δ . Taking the top exterior power and using $\det(M^\vee) = \det(M)^\vee$ gives the last two identities in (35). $\square$

For a point coefficient define the following two complexes. Over a split base, the star expression uses the full selected adic source; the coefficient-dual partner and its adic properties below use the lisse-perfect subspace of Lemma A.12(ii). All general-base expressions are interpreted levelwise in that model:

$$\mathbf{K}_{M,*}^{\text{ad}}(V_{\text{cont}}) := R\text{Hom}_{\Lambda_M^{\text{Iw}}}^{\text{cts}}(\mathbf{K}_M^{\text{Iw}}, V_{\text{cont}}), \quad (36)$$

$$\mathbf{K}_{M,!}^{\text{ad}}(V_{\text{cont}}) := (\mathbf{K}_{M,*}^{\text{ad}}(V_{\text{cont}}^\vee))^{\vee_0} \otimes_{\mathbf{Z}_\ell} \mathbf{Z}_\ell(-d-e)[-2(d+e)]. \quad (37)$$

Their rationalizations are denoted by $\mathbf{K}_{M,\bullet}(V)$. Over a base the displayed !-expression is a coefficient-linear partner, with dualizing object the constant coefficient unit. The Verdier partner in Lemma A.12(iii) is instead this expression tensored with $\omega_{\mathcal{S},\bullet}$. At a geometric point these coincide. Neither is by itself a construction of a general-base geometric supported functor. The factor $(-d-e)[-2(d+e)]$ is the inverse of the relative dualizing factor $(d+e)[2(d+e)]$ inside the product-chart identity

$$\mathbb{D}_X(f^* \mathcal{L} \otimes \tilde{V}) \simeq f^* \mathbb{D}_{\mathcal{S}}(\mathcal{L}) \otimes \widetilde{V}^\vee \otimes \omega_{X/\mathcal{S}}.$$

In this product identity $\tilde{V}$ is tensor-dualizable, $\mathcal{L}$ lies in the stated Verdier domain, and the smooth product projection supplies the relative purity factor. It is not an additional orientation correction. On the third realization it is the compact-support class of the actual rank- $(d+e)$ affine carrier (21); it is no longer attached to the bare central fibre by normalization. If V_{cont} is trivial in the characteristic-torus direction, Iwasawa–Koszul self-duality gives

$$\mathbf{K}_{M,!}^{\text{ad}}(V_{\text{cont}}) \simeq \mathbf{K}_{M,*}^{\text{ad}}(V_{\text{cont}}) \otimes \det(M \otimes \mathbf{Z}_\ell)[-d] \otimes \mathbf{Z}_\ell(-e)[-2e], \quad (38)$$

and hence, after tensoring with $\mathbf{Q}_\ell$,

$$\mathbf{K}_{M,!}(V) \simeq \mathbf{K}_{M,*}(V) \otimes \det(M \otimes \mathbf{Q}_\ell)[-d] \otimes \mathbf{Q}_\ell(-e)[-2e]. \quad (39)$$

Thus the determinant line gives the stated factor in the adic category.

Proposition D.2 (The base-linear continuous dualizing bimodule). *Put*

$$A_M^{*,\text{ad}} := C_{\text{cts}}^*(\Gamma_M, \mathbf{Z}_\ell), \quad A_{M,e}^{\text{l,ad}} := (A_M^{*,\text{ad}})^{\vee_0} \otimes \mathbf{Z}_\ell(-d-e)[-2(d+e)]. \quad (40)$$

Their rationalizations are denoted $A_M^* := A_M^{*,\text{ad}}[1/\ell]$ and $A_{M,e}^{\text{l}} := A_{M,e}^{\text{l,ad}}[1/\ell]$; over $\mathcal{S}$ these are understood with their specified lattice and finite descent data.

The cup product makes $A_M^{*,\text{ad}}$ an E_∞ -algebra. Dualizing its left and right regular actions makes $A_{M,e}^{\text{l,ad}}$ an $(A_M^{*,\text{ad}})^e$ -module, where

$$(A_M^{*,\text{ad}})^e := A_M^{*,\text{ad}} \widehat{\otimes}_{\mathbf{Z}_\ell}^{\mathbf{L}} (A_M^{*,\text{ad}})^{\text{op}}.$$

The evaluation map

$$A_{M,e}^{\text{l,ad}} \widehat{\otimes}_{\mathbf{Z}_\ell} A_M^{*,\text{ad}} \longrightarrow \mathbf{Z}_\ell(-d-e)[-2(d+e)] \quad (41)$$

is balanced for the two cup actions. Because the Iwasawa–Koszul model is a bounded finite free $\mathbf{Z}_\ell$ -complex, it is dualizable over the coefficient base. Put $L = \mathbf{Z}_\ell(-d-e)[-2(d+e)]$. The displayed evaluation and the coevaluation $L \rightarrow A_M^{*,\text{ad}} \widehat{\otimes}_{\mathbf{Z}_\ell}^{\text{l,ad}} A_{M,e}^{\text{l,ad}}$ are the ordinary base-duality maps tensored with L . After removing this invertible twist, the pair $(A_M^{*,\text{ad}}, (A_M^{*,\text{ad}})^{\vee_0})$ obeys the usual triangle identities. Balancing and dinaturality with respect to regular multiplication are asserted; no A^e -linear coevaluation from the base unit with an augmentation action is asserted. No assertion that $A_M^{*,\text{ad}}$ is perfect as a module over its enveloping algebra is used. Thus $A_{M,e}^{\text{l,ad}}$ is the local candidate compact-support dualizing bimodule, identified geometrically in Corollary 5.7 on its geometric domain. For every coefficient in the full selected source, $\mathbf{K}_{M,*}^{\text{ad}}(V_{\text{cont}})$ is naturally an $A_M^{*,\text{ad}}$ -module. On the coefficient-duality domain specified above, $\mathbf{K}_{M,!}^{\text{ad}}(V_{\text{cont}})$ has the dual action; all geometric comparison maps preserve these actions under the multiplicative star and supported comparisons below. Formula (38) is the constant-character calculation of this bimodule. In particular, this proposition does not assert an A^e -linear equivalence $A_M^{*,\text{ad}} \simeq A_{M,e}^{\text{l,ad}}$ and does not equip $A_M^{*,\text{ad}}$ with a Frobenius or Calabi–Yau structure. The proven object is the base-linear dualizing A^e -module together with its balanced evaluation and base-linear coevaluation.

Proof. Continuous invariants are the derived right adjoint to the strong symmetric-monoidal trivial-action functor. Their canonical lax symmetric-monoidal structure sends the unit to the stated E_∞ -algebra and defines its regular actions, before a lattice basis is chosen. The completed normalized bar with its Alexander–Whitney diagonal models the binary cup product; that diagonal alone is not being used to supply all E_∞ cells. The finite free Iwasawa–Koszul replacement proves that the resulting augmented cochain object is perfect over $\mathbf{Z}_\ell$. It is therefore dualizable in the enhanced coefficient category. Dualizing the two regular actions gives the displayed bimodule; evaluation is balanced by associativity of cup product, and the ordinary dualizability coevaluation has source $\mathbf{Z}_\ell(-d-e)[-2(d+e)]$ and target $A_M^{*,\text{ad}} \widehat{\otimes}_{\mathbf{Z}_\ell}^{\text{l,ad}} A_{M,e}^{\text{l,ad}}$. After removing the factor L , these are the ordinary triangle identities for A and $A^{\vee_0}$. Under $A \otimes A^{\vee_0} \simeq R\text{Hom}_{\mathbf{Z}_\ell}(A, A)$, coevaluation corresponds to the identity. Its dinaturality for left and right multiplication, and evaluation’s balancing $\text{ev}(\phi a, x) = \text{ev}(\phi, ax)$ (with the graded conventions), are the required action identities. Every construction commutes with strict-étale descent and lattice automorphisms; hence the bimodule and its trace descend over $\mathcal{S}$ with the Δ -action. Applying the same bar action to a coefficient module proves the final assertion. $\square$

For the dual Kummer local system $V_M^{\text{Kum}} := M^\vee \otimes \mathbf{Q}_\ell(-1)$ and $\mathfrak{o}_M := \det(M \otimes \mathbf{Q}_\ell)$, the determinant bookkeeping is

$$\det V_M^{\text{Kum}} = \mathfrak{o}_M^\vee(-d), \quad (\det V_M^{\text{Kum}})^\vee = \mathfrak{o}_M(d).$$

This is the convention used when the orientation factor is rewritten in terms of the torus orientation line.

Remark D.3 (Rank-one constant-coefficient check). For $d = 1$, $e = 0$, and the constant coefficient $V = \mathbf{Q}_\ell$, the two-term model gives

$$\mathbf{K}_{M,*}(\mathbf{Q}_\ell) \simeq \mathbf{Q}_\ell \oplus \mathbf{Q}_\ell(-1)[-1], \quad \mathbf{K}_{M,!}(\mathbf{Q}_\ell) \simeq \mathbf{Q}_\ell[-1] \oplus \mathbf{Q}_\ell(-1)[-2].$$

The second identity is the normalized continuous Verdier/Borel–Moore mate, including the rank-one orientation shift. By Theorem 4.2 and Corollary 5.7, the same two identities hold for the rational geometric complexes $(Rg_{*,\text{ad}}^{\text{act}} \mathbf{Z}_\ell)[1/\ell]$ and $\mathbf{G}_!^{\text{tr}}(\mathbf{Z}_\ell)$ on a geometric-point chart satisfying the cited hypotheses, with its specified rank-one carrier. The latter is the finite-carrier trace realization.

E Geometric comparison tools

This appendix supplies the geometric and site inputs for the realization-induced mapping comparison in Section 4. We specify specialization sites, construct the log-lisse Kummer source, and compare logarithmic, ordinary and tame nearby cohomology. The completion and analytic-site comparisons preserve the products used in that mapping comparison.

Definition E.1 (The actual analytic specialization sites). For the geometric tube T of a special formal scheme with reduction Y , Berkovich’s construction uses morphisms of topoi

$$T_{\acute{e}t} \xleftarrow{\mu} T_{q\acute{e}t} \xrightarrow{\nu} Y_{\acute{e}t}, \quad R\Theta(F) = R\nu_*\mu^*F.$$

Here $q\acute{e}t$ denotes the quasi-etale site. Formal etale lifting followed by generic fibre is quasi-etale, and does not in general define a morphism from the ordinary analytic etale topos to $Y_{\acute{e}t}$. This is the construction of [8, §2, Proposition 2.1 and the definition of Θ_K following it]: formal etale lifting defines ν , and the inclusion of ordinary etale objects in the quasi-etale site defines μ .

In all analytic star formulas in this paper, $R\text{sp}_*F$ is shorthand for $R\Theta(F)$ with this factorization. The analytic coefficient F itself is on $T_{\acute{e}t}$. Pullback of a general sheaf L on Y gives ν^*L on $T_{q\acute{e}t}$; it is not thereby an algebraically constructible generic coefficient, nor has a lift to $T_{\acute{e}t}$ been specified. For finite-torsor descent of the formal star functor one works on $T_{q\acute{e}t}$ with the actual coefficient μ^*F and applies $R\nu_*$. This requires no interchange of μ^* with an arbitrary infinite totalization.

E.1 Logarithmic specialization and tube cohomology

The following results apply to vertical log-smooth charts. Their specialization to the node is used in Section 4.

Definition E.2 (The vertical logarithmic coefficient domain). Let X/R be a finite-type fs saturated log-smooth scheme, vertical near a noetherian locally closed stratum Y : the generic log structure is trivial on that neighbourhood. On Y fix a split characteristic chart $\mathbf{N}\rho \rightarrow P$ with ρ primitive and $L = P^{\text{gp}}/\mathbf{Z}\rho$ free of rank d . Use the actual special formal completion $\widehat{X}_Y$ and its geometric tube, without assuming an entire-torus product. For a supplied generic open $j : U \hookrightarrow X_\eta$, require $Y \cap \overline{X_\eta} \setminus \overline{U} = \emptyset$.

A coefficient for the theorem below is a bounded complex $\mathcal{L}_r$ on X_{ket} , in a neighbourhood of Y , with finite locally constant cohomology and the selected underlying R_r -perfectness. Its generic restriction, on the ordinary etale site by verticality, is $\mathcal{E}_r$. Its restriction E_r to the geometric relative log fibre is required to be trivial on the non- ℓ relative tame factor. For adic

coefficients require a derived-cartesian system with the uniform amplitude conditions already specified in Theorem B.13. If arithmetic descent is retained, it is the descent supplied by this actual prime-to- p log-Kummer source and the chosen geometric base-root lift. We allow no additional wild base-inertia representation on the coefficient. This restriction is imposed on the logarithmic neighbourhood of Y . The non- ℓ relative tame factor is discarded by exact invariants; external tame base inertia, including its prime-to- ℓ part, remains separate action data and is not absorbed into Γ . The same convention applies to dual coefficients and to log-lisse sources used below. The extension $\mathcal{L}_r$ is part of the source data. The following statements take values on $Y_{\text{ét}}$ itself; application of a further $Rf_{Y,*}$ is a separate operation on both sides.

Lemma E.3 (Slices of the geometric logarithmic limit sites). *Let X be an fs log scheme over the logarithmic trait. Use the finite logarithmic base extensions and the limiting Kummer étale sites defining the geometric total, special and generic topoi in [13, §8.2, (8.2.1)]. If $q : X' \rightarrow X$ is Kummer étale, its base changes determine objects Q_α of these three topoi $\mathcal{X}_\alpha$, and canonically*

$$\mathcal{X}'_\alpha \simeq \mathcal{X}_\alpha / Q_\alpha, \quad \alpha \in \{\text{tot}, \bar{s}, \bar{\eta}\}.$$

Under these equivalences the generic and special structural maps for X' are the localizations of those for X . The identifications commute with further base extension, composition of Kummer étale maps, and transport by geometric base automorphisms.

Proof. Work on quasi-compact chart opens and use the finite-presentation basis of the Kummer étale sites. A general Kummer étale object is covered by these opens, so the resulting statement glues. Let λ index the finite logarithmic base extensions. At each level use the fs fibre products and denote the corresponding site by $\mathcal{C}_{\lambda,\alpha}$ and the base change of q by $Q_{\lambda,\alpha}$. Subcanonicity and stability of Kummer étale maps under composition and fs base change identify the site over X'_λ with $\mathcal{C}_{\lambda,\alpha}/Q_{\lambda,\alpha}$. Its coverings are precisely the induced Kummer étale coverings.

Here is the passage to the limit explicitly. The limiting site is presented by the filtered category $\mathcal{C}_{\infty,\alpha}$ of finite-level objects, with morphisms taken after further base extension and the topology generated by finite-level coverings. This is the limiting-site convention in the cited definition, not the small site of an unqualified underlying scheme limit. There is a functor

$$\text{colim}_\lambda (\mathcal{C}_{\lambda,\alpha} / Q_{\lambda,\alpha}) \longrightarrow \mathcal{C}_{\infty,\alpha} / Q_\alpha.$$

An object on the right is a finite-level object U_λ together with an arrow to Q_α . That arrow is represented at a later level, so it gives an object on the left. A morphism over Q_α is a finite-level morphism and one triangle equality; both are represented at a common later level. The same observation detects equality of two morphisms. This proves equivalence of the categories. For a finite covering family all arrows and their triangles can be represented at one level. The induced covering topology is the same on both sides, since both are generated by pullbacks of the finite-level coverings. On the quasi-compact basis finite coverings suffice; arbitrary opens and coverings are obtained by gluing. Sheafification identifies the topos of this slice site with the slice by its representable sheaf, proving the assertion.

The generic and special maps are induced at every level by the same fs pullbacks. Pulling back an arrow $U \rightarrow Q$ gives the arrow between the corresponding pullbacks, with its triangle unchanged. Thus the three site equivalences intertwine those structural maps, together with their site functors. Successive pullbacks supply the composition identifications; their associativity follows from the universal property of fibre product. A base automorphism transports these same diagrams (and reindexes the finite extensions), which gives the asserted geometric-action compatibility. No formula interchanging a derived direct image and a filtered colimit is used. $\square$

Proposition E.4 (Kummer localization of logarithmic specialization). *Let X be an fs log scheme over the logarithmic trait and $q : X' \rightarrow X$ a Kummer étale morphism. Use the geometric logarithmic base change defining nearby cycles, with the integral saturated fibre products and limiting Kummer étale topoi of [13, §8.2, (8.2.1)]. For $\mathcal{L} \in D^+(X_{\text{ket}}, R_r)$ there is a canonical exchange*

$$q_s^* R\Psi_X^{\log} \mathcal{L} \xrightarrow{\sim} R\Psi_{X'}^{\log}(q^* \mathcal{L}).$$

It makes the specialization-unit square commute:

$$\begin{array}{ccc} q_s^*(\mathcal{L}|_{X_s^{\log}}) & \xrightarrow{q_s^* \text{sp}_{\mathcal{L}}} & q_s^* R\Psi_X^{\log} \mathcal{L} \\ \parallel & & \downarrow \sim \\ (q^* \mathcal{L})|_{(X')_s^{\log}} & \xrightarrow{\text{sp}_{q^* \mathcal{L}}} & R\Psi_{X'}^{\log}(q^* \mathcal{L}). \end{array}$$

For a covering family of such q , the pullbacks on the geometric logarithmic special fibre are jointly conservative. The exchanges are compatible with morphisms of coefficients, composition of covers, and the retained inertia action.

Proof. The Kummer étale topology is subcanonical; composition and fs base change preserve Kummer étale maps and their coverings [13, §2.1–2.3 and Theorem 2.6]. Consequently $(X')_{\text{ket}}$ is the slice of X_{ket} by the representable object X' : its objects are Kummer étale objects over X with a map to X' , and its coverings are the induced coverings. This is an equality of the localized Kummer étale sites.

Let $\mathcal{X}$, $\mathcal{X}_s$ and $\mathcal{X}_{\bar{\eta}}$ be the logarithmic total, special and generic topoi in the geometric base change, and write i, j for their structural morphisms. The defining expression is $R\Psi_X^{\log} \mathcal{L} = i^* Rj_*(\mathcal{L}|_{\mathcal{X}_{\bar{\eta}}})$. Lemma E.3 identifies the three topoi for X' with the slices by the compatible base changes of q , including their generic and special structural maps.

Apply Lemma E.16 to j and this object of $\mathcal{X}$. Commutation of inverse images in the special-fibre square then gives

$$q_s^* i^* Rj_* \simeq i'^* q_{\mathcal{X}}^* Rj_* \simeq i'^* Rj'_* q_{\bar{\eta}}^*.$$

The specialization map is i^* applied to the unit $\mathcal{L}|_{\mathcal{X}} \rightarrow Rj_* j^*(\mathcal{L}|_{\mathcal{X}})$. The unit clause of the lemma therefore gives the displayed square with this very map. Coverings remain coverings under the geometric log base change, so exact inverse images along them are jointly conservative on cohomology sheaves. This proves the descent assertion. Composition and inertia compatibility follow from the same site pullbacks and adjunction units before forgetting the action. $\square$

Lemma E.5 (Logarithmic specialization for the actual lisse source). *For an fs log-smooth scheme X over the logarithmic trait and a bounded R_r -complex $\mathcal{L}$ on X_{ket} with finite locally constant cohomology, the log-specialization unit*

$$\mathcal{L}|_{X_s^{\log}} \xrightarrow{\sim} R\Psi_X^{\log} \mathcal{L}$$

is an equivalence. In addition there is a natural exchange

$$R\Psi_X(R\varepsilon_{\eta,*} \mathcal{L}_{\eta}) \simeq R\varepsilon_{s,\text{rel},*} R\Psi_X^{\log} \mathcal{L}.$$

These use the geometric logarithmic base change, not the special fibre with its absolute log structure. In the vertical case the generic logarithmic forgetful map ε_{η} is the identity. The maps are the specialization unit and the forgetful exchange on the indicated actual topoi.

Proof. The external local-acyclicity input is the constant-coefficient statement of [12, Theorem 3.2], also stated in [13, Theorem 8.3]. It applies to $\mathbf{Z}/\ell^s$ for every s . A finite constant R_r -module is a finite direct sum of such cyclic modules, so its unit is an equivalence as well. This is the only coefficient case taken from that theorem.

There are three separate steps for the lisse extension of this input. First choose a Kummer-etale covering family trivializing a finite locally constant coefficient. Finitely many such coefficients or specified maps can be handled on the common refinement obtained by finite fibre products of covering families; no single cover for an infinite coefficient system is used. Second, each covering fs log scheme remains log smooth over the trait, since Kummer etale is log etale and log smoothness is stable under composition. Saturation over the trait is not an extra hypothesis of the cited theorem. Third, Proposition E.4 identifies the pullback of the actual specialization unit with the unit on that covering scheme. The base-changed Kummer-etale covering of the geometric logarithmic special fibre is jointly conservative, so the unit on the original coefficient is invertible. Together with source descent, this step extends the constant-coefficient comparison to the selected source.

Finite truncation triangles apply this natural unit to a bounded complex with locally constant cohomology. No splitting of its Postnikov extensions or simultaneous trivialization of the whole complex is required.

The forgetful exchange is the derived natural transformation of [13, §8.2, (8.2.3)], which is stated there for D^+ coefficients. Its construction uses strict closed base change for the forgetful map and the geometric logarithmic base limit. Apply that exchange to the present source. Its unit and pullback naturality are retained under the localizations above. This establishes lisse specialization with its natural comparison maps. $\square$

Theorem E.6 (Tube cohomology of vertical log-lisse Kummer coefficients). *For the actual prime-to- p log-Kummer source of Definition E.2, with no additional wild base-inertia action and with the full retained inertia action, write $s :]Y[_{\bar{\eta}} \rightarrow Y$ for geometric specialization and $\Gamma = \text{Hom}(L, \mathbf{Z}_\ell(1))$. There is a natural zigzag of equivalences*

$$C_{\text{cts}}^*(\Gamma, E_r) \xrightarrow{\gamma} i_Y^* R\Psi_X(Rj_* \mathcal{E}_r) \xrightarrow{\beta} Rs_* \mathcal{E}_r^{\text{an}}, \quad i_Y^* R\Psi_X^t(Rj_* \mathcal{E}_r) \xrightarrow{\sim} i_Y^* R\Psi_X(Rj_* \mathcal{E}_r).$$

Here E_r is interpreted on the actual root fibre by Proposition C.23. The first zigzag respects the commutative algebra of the unit and its coefficient modules, and the tame comparison preserves these structures. The maps are natural for algebraized Kummer refinements, strict-etale restrictions, and supplied finite semilinear descent. They identify actual functors on the indicated coefficient source. Their selected adic outputs are uniformly bounded and derived cartesian.

This closes the local star comparison for this vertical lisse domain. It does not assert essential surjectivity from every abstract cross-stratum action coefficient to $\mathcal{L}_r$, and supplies no exceptional or incidence comparison.

Proof. Remove the closure of the generic complement near Y , as in Proposition E.13. The source there is the actual generic restriction of $\mathcal{L}_r$. Let E_r be its restriction to the relative geometric log fibre. Apply $R\varepsilon_{Y, \text{rel}, *}$ to the specialization unit

$$E_r \longrightarrow (R\Psi^{\log} \mathcal{L}_r)|_{\tilde{Y}}.$$

Lemma E.5 makes this an equivalence and identifies its target with the restricted ordinary nearby complex. Proposition C.23 identifies its source with continuous invariants and with the actual root structural star. This constructs γ on the entire face etale topos, including its maps; it is not a reconstruction from equal stalk dimensions. Proposition E.13 constructs the actual completion comparison β and proves it is an equivalence. The rightmost term

is therefore cohomology on the actual tube, even when that tube is an annulus or a higher toric domain.

For tameness, the geometric logarithmic special fibre is constructed using prime-to- p roots of the base parameter. Its descent action, and that of the restriction of the given X_{ket} coefficient, factor through tame base inertia. Consequently wild inertia acts trivially on the cohomology of $R\varepsilon_{Y,\text{rel},*}E_r$. The ordinary comparison just constructed is equivariant for the full retained inertia action: the specialization unit and logarithmic forgetful exchange are formed before forgetting that action and are natural for base automorphisms. Thus it proves this assertion for the actual nearby complex. Lemma E.15 now makes its tame-invariant inclusion an equivalence. Here the log extension and verticality ensure the required wild inertia triviality.

The specialization unit is the unit of a geometric pullback/right-adjoint pair. The forgetful exchange in the displayed log/ordinary comparison is the mate of the strict pullback comparison of those same sites. Both are morphisms of the induced lax symmetric-monoidal functors: their adjoints are composites of strong tensor inverse images and the monoidal counits. Their invertibility on the present source was proved before taking the inverse exchange used in γ . Proposition E.14 gives precisely this structure for β . Invariants have the analogous lax structure and their inclusion is monoidal. These constructions prove the algebra/module claims for the specified zigzag, including higher multiplication cells. They also give the stated naturality under the geometric maps and coefficient morphisms.

Finally the continuous augmentation resolution has $d + 1$ finite free terms. On an input of amplitude $[a, b]$ it gives output amplitude contained in $[a, b + d]$, independent of r . The construction commutes with derived coefficient reduction on the selected class. The natural comparison zigzag transports this property to the actual analytic and nearby outputs. Only after this recovery do we take the inverse limit in r . No finite group bar has been replaced by a bounded complex before continuous passage. A subsequent direct image from Y to another base requires its own boundedness/preservation assertion. $\square$

Corollary E.7 (Actual tube cohomology and geometric refinement weights). *For the unit coefficient in Theorem E.6,*

$$R^j s_* R_r \simeq \bigwedge^j (L \otimes R_r)(-j), \quad 0 \leq j \leq d,$$

with zero in other degrees. Suppose the supplied compatible fs Kummer models over chosen tame base roots induce multiplication by ℓ^{m-n} on their relative characteristic lattices. Their actual tube pullbacks act on these sheaves by $\ell^{j(m-n)}$. These are geometric refinement maps between the covers.

Proof. The theorem identifies the actual derived output with continuous invariants by a natural geometric map. For the trivial coefficient, its degree-one cohomology is $\text{Hom}_{\text{cts}}(\Gamma, R_r) = L \otimes R_r(-1)$. Cup products compute its exterior powers by the augmentation Koszul resolution. The character pairing in the finite Kummer charts sends a root refinement to restriction along multiplication by ℓ^{m-n} on relative inertia, hence to the stated degree-one weight and its exterior powers. Naturality of γ and β identifies these maps with actual tube pullbacks. The bounded degree range and eventual vanishing of positive weights at fixed r are therefore now geometric assertions in this domain. They can be used in the actual finite-torsor filtration of Proposition E.10. $\square$

Remark E.8 (A horizontal-boundary example). Consider $X = \text{Spec } R[t]$ with the log structure of $\pi t = 0$ and $Y = \{\pi = t = 0\}$. Its chart is $P = \mathbf{N}^2$, $\rho = (1, 0)$; the base lift is primitive and the relative characteristic group has rank one. Nevertheless, for the constant

coefficient on the whole generic fibre, smooth base change gives $i_Y^* R\Psi_X R_r = R_r$ in degree zero. Proposition E.13 identifies this with the cohomology of the actual open-disc tube. Continuous invariants of the rank-one relative log inertia instead have a nonzero degree-one term $R_r(-1)$.

This log structure has a horizontal boundary and falls outside the vertical domain of Theorem E.6. The logarithmic generic coefficient corresponds on the ordinary generic fibre to $Rj_* R_r$ for $j : \mathbf{G}_m \rightarrow \mathbf{A}^1$, not to the constant sheaf on $\mathbf{A}^1$. The horizontal generic boundary also meets Y . Thus neither this forgetful direct image nor the open-boundary condition may be dropped.

E.2 Geometric sources and comparison maps

Fix a geometric generic point $\bar{\eta}$ and a completed algebraic closure $\widehat{K}$. Let $\mathrm{sp}_{\bar{\eta}} :]S_{\bar{\eta}}^{\mathrm{an}}/\Delta \rightarrow \mathcal{S}$ be geometric specialization. The analytic functors use the full geometric étale site, not arithmetic K -analytic cohomology. The actual log source can carry external tame transport, which its geometric star comparisons retain. This supplies neither an equivariant structure on every abstract root coefficient nor an infinite tame-equivariant finite-carrier trace target. Additional finite descent is treated with its supplied action diagram. For the integral Kummer local system $E_\bullet = (E_r)_r$, set

$$A_{*,r}(E_r) := R\mathrm{sp}_{\bar{\eta},*} E_r.$$

For the supported comparisons in this paper the target is a geometric point. Analytic compact support is the actual functor

$$A_{!,r}(E_r) := R\Gamma_c(T, E_r). \quad (42)$$

When computed through a compactification, extension by zero is followed by the proper analytic structural direct image. This formula defines neither $\nu_!$ nor an exceptional functor for relative specialization. Define $A_{*,\mathrm{ad}} = R\varprojlim_r A_{*,r}$ only with its stated coefficient system; its bounded constructible adic recovery is a conclusion of the relevant comparison, not automatic. Write $A_\star = A_{*,\mathrm{ad}}[1/\ell]$ afterwards.

Let $\bar{\mathcal{U}}$ be the fixed proper log compactification, $\bar{j}_\eta : \mathcal{U}_\eta \hookrightarrow \bar{\mathcal{U}}_\eta$ its generic open immersion, $\bar{i}_Y : Y \hookrightarrow \bar{\mathcal{U}}_s$ the selected locally closed face, and $f_Y : Y \rightarrow \mathcal{S}$ its structural map. The face need not be proper: being locally closed in a proper model does not imply properness. Require f_Y to be in the compactifiable domain when using $Rf_{Y,!}$, and otherwise leave the supported functor undefined. For shorthand put $\Psi_r = R\Psi_{\bar{\mathcal{U}}}^t$ at precision r . The two geometrically typed candidates are

$$\begin{aligned} N_{*,r}(E_r) &:= Rf_{Y,*} \bar{i}_Y^* \Psi_r(R\bar{j}_{\eta,*} E_r), \\ N_{!,r}^{\mathrm{cost}}(E_r) &:= Rf_{Y,!} \bar{i}_Y^! \Psi_r(R\bar{j}_{\eta,!} E_r). \end{aligned} \quad (43)$$

We use $N_{!,r}$ for the second, costalk construction. The superscript may be restored when comparing conventions. Its identification with analytic compact support on the local supported domain is Theorem 5.5; the needed nearby-cycle duality is Proposition 5.2. If f_Y is proper, only the outer $!$ may be replaced by $*$.

Definition E.9 (Kummer-basic source and relative tame inertia). Let $\mathcal{L}$ be a stratified-perfect constructible adic object on $\mathcal{S}$, and let V_{cont} be an Iwasawa-admissible coefficient. The notation $\mathcal{L} \boxtimes V$ denotes its pullback to a realization chart tensored with the Kummer local system defined by V . On the root/action side this is defined intrinsically. On the analytic/nearby side it is only notation for an actually supplied common source realization; it is not a construction for every constructible $\mathcal{L}$. In particular, $\nu^* \mathcal{L}$ is a quasi-étale coefficient

and need not come from an algebraic generic object, by Proposition F.2. The local theorem supplies that realization in its stated domain. For a general statement, a *Kummer-basic common source category* means a specified enhanced category with the actual realization functors and coefficient maps. Closure under finite cones and retracts uses maps in that source category; tensoring by a perfect base object requires its compatible actual realization as well. This does not make a stratified-perfect base object tensor-dualizable. Coefficient-dual cancellation uses a globally lisse-perfect tensor factor and the Verdier-reflexive domain of Definition A.10; the general-base adic duality conclusions use Lemma A.12. No such extra condition is added to the constructible star source. It is not automatically the full stable root subcategory generated by the displayed object values. In Theorem 4.2, Corollary 4.5 proves the required full mapping and cone closure. On the full proper-image source, Proposition F.1 prevents that automatic identification. This qualification is part of the source typing, not an assumed theorem lifting arbitrary root maps.

On the tame nearby side, the chosen base-root section singles out the relative characteristic kernel of the finite deck extension in Lemma C.14; on a split geometric fibre its limit is Γ_M . We write $R\Psi_{\text{rel}}^t$ for restriction to this relative inertia, not for invariants under base tame inertia. Nearby cycles retain their own geometric tame action. No global weak base-inertia action on the root coefficient category, or comparison with such an action, is asserted here. Additional equivariance requires a supplied action diagram on the chart, its compactification and the coefficients.

Proposition E.10 (Actual finite Galois descent and its comparison map). *Let $\pi : U \rightarrow S$ be a morphism of the actual coefficient topoi under consideration, and let $q : V \rightarrow U$ be an actual finite étale K -torsor, with K finite. For a bounded-below coefficient $\mathcal{E}$, writing $p = \pi q$, there is a canonical equivalence*

$$R\pi_* \mathcal{E} \xrightarrow{\sim} \text{Tot } C^*(K, Rp_* q^* \mathcal{E}).$$

It is natural for maps of these torsors and their quotient groups. On the unit coefficient it is multiplicative in the enhanced sense; coefficient comparisons are module comparisons over it. If $q^ \mathcal{E} = p^* E$ is a specified equivariant trivialization, the adjunction unit further supplies a natural map*

$$C^*(K, E) \longrightarrow \text{Tot } C^*(K, Rp_* p^* E) \simeq R\pi_* \mathcal{E}.$$

The second map is the coefficient comparison used in the subsequent continuous calculation.

Proof. The augmented Čech diagram of the finite étale cover resolves $\mathcal{E}$. This may be checked after the exact conservative pullback q^* , where the cover has a section and the augmented diagram is split. Its totalization is therefore $\mathcal{E}$ in the bounded-below derived category. Applying the right adjoint $R\pi_*$ commutes with this limit. The torsor identifications $V \times_U \cdots \times_U V = V \times K^a$ identify the resulting cosimplicial terms and all face and degeneracy maps with the full group-cochain diagram on $Rp_* q^* \mathcal{E}$. This proves the first formula with the full complex $Rp_* q^* \mathcal{E}$.

The map in the second formula is induced by the equivariant unit $E \rightarrow Rp_* p^* E$. All maps come from the actual cover and the pullback–pushforward adjunction. Consequently a refinement of torsors gives a commuting diagram, including its composition cells. Pullback is symmetric monoidal and its right adjoint has its canonical lax symmetric-monoidal structure; the unit and Čech maps respect these structures. The same augmented descent argument in commutative algebra objects and their modules gives the claimed multiplicative/module comparisons. Thus the enhancement belongs to the constructed map. There is no assertion here that an unbounded finite group bar commutes with derived coefficient reduction. $\square$

Lemma E.11 (Algebraic torus cohomology and root weights). *Let S be noetherian with ℓ invertible and use a split Kummer tower. Let $\pi : T \times_S \mathbf{A}_S^e \rightarrow S$, where $T = \mathbb{T}_{M,S}$. If E is a constructible R_r -sheaf on S , then*

$$R^j \pi_* \pi^* E \simeq E \otimes_{R_r} \bigwedge^j (M^\vee \otimes R_r)(-j), \quad 0 \leq j \leq d,$$

with zero in the other degrees. The identification is natural in E and base pullback. On these sheaves, Kummer refinement by $[\ell^{m-n}]$ acts as $\ell^{j(m-n)}$ in degree j ; deck translations act trivially on the exterior factor. This is a statement about relative cohomology sheaves, not a trivialization of the entire derived deck action.

Proof. The affine-space factor has ordinary cohomology E by homotopy invariance. The same fact and the one-dimensional torus calculation can be obtained uniformly from $\mathbf{P}_S^1$. If $p : \mathbf{P}_S^1 \rightarrow S$ and i is one of its standard sections, smooth purity and $pi = \text{id}$ give $i^! p^* E = E(-1)[-2]$. The projective bundle formula gives $Rp_* p^* E = E \oplus E(-1)[-2]$. The localization triangle for the complement of infinity cancels the second summand and gives the assertion for $\mathbf{A}_S^1$. Removing both zero and infinity instead gives a map from two copies of $E(-1)[-2]$ to this second summand; both section classes have degree one. Its cofiber has cohomology E in degree zero and $E(-1)$ in degree one. The relative degree-one generator is the Kummer class of the coordinate t . These calculations use the actual proper compactification and its purity maps, so apply to coefficients pulled from S and commute with base change.

Iterating the one-dimensional calculation and the projection formula gives the displayed exterior cohomology sheaves. No cohomology sheaf of the coefficient is assumed flat: the exterior factors are finite free, and the projective-bundle and localization computations are carried out on the coefficient itself. The Kummer sequence sends t^a to a times the class of t , proving the stated degree-one weight. Cup products of the coordinate classes give its j th exterior power. Translation $t \mapsto \zeta t$ changes the Kummer class only by a class from the base, which vanishes in relative $R^1 \pi_*$. It therefore acts trivially on the exterior cohomology factor. This last observation does not erase the higher derived coherence of the deck action retained in Proposition E.10. $\square$

Theorem E.12 (An actual algebraic torus star comparison). *Use the algebraic product of Lemma E.11. Let $\mathcal{E}$ be the Kummer local system associated to a finite K_{n_0} -action on a constructible base coefficient E . The actual geometric torsors $q_n : U_n \rightarrow U$ supply a canonical comparison*

$$C_{\text{cts}}^*(\Gamma, E) \xrightarrow{\sim} R\pi_* \mathcal{E}.$$

It extends to bounded derived Kummer coefficients by finite derived action descent and is compatible with their sheaf attaching maps. On the unit it is a comparison of commutative algebra objects; on coefficients it is a module comparison. On selected cartesian adic systems the outputs have bounded derived-cartesian recovery. This theorem is for the algebraic product $T \times \mathbf{A}^e$; it does not identify an analytic polydisc specialization with that algebraic projection.

Proof. For $n \geq n_0$, torsor descent gives the equivariant trivialization $q_n^* \mathcal{E} = p_n^* E_n$. Proposition E.10 constructs compatible maps

$$C^*(K_n, E_n) \longrightarrow F_n := \text{Tot } C^*(K_n, Rp_{n,*} p_n^* E_n) \simeq R\pi_* \mathcal{E}.$$

Take their filtered colimit. The left side is continuous cochains; the right sides are a compatible diagram of equivalences with the fixed actual direct image. We now prove that this comparison map is an equivalence.

Use the quotient Postnikov tower $\tau_{\geq j}$ of $Rp_{n,*}p_n^*E_n$, with arrows from j to $j+1$. It has length $d+1$ by Lemma E.11. After derived invariants the fibres of its successive arrows are

$$C^*(K_n, E_n \otimes \bigwedge^j (M^\vee \otimes R_r)(-j))[-j].$$

The transition on the positive- j piece is inflation followed by multiplication by $\ell^{j(m-n)}$, and is zero after sufficiently large refinement at fixed r . The degree-zero comparison induced by the unit is the identity on E_n . Exactness of filtered colimits and the finite length of this filtration therefore prove that the constructed map is an equivalence. This argument uses neither formality of the finite derived deck complex nor interchange with an arbitrary infinite totalization.

The associated-sheaf functors at finite level are exact on the coefficient abelian categories and compatible with inflation. The finite derived descent of Theorem B.5 therefore extends the construction to the stated derived source, including its global sheaf maps. Equivalence on bounded objects follows by finite truncation triangles in that larger constructible category, without requiring their cohomology objects to be perfect. The multiplicative/module structure is inherited from the actual unit maps and descent diagrams of Proposition E.10. Filtered colimits preserve these commutative algebra and module structures. Finally the continuous augmentation Koszul calculation identifies the already compared actual output with a finite construction on E_r , proving uniform boundedness and derived coefficient recovery on the selected adic class. Rationalization is performed only afterwards. $\square$

E.3 Completion comparison and its tensor structure

Proposition E.13 (Ordinary nearby comparison away from the generic boundary). *Let X/R be the chosen algebraization, $Y \subset X_s$ the selected locally closed subscheme, and $j : U \hookrightarrow X_\eta$ the supplied open. Assume*

$$Y \cap \overline{X_\eta \setminus U} = \emptyset \quad \text{inside } X.$$

Let $\mathcal{E}$ be a bounded constructible R_r -complex on U . For ordinary geometric nearby cycles, retaining the full inertia action, there is a canonical comparison equivalence

$$i_Y^* R\Psi_X(Rj_*\mathcal{E}) \xrightarrow{\sim} R\mathrm{sp}_{Y, \bar{\eta},*}(\mathcal{E}^{\mathrm{an}}|_{Y[\bar{\eta}]}).$$

After the structural direct image from Y to the face base this compares the analytic star functor with the ordinary nearby star functor. It is natural on algebraized finite Kummer diagrams for which the displayed boundary condition holds. It does not by itself replace ordinary nearby cycles by tame nearby cycles, and it asserts no compact-support or incidence comparison.

Proof. Remove the closure of $X_\eta \setminus U$ from X , then choose an open neighbourhood on which Y is closed. This leaves the formal completion along Y unchanged, and the entire generic fibre of this neighbourhood lies in U . On this neighbourhood the coefficient $Rj_*\mathcal{E}$ restricts to $\mathcal{E}$ on the generic fibre. This uses the boundary condition while keeping the coefficient on the specified generic open U .

Use the standard comparison morphism from algebraic nearby cycles restricted to Y to the formal nearby cycles of the completion $\hat{X}_Y$. It is induced by restricting algebraic etale neighbourhoods to their completions and then to their analytic generic fibres. Those restriction maps give compatible maps of coefficient complexes and hence a derived comparison morphism. Berkovich's comparison theorem for completion along a subscheme [8, Theorem 3.1] identifies its cohomology sheaves for prime-to-residue-characteristic constructible torsion coefficients. Here the algebraization is finite type over the complete valuation ring R ,

the localized Y is closed in its special fibre, and $\widehat{X}/_Y$ is a special formal completion, as in that theorem. The coefficient is algebraic constructible R_r -torsion on the localized generic fibre, so its torsion order is invertible in k . The theorem identifies the canonical map from restricted algebraic nearby cycles to the formal nearby functor on this completion. Proposition 1.3 identifies its generic fibre with the tube. Geometric base change to the chosen completed algebraic closure gives the displayed geometric version. Finite truncations give bounded complexes.

The formal nearby functor is the actual geometric specialization direct image on the generic fibre of $\widehat{X}/_Y$, and that fibre is the tube $]Y[_{\bar{\eta}}$. This identifies the target in the display. Naturality is the naturality of the restriction maps used to construct the comparison, so applies to the stated finite Kummer diagrams. Applying $Rf_{Y,*}$ proves the facewise assertion. The proposition uses this published ordinary comparison in its specified domain. Proposition E.14 identifies this construction with a tensor-compatible specialization mate and supplies its algebra/module enhancement. $\square$

Proposition E.14 (The tensor structure of completion comparison). *The ordinary comparison of Proposition E.13 is a morphism of lax symmetric-monoidal derived functors with their coefficient-module actions. It is natural for the algebraized finite Kummer diagrams in that proposition and for coefficient change wherever the derived comparison is defined. This statement supplies the multiplicative clause of Proposition E.13 in its stated geometric domain.*

Proof. Topoi and the inverse-image transformation. Localize as in Proposition E.13, so that Y is closed and the generic boundary is absent. Put $A = R_r$. All categories in this proof are the stable localizations $D_A(\mathcal{Z})$ of complexes of A -module sheaves on the indicated topoi; functors are derived unless they are exact inverse images. The directions of the geometric morphisms are

$$\begin{array}{ccc} (X_{\bar{\eta}})_{\acute{e}t} & \xrightarrow{a} & X_{\acute{e}t} \xleftarrow{i} Y_{\acute{e}t}, \\ T_{\acute{e}t} & \xrightarrow{c} & (X_{\bar{\eta}})_{\acute{e}t}, \\ T_{\acute{e}t} & \xleftarrow{\mu} & T_{q\acute{e}t} \xrightarrow{\nu} Y_{\acute{e}t}. \end{array}$$

Here a is geometric generic inclusion, c is analytification restricted to the tube, and (μ, ν) is exactly Definition E.1. Thus the two inverse-image functors below both have domain $D_A(X_{\acute{e}t})$ and target $D_A(T_{q\acute{e}t})$:

$$\sigma : \nu^* i^* \longrightarrow \mu^* c^* a^*.$$

We specify its monoidal structure before taking a mate. For an etale $V \rightarrow X$, complete V along V_Y . Formal etale lifting identifies its generic fibre over T with the quasi-etale object defining $\nu^* h_{V_Y}$. The canonical map from that generic fibre to $V_{\bar{\eta}}^{\text{an}}$ defines σ_{h_V} . For a finite family $V_1, \dots, V_t$, completion and generic fibre identify the lift of $\prod_X V_j$ with the product of the lifts. The restriction map for this product is the product of the restriction maps, and the map for $V = X$ is the unit map. For a permutation or a substitution of finite families, both composites are this same restriction of etale objects. This gives the finite-arity symmetry, unit and substitution identities.

These constructions respect etale coverings and morphisms, so they extend by sheafification to inverse images on sheaves of sets. On free module generators they give

$$\sigma_{A[h_{V_1 \times_X \dots \times_X V_t}]} = \bigotimes_{j=1}^t \sigma_{A[h_{V_j}]}.$$

This is a natural transformation of exact inverse-image functors on module sheaves. Applying it degreewise to complexes and localizing at quasi-isomorphisms gives a specified transformation in the chosen derived enhancement: both functors preserve quasi-isomorphisms, and the

naturality squares descend with them. To compute its tensor structure, resolve a complex of presheaves by a semi-free complex, with cells sums of shifts of free representables, and then sheafify. Exact sheafification gives a resolution of the original sheaf complex. Its cell filtration makes it K-flat: free representable sheaves are flat, tensoring an acyclic complex with each cell is acyclic, and filtered colimits of module sheaves are exact. The inverse images of these resolutions have the same property, since inverse image preserves free modules, sums and these filtrations. Thus the sheaf-level finite-arity identities above compute the derived tensor identities on these resolutions. They are compatible with quasi-isomorphisms and hence descend, together with symmetry and substitution, to the localization. This defines the symmetric-monoidal transformation σ ; it need not be an equivalence.

The mate, including its multiplication. Write $B = \mu^*c^*$, $Q = i^*Ra_*$ and $P = R\nu_*B$, functors from $D_A((X_{\bar{\eta}})_{\acute{e}t})$ to $D_A(Y_{\acute{e}t})$ for Q, P . The composite in the quasi-etale category is

$$\psi_F : \nu^*QF \xrightarrow{\sigma_{Ra_*F}} Ba^*Ra_*F \xrightarrow{B\epsilon_{a,F}} BF.$$

Its adjoint is $\tau_F : QF \rightarrow PF$. The lax products on Ra_* and $R\nu_*$ are the mates of their strong-monoidal inverse images; explicitly they are adjoint to the tensor products of the corresponding counits. For a finite family $F_1, \dots, F_t$, the two maps

$$\bigotimes_j \nu^*QF_j \longrightarrow B(\bigotimes_j F_j)$$

obtained either by multiplying in Q and then using ψ , or by using the individual ψ_{F_j} and multiplying in B , agree. Indeed substitute the formula for ψ : the σ portions agree by its finite-arity product identity, and the remaining identity is the defining counit formula for the lax product of Ra_* . The empty family checks the unit in the same way. Adjunction with ν^* turns these equalities into exactly the lax-monoidal squares for τ . Permutations and successive substitutions give the associativity and symmetry cells because their inverse-image identities were already the same restriction maps. This is also the monoidal calculus of mates of [2, Theorem 3.4.7], which preserves composition of these transformations. It gives algebra and coefficient-module compatibility, with no appeal to equality of cup products on cohomology.

Identification of the map to which comparison applies. For each etale neighbourhood $V \rightarrow X$, the map defining ψ restricts a section on $V_{\bar{\eta}}$ to the generic fibre of its completion along V_Y . The last arrow is precisely evaluation of the generic pullback/direct-image counit on that section. More explicitly, choose a K-injective representative I of F . Since a^* is exact, a_*I computes Ra_*F . The chain map

$$\nu^*i^*a_*I \xrightarrow{\sigma} Ba^*a_*I \xrightarrow{B\epsilon} BI$$

represents ψ_F . Resolve $BI \rightarrow J$ by a K-injective complex on the quasi-etale topos and take the ordinary $\nu^* \dashv \nu_*$ adjoint of this composite. The resulting map $i^*a_*I \rightarrow \nu_*J$ represents τ_F . No preservation of K-injectives by inverse image is needed. On etale neighbourhoods it is exactly restriction to the completed generic fibre followed by the resolution map. These are the restriction maps defining the algebraic-to-formal nearby comparison in [8, §§2–3, Theorem 3.1]; the construction uses the same generic pullback and specialization topoi. Derived adjunction identifies their mate with τ , independently of resolutions. Thus the comparison theorem applies to this specified map. That theorem supplies invertibility on the stated constructible prime-to-residue-characteristic source; the preceding construction supplies its multiplicative enhancement.

For a morphism of the supplied algebraized finite Kummer diagrams, restriction to completion composes with the induced map of generic fibres. The corresponding σ and counit

squares therefore paste to the same ψ , and taking mates gives naturality of τ . Coefficient extension carries free module restriction maps and evaluation to their coefficient-changed versions. The same argument gives the coefficient-change squares, without declaring arbitrary right-direct-image base-change maps invertible. Neither an invertible generic/closed base-change square nor an exceptional specialization functor is asserted. $\square$

Lemma E.15 (The precise passage from ordinary to tame nearby cycles). *Let P be wild inertia, a pro- p group, and $\ell \neq p$. On smooth P -action sheaves of R_r -modules, invariants are exact. For a bounded complex C with such an action, the canonical map*

$$C^P \longrightarrow C$$

is an equivalence if P acts trivially on every cohomology sheaf of C . Accordingly the ordinary comparison in Proposition E.13 gives the desired tame comparison only after this wild-triviality condition is proved for the relevant nearby complex (or if no wild inertia is present).

Proof. To lift an invariant local section across an epimorphism, first choose a local lift. Smoothness makes its orbit factor locally through a finite p -group. Averaging that lift over this group is allowed in R_r , since its order is invertible, and gives an invariant lift. This proves exactness as a sheaf statement. Thus $H^j(C^P) = H^j(C)^P$, and the assumed triviality makes the canonical map an isomorphism on every cohomology sheaf. This proves the criterion. It does not prove wild-triviality from the word “Kummer” or from a coincidence of relative inertia groups. $\square$

Lemma E.16 (Derived direct image under localization of topoi). *Work with the étale and Kummer-étale sheaf topoi and their limiting localizations used here. Let $f : \mathcal{E} \rightarrow \mathcal{F}$ be a morphism of these topoi, and let U be an object of $\mathcal{F}$, and use constant coefficients R_r . Write $p : \mathcal{F}/U \rightarrow \mathcal{F}$ and $p' : \mathcal{E}/f^*U \rightarrow \mathcal{E}$ for the localizations and $f_U : \mathcal{E}/f^*U \rightarrow \mathcal{F}/U$ for the induced morphism. For $K \in D^+(\mathcal{E}, R_r)$ the canonical base-change map is an isomorphism*

$$p^* Rf_* K \xrightarrow{\sim} Rf_{U,*} p'^* K.$$

It identifies the pullback of the adjunction unit of f with that of f_U and is compatible with further localization.

Proof. Evaluation on $V \rightarrow U$ identifies both underived direct images with evaluation on f^*V . The exact restriction p'^* preserves injectives because its left adjoint $p'_!$, the sheafified direct sum over the slicing object, is exact. Restricting a bounded-below injective resolution therefore proves the displayed derived exchange. The same evaluation identity identifies units; iterated slicing gives compatibility with composition and its cells. $\square$

Lemma E.17 (The derived analytic site model and its comparison map). *Let T be the analytic tube in the stated geometric source domain. At precision r use the stable infinity-categories $D(\mathrm{Sh}(T_{\acute{e}t}, R_r))$ and $D(\mathrm{Sh}(T_{q\acute{e}t}, R_r))$, obtained by localizing complexes at quasi-isomorphisms. K -injective representatives compute right adjoints and derived Hom; K -flat representatives compute tensor. For bounded constructible F the canonical unit map*

$$R\Gamma(T_{\acute{e}t}, F) \longrightarrow R\Gamma(T_{q\acute{e}t}, \mu^* F)$$

is an equivalence in $D(R_r)$. It respects evaluation for dualizable coefficients and pullback by finite étale torsors.

Proof. The exact strong tensor functor μ^* induces a functor on the stable localizations with its derived adjunction unit. For an ordinary étale abelian sheaf, the canonical comparison on sections and cohomology is [9, Theorem 3.3(i)–(ii)], applied to the identity quasi-étale map $T \rightarrow T$. This published theorem applies to analytic spaces without a compactness assumption. Its degree-zero map is the inverse-image unit; the higher maps are its derived cohomology maps. Finite truncation triangles therefore give the asserted equivalence for bounded complexes. The tensor and mapping assertions use the following argument in the selected derived enhancement. For dualizable E , $R\mathcal{H}om(E, F) = E^\vee \otimes^L F$, and $R\mathcal{H}om(E, F) = R\Gamma R\mathcal{H}om(E, F)$ determines the mapping spectrum. Strong tensor inverse images preserve evaluation. Torsor site maps, their units, and all Čech face and degeneracy maps pass to these same localizations. The exact tensor associated-sheaf functors likewise derive and commute with inflation before any full-faithfulness claim. Finite-action descent defines their continuous realization; Lemma 4.3 checks its mapping map. For cartesian adic systems, the universal property of the category limit computes mapping spectra as limits of the finite-precision mapping spectra. Its selected full subcategory has the same maps. $\square$

F Examples delimiting the coefficient source

Proposition F.1 (Proper-image presentations cannot lift all root morphisms). *Proper-image coefficients need not admit arbitrary root-morphism lifting to generic or analytic coefficient maps, even for one fixed strict proper map. In particular the comparison of star outputs cannot be promoted to a fully faithful identification of the full realized analytic coefficient category with its root outputs while retaining all these presentations.*

Proof. Let $X = \mathbf{A}_R^1$, with only the vertical log structure, and set

$$X' = \mathrm{Spec} R \amalg \mathrm{Spec} R, \quad h = (t = 0) \amalg (t = \pi) : X' \longrightarrow X.$$

Each component is a closed immersion. Their disjoint union is a finite, strict proper log morphism; X' is fs saturated log smooth over the log trait. Take $Y = X_s = \mathbf{A}_k^1$. Its relative inertia rank is zero. Both components of Y' map to $0 \in Y$.

The two log-lisse perfect inputs $(R_r, 0)$ and $(0, R_r)$ have root outputs

$$E_0 = i_{0,s,*} R_r = E_\pi$$

and actual generic outputs

$$F_0 = i_{0,\bar{\eta},*} R_r, \quad F_\pi = i_{\pi,\bar{\eta},*} R_r.$$

Their analytifications on the closed unit-disc tube have disjoint closed point supports, since $0 \neq \pi$ in $\bar{K}$. For disjoint closed immersions i_a, i_b , exceptional restriction satisfies $i_a^! i_{b,*} = 0$; adjunction therefore gives

$$R\mathrm{Hom}_{T_{\acute{e}t}}(F_0^{\mathrm{an}}, F_\pi^{\mathrm{an}}) = 0.$$

The same vanishing holds on the algebraic generic fibre. On the root side, the identity of the common nonzero coefficient gives

$$\mathrm{Hom}_{Y_{\acute{e}t}}(E_0, E_\pi) \supseteq R_r \neq 0.$$

Hence that identity cannot lift to a generic or analytic morphism between the two prescribed realizations. Already at degree zero full faithfulness and arbitrary morphism lifting are impossible. The same obstruction holds for the cartesian integral systems; it persists after their $\mathbf{Q}_\ell$ -realization. No individual R_r -coefficient is rationalized.

Proper base change still computes the star images in this example; the loss occurs in passing from the coefficient to its star image. $\square$

Proposition F.2 (An obstruction to the unrestricted common coefficient source). *Let $X = \operatorname{Spec} R[t]$, $Y = X_s = \mathbf{A}_k^1$, with k algebraically closed, and let T be the geometric closed unit-disc generic fibre of $\widehat{X}/Y$. Set $L = i_{0,*}R_r$ on $Y_{\text{ét}}$. This is a constructible coefficient of finite Tor-amplitude, allowed by the earlier stratified coefficient condition. There is no algebraically constructible bounded complex F on $X_{\bar{\eta}}$ satisfying*

$$\mu^*(F^{\text{an}}|_T) \simeq \nu^*L \quad \text{on } T_{q\text{ét}}.$$

Thus the unrestricted prescription “pull back every constructible base coefficient” cannot also require that same analytic coefficient to come from a common algebraically constructible generic source.

Proof. At a classical geometric point $t = a$ with $|a| \leq 1$, the right side has degree-zero stalk R_r if $\bar{a} = 0$, and zero if $\bar{a} \neq 0$. These are the stalks of the actual inverse image under ν . If the displayed equivalence existed, exactness of the inverse images would impose these stalk values on the constructible sheaf $H^0(F)$. A constructible sheaf on the algebraic affine line is locally constant on the complement of a finite set. That complement is connected, so the cardinality of its stalks is constant. However, both sequences

$$a_m = \pi^m, \quad b_m = 1 + \pi^m, \quad m \geq 1,$$

lie in T and contain infinitely many distinct points. Outside any finite exceptional set their required stalks have respectively cardinality $|R_r|$ and 1. This is a contradiction. The argument uses only classical stalks as a necessary condition; it does not claim that they alone classify all quasi-étale sheaves.

This is a source obstruction, not a counterexample to the already proved star theorem on an algebraized lisse source. Choosing instead an algebraic skyscraper at $t = 0$ changes the analytic coefficient: its support is a single generic point rather than the whole residue disc. Equal nearby outputs would not identify those two sources. Nor does the proposition assert that a comparison formulated entirely on formal or quasi-étale coefficients is impossible; that is a different realization datum which must be defined and proved separately. $\square$

Proposition F.3 (The standard Artin fan is not a third realization). *The local comparison cannot be extended by replacing the relative Kummer face with the ordinary unrigidified Artin fan. For $P = \mathbf{N}^2$ and base direction $\rho = e_1 + e_2$, the closed face of $[\mathbf{A}^2/\mathbf{G}_m^2]$ retains a residual $B\mathbf{G}_m$ after stacky quotienting by the base diagonal. At a strict geometric point $\bar{s} \rightarrow S$, its lisse-étale ℓ -adic cohomology is*

$$H_{\text{lisé}}^{2j}(B_{\bar{s}}\mathbf{G}_m, \mathbf{Q}_{\ell}) \simeq \mathbf{Q}_{\ell}(-j), \quad H_{\text{lisé}}^{2j+1} = 0,$$

with the universal Chern class in $H^2(B\mathbf{G}_m, \mathbf{Q}_{\ell}(1))$. In contrast, the relative Kummer cochain complex at $\bar{s}$ is finite:

$$\operatorname{Sym}((\mathbf{Q}_{\ell}(-1))[-1]) \simeq \mathbf{Q}_{\ell} \oplus \mathbf{Q}_{\ell}(-1)[-1].$$

Thus the ordinary Artin-fan leg is not equivalent to the analytic or tame nearby leg, even on the constant coefficient.

Proof. The quotient description leaves the residual diagonalizable stabilizer $\mathbf{G}_m$. Its ordinary classifying-stack cohomology is polynomial in the universal Chern class, hence unbounded, while the Kummer cochain model of the relative characteristic lattice has cohomological dimension one. The graded dimensions differ, so no equivalence can exist. Rigidifying the base direction and retaining the Kummer site is exactly what removes this obstruction in the relative root construction. $\square$

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